CAMP MACARIO SAKAY WEB SECURITY SYSTEM: IMPLEMENTATION OF FACIAL IDENTIFICATION AND RECOGNITION USING CONVOLUTIONAL NEURAL NETWORK (CNN)

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In Partial Fulfillment of the Requirement for the Degree Bachelor of Science in Information Technology Specialized in Web and Mobile Application Development

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The Researchers

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ABSTRACT

This study developed and evaluated a Web-Based Facial Recognition Security System for Camp Macario Sakay to strengthen identity verification, improve monitoring accuracy, and modernize security operations within the camp. The system incorporates real-time facial detection, automated logging, and a web-based interface designed to replace traditional manual verification methods. To measure its technical quality and user acceptance, the study applied the ISO 25010 Software Quality Model and the Technology Acceptance Model (TAM). IT experts rated the system as excellent in functional suitability, reliability, performance efficiency, and portability, with usability and security identified as areas needing further enhancement. End-user evaluation through TAM revealed strong agreement regarding the system's usefulness, ease of use, positive attitude toward use, and intention to adopt the platform. Overall, the results confirm that the system is effective, user-friendly, and capable of supporting improved and efficient camp security operations. Recommendations for refining security and usability features are provided to guide future system development.

Keywords: *Facial Recognition, Web-Based System, ISO 25010, Technology Acceptance Model, Security System, User Evaluation, Camp Macario Sakay.*

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DEFINITION OF TERMS

Automated Logging	A feature of the system that automatically records entry data such as detected faces, timestamps, and access events without requiring manual input from security personnel.
Attitude Towards Use	The overall perception or feeling of users regarding their experience in using the system, whether positive or negative.
Behavioral Intention to Use	The capability of the system to operate reliably with other existing platforms, devices, or technologies.
Compatibility	The capability of the system to operate reliably with other existing platforms, devices, or technologies.
Facial Recognition	A biometric method that identifies or verifies individuals based on unique facial features. In this study, it is used for secure identity verification within the camp.
ISO 25010	An international standard for software quality evaluation, assessing attributes such as functionality, reliability, usability, efficiency, maintainability, compatibility, security, and portability.
Maintainability	The system's ability to be updated, modified, or repaired with ease in order to improve performance or fix issue

Perceived Ease of Use	The degree to which users find the system straightforward, simple to learn, and easy to operate.
Perceived Usefulness	The extent to which users believe the system enhances performance and contributes positively to security operations.
Portability	The ability of the system to be transferred or used across various hardware or software environments with minimal adjustments.
Real-Time Monitoring	The capability of the system to detect, track, and record individuals instantly as they enter the premises, supporting immediate security actions.
Security System	A digital solution designed to protect facilities from unauthorized access. In this study, it includes automated facial recognition and monitoring features.
Technology Acceptance Model	A framework used to assess user acceptance of the system based on perceived usefulness, perceived ease of use, attitude toward use, and behavioral intention to adopt the technology.
Technology Acceptance Model	A measure of how user-friendly, accessible, and intuitive the system is for security staff and other end users

Web-Based System	A software application accessible through an internet browser, requiring no local installation. It enables online verification, data access, and system control.

CHAPTER I

INTRODUCTION

Project Context

In today's world, security systems are essential for protecting an establishment from potential threats, ensuring the safety of individuals and assets, and maintaining smooth business operations (Martins Ajaero, 2020). Information Technologies (IT) and electronics opened up new opportunities for surveillance, increasing its effectiveness and resilience. There are cameras everywhere. They can be found, among other places, at retail establishment, manufacturing floors, residences, apartments, landbank, schools, financial organizations, and transit hubs. This security camera uses include anything from quality control monitoring to basic security safety. (Arshiya Jabeen, 2013).

The Philippines had one of the highest rates of the imprisonment per hundred thousand people. Poorer neighborhoods and locations with more inhabitants and higher unemployment rates have notably high crime rates. By Christy Balita on Statista (2023). According to Vito Barcelo from Manila Standard (2024) Filipino families and other establishments who were victimized by common crimes increased to 6.1% in September 2024, according to the latest Social Weather Station survey. The survey said that 6.1 percent of Filipinos reported to be victimized by any of the common crimes (pickpocketing or robbery of personal property, break- ins, carnapping, and physical violence) within the past six months. This is 2.3 points above the 3.8% in June 2024 and the highest since the 8.1% in

September 2023, the survey said by Social Weather Stations (SWS, 2024).

Nowadays, every establishment is facing serious security concerns, to achieve the level of security they require many staff with specialized training, according to the Philippine Association of Detective & Protective Agency Operators (PADPAO, 2021). Because they are human , these employees make mistakes that might compromise the security level.

A Face Recognition Security System, which can identify intrusions into restricted or high security locations and reduce human error, is one suggested solution to the aforementioned problem. (Michel Owayjan, et al., 2023). Face recognition is an important function of surveillance systems, with the help of Convolutional Neural Network (CNN). Verifying and identifying individuals who appear in a scene that is frequently caught by a dispersed network of cameras, where the identity can now be precisely determined (Yahia Said, et al., 2020). Using Convolutional Neural Network (CNN) as In essence, face recognition software compares an individual's geometric and facial attributes. The conventional approach stores people's photos in a database so they may be compared to photos that were taken. A variety of characteristics were retrieved and utilized as face features, such as corners, edges, texture descriptors, etc. For categorization, these characteristics were used with machine learning methods. (Ranjeeth K., et al., 2023).

The Camp Macario Sakay faces challenges when it comes to identifying every person that was entering the establishment. These challenges include the lack of a centralized identification system, limited manpower for manual verification, the possibility of human error in recognizing individuals, and the absence of real-time monitoring capabilities.

Having an open gate for everyone, the camp wants to have another layer of security to accurately distinguish between employees, civilians, and even recorded criminals to prevent entry by unrecognized or unverified individuals and ensure the safety and integrity of the premises.

The development of a web security system using Convolutional Neural Network (CNN) for facial identification and recognition adds another layer of security in Camp Macario Sakay by providing an intelligent, automated solution for identifying and categorizing individuals entering the premises. This system serves as a reliable tool to distinguish employees, civilians, and recorded criminals in real time, helping security personnel make informed decisions quickly. With the use of a standard webcam installed at the entrance, the system can seamlessly scan faces. By integrating this technology, the camp enhances its security infrastructure, ensures efficient access management, and minimizes potential threats through advanced Convolutional Neural Network-based surveillance.

Project Objectives

The main objective of this study is to develop a web-based facial recognition security system for Camp Macario Sakay aiming to enhance the camp's security by providing accurate identification and recognition on every person entering the premises. Furthermore, the system specifically aims to:

1. To develop a web-based Facial Recognition security system using Convolutional Neural Network (CNN) that can automatically detect, identify, and classify individuals at the camp entrance.
2. To integrate an additional security surveillance feature that

displays, tracks, and logs facial data as individuals are identified.

3. To evaluate the system's software quality through the ISO 25010 and Technology Acceptance Model (TAM), focusing on attributes such as functionality, usability, reliability, and maintainability.

Project Purpose

The purpose of this project is to enhance the security infrastructure of Camp Macario Sakay through a web-based facial recognition system using Convolutional Neural Network (CNN) technology. This system aims to support the security sector by providing an intelligent solution for accurate identity verification and access control. It identifies employees, civilian, and recorded criminals in real time using a standard webcam, helping to create a safer environment. This project contributes to SDG 9: Industry, Innovation, and Infrastructure by promoting the use of advanced technologies in public security systems. It also aligns with SDG 11: Sustainable Cities and Communities by supporting safer and more secure public environments. Additionally, it supports SDG 16: Peace, Justice, and Strong Institutions by ensuring efficient, transparent, and secure facility operations. Through these goals, the system strengthens both physical and digital security. Ultimately, the project encourages innovation and responsible technology use in the public safety sector.

Safety and Security Sector. This study aims to promote enhanced safety and security within institutional environments. By utilizing advanced facial identification technology, the system provides an intelligent and automated approach to access control, allowing for accurate identification of

individuals such as employees, civilians, and recorded criminals. This contributes to creating a safer and more organized environment, where entry is efficiently monitored and managed using a standard webcam setup.

Local Government Units. This study promotes new methods of enhancing public safety and overall facility security through the integration of advanced facial recognition technology. By adopting this system, LGUs can improve access control in government facilities and public spaces by accurately distinguishing individuals as employees, civilians, or persons of interest. This not only strengthens community trust but also encourages the modernization of local security infrastructure.

Future Researchers. This study is beneficial for future researchers, providing insights into the development and implementation of web-based facial recognition systems using Convolutional Neural Networks (CNN). It serves as a reference for building similar security systems that prioritize accuracy, efficiency, and innovation. Additionally, it opens opportunities for further enhancements, such as emotion detection, multi-camera integration, or broader applications in other sectors like education, healthcare, and transportation.

Scope and Limitation of the Study

The scope of this study is to design and develop a web-based security system using Convolutional Neural Network (CNN) for facial identification and recognition, for enhancing the monitoring and security measures. This project is implemented and tested at Camp Macario Sakay, which is located in Los Baños, Laguna. The system consist of four modules:

Login Module. This module lets the Admin and Super Admin securely sign in to access the system. It verifies credentials, starts a user session, and routes each user to a role-based dashboard. Basic safeguards (hashed passwords, input checks, and login attempt logging) help keep access secure.

Dashboard Module. This module serves as the system's main overview. It displays key counters for Employees, Persons of Interest, Total Logs, and Detections Today (including civilians), and shows quick charts for daily detections and top detected individuals. It also highlights the user's role and provides easy access to other pages for further review.

Monitoring Module. This module allows the security guard to monitor the individuals entering Camp Macario Sakay through a real-time facial recognition interface. The system displays the identity of each person detected by camera, automatically classifying them as an employee or recorded criminal while the unrecorded faces is marked as civilian, based on the data stored in the system.

Face Recognition. This system in this project aims to provide automated identification and classification of individuals entering Camp Macario Sakay. The system using a Convolutional Neural Network (CNN) is designed to accurately recognize faces captured through a webcam and classify them into categories such as employees, visitors, or recorded criminals. This process involves real-time facial feature extraction and matching against a pre-registered database, ensuring secure access control at the entrance of the facility. The face recognition module also integrates with other system components, such as logging and notifications, to notify

security personnel of detected individuals.

Convolutional Neural Network. It is responsible for processing and analyzing facial images captured by the webcam in real time. It extracts key facial features and compares them with the stored data to determine whether the individual is an employee, a recorded criminal, or a civilian. The CNN enhances the accuracy of facial recognition by learning from patterns in the dataset and continuously improving its classification performance. The CNN plays a critical role in ensuring that the system can automatically and intelligently identify individuals as they enter Camp Macario Sakay.

Logs Module. This module is responsible for automatically recording data after successfully identifies a person who were entering Camp Macario Sakay. Once an individual is classified as a civilian or criminal, the system logs essential details such as classification, date, time of entry, and ID. These records are stored in a secure database and can be accessed through a web-based dashboard by the administrator for review, auditing, or investigation purposes. This module helps maintain an organized and tamper proof record of entry activity within the premises.

Catalogue Module. This module manages the registration and listing of employees and recorded criminals. Authorized personnel can add individuals by capturing facial data and entering personal details like name, email, ID, contact, and classification. Once added, they become part of the system’s database for recognition by the Monitoring Module. The module also offers an organized list view for easy verification, updates, and record monitoring through a user-friendly interface.

Notification Module. This module is responsible for alerting security personnel when a civilian or a recorded criminal is detected by the system. Once the CNN- powered Monitoring Module identifies an individual, the Notification Module automatically sends a real-time alert to the security guard. There are two types of notifications: one for civilians, which serves as a standard entry alert, and another for criminals, which triggers a high-priority warning requiring immediate attention. These notifications are displayed through the system's web interface and are designed to enhance situational awareness and response time at the entrance gate.

Reports Module. This module provides a clear summary of activity for any selected date range. It displays key totals Employees detected, Persons of Interest (POI), Civilians, and Total Sessions followed by two side-by-side tables that list daily detections for employees and POIs. A "Recent POI Encounters" section includes larger snapshot previews for quick visual review. All results can be exported as a well- formatted PDF for filing, sharing, or audits, making it easy to track patterns and document incidents.

Limitations

The web-based security system for Camp Macario Sakay has several limitations. It heavily relies on the quality of the webcam feed and lighting conditions, which can affect facial recognition accuracy in low-light or unclear environments. Any facial covering, such as a mask, sunglasses, or full-face helmet, can obstruct key facial features, making it difficult for the system to recognize individuals. The Convolutional Neural Network (CNN) used for facial recognition is limited to the faces in its trained dataset, and may misclassify unfamiliar faces or individuals with similar features, such as twins.

The system cannot instantly adapt to changes in appearance, like aging, without retraining the model. The Logs Module does not record sensitive personal information like birthdates or addresses, nor does it support exporting data to formats like Excel or PDF. The Catalogue Module requires manual, individual registration, and does not support bulk uploads or batch processing, which could slow down data management. Updates in the Catalogue Module are limited to authorized personnel only, potentially delaying necessary changes. The Notification Module is restricted to on-screen alerts and lacks multi-channel notifications like SMS, email, or push notifications. Finally, the system does not offer advanced analytics, cross-device support, or mobile access, focusing instead on core desktop-based functionality for security personnel.

CHAPTER II

THEORETICAL FRAMEWORK OF THE PROJECT

Review of Related Literature, Studies, and Systems

The following information presented in this chapter were brought together from various writers and sources. Using these resources has significantly contributed to the development and completion of this capstone project. The information in the following relevant literature has helped the researcher conduct the study in more comprehensive and informative manner.

Face Recognition Security

Based on the research of (P. B. Balla and K. T. Jadhao, 2018), computers have been a major factor in the constant effort for higher security standards. Significant the field of facial recognition has made strides because of their accuracy, vast memory banks, and high processing power. In many face recognition tasks, computers are currently more proficient than humans. There is a limit to how many faces a human can recall. However, a computer has no limitations, thus it can be employed in situations where big databases of facial information are required. Numerous possible uses for such a facial recognition system exist, such as enhanced human-computer interaction, private security, and surveillance of crowds and airports. Such a technology is ideal for addressing security concerns and providing control flexibility for smart homes. It may also be used to identify desired individuals in in various setting, such as offices, business, and even airport. Between the other biometric one major benefit of facial algorithms is their user friendless.

According to (Danish Ali Chowdhry, et al., 2013), many applications,

such as biometric, security, surveillance, identification and authentication depend on facial recognition. Face recognition is computationally demanding since it typically uses high-dimensional information. Furthermore, incorrectly identified characteristics may slow down the recognition process even more. Many facial recognition algorithms have been developed to address these issues. (K. I. Kim, K. Jung, and H. J. Kim, 2012) adopted kernel principal component analysis (PCA) to extract the facial features. This technique produces good performance by using a polynomial kernel. In 2012, (J. Mazanec, and M. Melisek) suggested a combination of PCA and linear discriminant analysis (LDA) methods with support vector machines (SVM). This approach produces interesting results in terms of recognition, success and robustness, but has limitations over the variation in light and pose.And in May, 2019 Prafful Mishra proposed a statement on datadriveninvestor.com that the convolutional neural network (CNN) are better suited for image classification tasks compared to other neural networks due to their ability to exploit the spatial hierarchies and local patterns in images. CNNs use convolutional layers that can automatically extract features from images, making them more effective for tasks involving visual data. CNN can also handle a larger number of features and require fewer parameters, allowing for faster training and the use of less data. For instance, in pedestrian detection, traditional methods might still outperform CNNs due to the complexity of the task, which involves not just identifying pedestrians but also distinguishing between adults, children, dogs, and other objects, and understanding their motion and behavior. That is why CNNs are preferred for image classification due to their ability to automatically extract relevant

features, handle large feature sets efficiently, and exploit the spatial hierarchies in images, making them superior to other neural networks and traditional machine learning algorithms for such tasks.

As mentioned by (Trinanjana Bagch, et al., 2022), security is a matter of prime concern, whether it shall pertain to the security of data, wealth, and/or human life. As technology advances, there has been a rise in using IoT and biometric security authentication. Face recognition technology has emerged to develop into a readily available, collectible, and ubiquitous biometric security system. The same identification used by people is the basis for biometric face recognition. Recent years have seen a rise in the use of control engineering systems in daily life due to significant advancements in digital sciences. As a result of this breakthrough, systems that can interpret debt data, including images and videos, are now available. Finding the face in the database that most closely resembles the person's face is the goal. The requirement for face recognition in security systems has led to an increase in commercial interest and, consequently, the creation of useful technologies to facilitate the development of face recognition.

As mentioned by (A. Alshammari and D. B. Rawat, 2019), However, one of the difficulties that many security system confront today the fact that they cannot enable the system to execute some duties without an administrator monitoring the progress of the running system. Humans cannot work around the clock. There is a limitation to the time that human security personnel can work when they are still awake. Humans always need rests. Even when they are still awake, they are unable to stay in one spot for extended periods of time without becoming distracted. That’s why there are

some of the unique aspects of the multi-camera video security system that creates great impacts in the security industry. The technology aspect of the multi-camera video security system is great and it can help many individuals overcome any security issues they encounter in their lives. Using footage from cameras installed on the wall to cover a target region, our method could identify and verify a human target. The suggested method entails tracking and identifying potentially dangerous targets. After that, the decision making process determines weather or not the individual poses a threat.

Indicated by (Vetter, V. et al., 2017), automated face image processing has grown to be a significant market for software and hardware. The most important factor in the economic success of facial recognition systems is how easy it is for those who use them to take pictures of their faces. When face recognition system is used in the real world, finding faces and separating faces into groups are just important as the recognition process itself. It uses a low-level model to make face candidates and a neural network to check the candidate areas in the end. The approaches adopted were determine within the context of an unrestricted government and real-time functionality on constrained hardware resources. The system's solidity was enhanced by the utilization of several picture attributes that complemented one another.

Surveillance Security System

Based on the findings of **(**Xu, Z., et al., 2016**),** in the military, customs, police, firefighting, airports, railroads, urban transportation and many other public spaces, video surveillance is a comprehensive system with powerful preventive capabilities. It's comprehensive, timely, and visible information

content makes it a crucial component of the security system. Due to its extensive, clear, and precise information, video surveillance has emerged as the primary instrument, however, issues like “useful information and clues can’t be found immediately with video big data” reduce detecting efficiency during crime prediction and public security governance due to the widespread deployment of video surveillance system worldwide. Parsing and organizing video footage is crucial due to the growing need for video-based applications. Nonethless, there is still a lack of precise semantic comprehension and management of video material. Manual or semi-automated approaches are unable to close the semantic gap between low level characteristics and high level semantics. This works initially introduces video structural description (VSD), a semantic based methodology for organizing and expressing video material. Using spatiotemporal segmentation, feature selection, object identification, and semantic web technologies, video structural description seeks to parse video footage into text information. The used of VSD in technology in surveillance footage for public safety is described. VSD-based intelligent analysis of people and cars from security footage is shown. Additionally provided is the cloud enhanced platform for managing security footage. Finally, VSD-based applications are presented.

As indicated by (A. Shifa, et al., 2020), Video Surveillance (VS) systems are widely used in real-time abnormal event detection and autonomous video analysis. The videos featuring identifiable personal information taken by surveillance cameras in real time require privacy protection, as well as possibly the location of surveillance and/or certain

sensitive information. Within the Surveillance System, such videos are analyzed and stored in different devices. The diversity in devices' processing and storage, coupled with their network demands, make real-time surveillance systems complicated and challenging. This paper presents a surveillance system, Multi-Level Video Security (MuLViS), intended to support privacy protection in cameras. Firstly, a Smart Surveillance Security Ontology (SSSO) is incorporated in MuLViS with the objective of autonomously determining the privacy level consistent with hardware specifications of the functional device. On the whole, in addition to device-specific protection, it results in fairly fast video indexing and retrieval in surveillance. Secondly, protection of information in videos is carried out during capturing, streaming, and storage phases with different levels of encryption. The appropriateness of the security level assignments has been shown by a thorough assessment of the system using visual inspection and statistical analysis of experimental video data, such as by the Encryption Space Ratio (ESR). The system is appropriate for protecting surveillance video and may be made compatible with the General Data Protection Regulation (GDPR), guaranteeing that legitimate data access respects people's right to privacy.

Based on the findings of (Al-Hashemi, et al., 2024), The quick development of object identification, tracking, and human activity analysis technologies has made video surveillance systems an essential instrument for protecting urban and industrial services. These systems' success depends on motion detection, backdrop analysis, and creative methods for processing and storing data in real time. IP-based systems, which offer

greater flexibility and remote control capabilities, have replaced analog configurations in these schemes. However, companies must contend with significant cybersecurity issues like high processing demands, bandwidth use, and data leaks. With an emphasis on the security issues and analytics in urban settings, this study investigates the technical developments in video surveillance systems. Additionally, it discusses methods for maximizing system efficiency using real-time human activity analysis and contemporary compression techniques.

As stated by (Kalbo, N., et al., 2020), video surveillance systems are considered part of the Internet of Things (IoT). Presently, IP-based video surveillance systems safeguard industrial plants, railways, gas stations, and even homes. But, as with any IoT application, inherent vulnerabilities exist that may result in serious privacy breaches for a particular user. This book review analysis examines attack possibilities for today's surveillance systems and list ways they might be breached. We have also identified threat actors, attack goals, attack vectors, and attack consequences. Finally, the researches examines recent methods for countering and conclude with the threat horizon. The objective of this book review analysis is to inform researchers and technologists about improved security possibilities for today's video surveillance systems.

According to (Gulve, S., et al., 2017), the IOT-based smart video surveillance application since it makes use of the Internet for many things. The suggested approach notifies everybody who may be present on the property and increases security by documenting their activities. The user enters a password to activate the system when departing the premises. The

system's operation begins with motion detection, which is refined to person detection. Next, it counts the number of people in the room and notifies the neighbor of their presence by setting off an alert. Additionally, the user receives notifications about the same by email and SMS. The Raspberry Pi and Arduino board provide the hardware implementation of the suggested system, while OpenCV (for video surveillance) and GSM module (for SMS alert and email notification) provide the software. In addition to security, the system is clever enough to optimize power consumption waste in the event that a user forgets to turn off any electronic gadgets by tailoring code for certain appliances.

In the words of (M. T. Awad, et al., 2019), due to the large-scale development of technology in the modern world, institutions, businesses and residential areas need to secure their facilities leading to the need for a security system. There are several ways to help increase security; one of these methods is to install a surveillance security system. Conventional surveillance systems have disadvantages where they are often human based, allocated a lot of video cameras to keep an eye on everything, and gathering a lot of video data necessitates a lot of computation and labor. In general, the recorded video files take up a lot of storage space, and someone has to constantly check to see whether anything happened in the monitoring area. In order to eliminate these problems a need to use an effective surveillance security system which only records when motion is detected, which is the purpose of this paper. The system analyzed an “area of the environment being monitored” through the Sum of Absolute Difference (SAD) algorithm to detect motion in real time. Where a web camera captures the

images which are analyzed to detect the motion (intervention) and generate alarm. The obtained results were satisfactory, where all intrusion scenes were detected without errors. Hence, storage requirements are reduced significantly.

Based on the findings of (S. N. Jyothi and K. V. Vardhan, 2016), there is an increase in video surveillance systems in public and private environments due to a heightened sense of security like CCTV and RFID. Numerous flaws exist in video surveillance systems, including blurry images, complicated structures, poor stability, the requirement for a large amount of storage space to store the surveillance data, and comparatively high costs. In this research, an IoT-based real-time security surveillance system is proposed. Python is the default programming language used in the system design for the Motion Detection method. The Raspberry Pi 2's low processing power CPU and Pi camera are being used to build the Motion Detection algorithm, which allows live video streaming with the ability to identify moving objects, get an alarm when motion is detected, and broadcast images and videos straight to a cloud server via the Pi camera. The data is saved locally on the Raspberry Pi when the cloud is unavailable, and it is sent when the connection is restored. The camera is installed on the motor, and the user controls its left and right movement via an IoT portal, giving them a better view of their surroundings.

According to (Andrei Costin, 2016), IP camera systems, closed-circuit television, and video surveillance have become practically ubiquitous and essential for many users, enterprises, and organizations. Their primary goals are to boost safety, deter crime, and offer physical security. As a result, they

become more complicated, with embedded hardware, nontrivial firmware, and several communication channels. However, the majority of research to far has mostly concentrated on the privacy elements of these systems and has not adequately addressed their problems with cyber-security in general and attacks on the visual layer, or imagery semantics, in particular. Based on publicly accessible data, the researchers did a comprehensive evaluation of the current and emerging dangers in IP camera systems, closed-circuit television, and video surveillance. The security and privacy issues connected to the creation, implementation, and usage of such systems can then be better understood and identified using the insights. We are looking into both new and current risks, along with possible or current mitigation strategies, in a condensed form that can be found in a table that is helpful in a real-world application of a cyber-security analysis of both new and existing CCTV systems. Additionally, we are offering suggestions for improving hardware, firmware, network connection, security and privacy levels, and the operation of video surveillance systems. The researchers are certain that findings are helpful in providing a pertinent body of information on the hazards that a specific kind of technology is exposed to.

In the words of (W. -T. Chen, et al., 2018), Gathering data on the actions and locations of potential targets in the sensing environment is a key objective of monitoring systems. These systems may be used for a variety of purposes, including smart homes, surveillance systems, and fire emergencies. The use of wireless sensor networks and video cameras in surveillance systems has grown in popularity recently. In classic video surveillance systems, the cost and efficacy of the system are determined by

the number of installed video cameras. In this work, we describe a real-time video surveillance system consisting of numerous low-cost sensors and a few wireless video cameras. The technology makes it possible for a group of cooperating sensor devices to track and recognize moving objects, relaying their positions to the sink node of the wireless sensor network. The IP cameras positioned in the sensing region are then used by the sink node to capture these occurrences and display the current circumstances. In order to facilitate inter-task handoff operations between cameras and establish the coverage distribution of cameras, we additionally provide a camera control system. With 16 sensor nodes and two IP cameras, we put the suggested system into practice and assessed its effectiveness. The outcome demonstrates that our surveillance system offers real-time information on the observed area and can adjust to different environments.

As stated by (Xu, Z. et al., 2016), The military, customs, police, firefighting, airports, railroads, urban transportation, and many other public spaces employ video surveillance, an integrated system with significant preventive capabilities. Its visible, precise, timely, and extensive information content makes it a crucial component of the security system. Because video surveillance provides abundant, accurate, and understandable information, it has become the primary tool. However, issues like "useful information and clues cannot be found immediately with video big data" reduce detection efficiency during crime prediction and public security governance due to the widespread installation of video surveillance systems worldwide. The growing need for video-based applications highlights the significance of organizing and parsing the content in videos. Nonetheless, there is still a lack

of precise semantic interpretation and management of video material. Both human and semi-automated approaches are unable to bridge the semantic gap between low level characteristics and high level semantics. This study first introduces a semantic-based paradigm for organizing and expressing video material called video structural description (VSD). Using spatiotemporal segmentation, feature selection, object identification, and semantic web technologies, video structural description attempts to parse video footage into text information. The use of VSD technology in surveillance footage for public safety is described. VSD-based intelligent analysis of people and cars from security footage is shown. Additionally provided is the cloud-enhanced platform for managing security footage. Finally, VSD-based applications are presented.

As documented by (Sun, K., et al., 2019), the attendance system is of great significance to the operating process of the modern enterprise, and it has always been of concern to the people to guarantee the security of the building. This paper, in view of networked surveillance videos, combines attendance and security together, integrating video image processing, deep learning, and face recognition to establish an intelligent attendance and security system. A new method of sliding average is introduced to judge identities of persons. The experimental results demonstrate that our method is of practical effectiveness. The false reject rate (FRR) of our system is 0.51%, false accept rate (FAR) is 2.52%, and the rate of right identification is 98.85%. This system is introduced in some video surveillance regions, which has dramatic benefits of nonintrusive attendance, passive attendance, and multiple persons' attendance simultaneously. In addition, the system reduces

the need for manual intervention, improving operational efficiency within the organization. It also minimizes errors commonly associated with traditional attendance logging methods. By leveraging deep learning algorithms, the system continuously improves its recognition accuracy over time. The integration of face recognition ensures a higher level of security by preventing unauthorized access. Furthermore, the use of networked surveillance allows for centralized monitoring and easier data management. Overall, the system demonstrates a practical solution for modern enterprises seeking to enhance both security and administrative operations.

According to (Sung, CS., & Park, J.Y. 2021), For many security-related applications, closed-circuit television video surveillance systems are becoming essential tools and is extensively utilized as a monitoring system in many regions due to the rise in crime and security threats worldwide. Nevertheless, after crimes and natural catastrophes, much of the data collected by a video surveillance system is used as evidence of objective data. Additionally, video surveillance systems are often used passively due to their high cost and labor requirements. However, in the case of an accident, the video surveillance system should immediately transmit data and aggressively detect crimes and accidents in advance. The creation of a sophisticated video surveillance system that doesn't require human input to actively watch in real time is the goal of this work. After constructing an artificial intelligence server and video surveillance camera, deep learning technology is applied to the data processing model design to display data for crime detection in order to address the issues with the current video surveillance system. In addition, this design suggests an intelligent

surveillance system that uses real-time processing to broadcast a video image and notification message to the web in order to swiftly and efficiently detect crimes.

According to (Li, H., et al., 2021), currently, many people are interested in how to properly choose the region of interest (RoI) and then use selective encryption to safeguard video privacy. Establishing a security framework from the study of the video surveillance system from the system and data life cycle, however, received very little attention. This paper establishes a safe surveillance framework by analyzing the video surveillance system's attack surface and surveillance video protection in a smart city. A secure video surveillance model and a secure authentication technique that is resistant to MITM and replay attacks have been presented in the secure framework. For the Key management for video encryption and the Chinese remainder theorem (CRT) has been introduced based on the group key management to ensure an efficient and secure key update process. Furthermore, a decryption environment has also been developed through transparent encryption to guarantee the confidentiality of the decryption environment. Our system can ensure the forward and backward security of the key update, as demonstrated by the security analysis. Our system's average decryption performance in the test setting is 91.47 Mb/s, which may satisfy realistic applications' real-time demands.

According to (T. Ko, 2018), These days, security cameras are widely available and reasonably priced, but the labor needed to keep an eye on and evaluate them is costly. As a result, the footage from these cameras is typically checked infrequently or not at all; after an event is known to have

occurred, the footage is frequently used just as an archive. If surveillance cameras can be utilized to identify situations that need attention as they occur and take action in real time, rather than just passively capturing film, they may be a significantly more helpful tool. The objective of automated visual surveillance is to describe what is occurring in a monitored area and then act appropriately based on that assessment. One of the most active areas of computer vision research is human video monitoring. It has several potential uses in homeland security. Systems for managing and interpreting videos have advanced significantly in recent years. In an era of growing public safety and security concerns, this study investigates how hardware and software might be combined to address surveillance issues. Generally speaking, a video surveillance system's framework consists of the following phases: environment modeling, motion detection, moving object categorization, tracking, behavior comprehension and description, and information fusion from many cameras. Reliable automated video monitoring is still a long way off, despite recent advancements in computer vision and related fields. In addition to analyzing the viability and difficulties of integrating motion analysis, behavior analysis, and standoff biometrics for the identification of known suspects, anomaly detection, and behavior comprehension, this study discusses advancements and general tactics of phases engaged in video surveillance.

As stated by (Valentín, L., et al., 2017), A smart city is one that employs digital technology to provide effective and user-friendly services while simultaneously improving people's quality of life and having a good effect on the environment individuals's safety and welfare are essential factors to take

into account in a smart city, therefore having a strong security system becomes essential as it enables us to recognize and identify possible danger scenarios and then make the right decisions to assist individuals or even stop criminal activity. In this paper, we present an architecture for automated video surveillance based on the cloud computing schema that can collect a video stream from a collection of network-connected cameras, process that data, automatically identify, label, and highlight security-relevant events, store the data, and provide situational awareness to minimize response time to take the necessary action.

Convolutional Neural Network

Based on the findings of (W. Rawat and Z. Wang, 2017), Since the late 1980s, convolutional neural networks, or CNNs, have been used for visual tasks. Nevertheless, they were dormant until the mid-2000s, when improvements in algorithms, computing power, and the availability of large amounts of labeled data helped them advance and put them at the forefront of a neural network renaissance that has seen rapid advancement since 2012. despite a few scattered applications. According to (A. A. M. Al-Saffar, 2017), As the era of enormous data advances, convolutional neural networks (CNNs) with more hidden layers have more complex network architectures and more powerful feature learning and feature expression capabilities than traditional machine learning approaches. Since its introduction, In many large-scale computer vision identification applications, the convolution neural network model trained by the deep learning method has shown remarkable performance.

In reference to (R. Chauhan, 2018), Deep learning algorithms are

designed to mimic the cerebral cortex of humans. These techniques describe deep neural networks, or neural networks with several hidden layers. Convolutional neural networks, a type of deep learning technology, can train enormous datasets with millions of parameters by utilizing 2D images as input and convolving them with filters to get the desired output. Deep neural network layers give numerous levels of abstraction by extracting different information. This is unable to extract or work on numerous features, in contrast to shallow networks. Convolutional neural networks are a potent deep learning technique that can handle millions of parameters and reduce processing costs by taking a 2D picture, convolving it with filters or a kernel, and then creating output volumes. Convolutional neural networks (CNNs) are an interesting method for adaptive image processing and act as a link between regular feed-forward neural networks and adaptive filters, as noted by Browne, M., and Ghidary, S.S., 2023. Two-dimensional CNNs consist of one or more layers of two-dimensional filters with possible non-linear activation functions and/or down sampling. Key characteristics of CNNs include spatially local connections (receptive fields) and translation invariance. We describe the design of the convolutional network and show how it can be used for real-world image processing on a mobile robot. An autonomous sewage inspection robot uses a CNN to identify and describe cracks. In every instance, 4x4 filters with non-linear activations between layers were utilized. From input to output, four feature maps were utilized in each of the three hidden layers. The network was trained using a dataset of 48x48 sub-regions that were recovered from 30 still image 320x240 pixel frames from a previously recorded sewage line inspection video. Fifteen

frames were used for training and fifteen for validating network performance. Although the development of a CNN system for civil use is still ongoing, the results support the notion that data-based adaptive image processing techniques, such as CNNs, are useful for image processing or other applications where the input arrays are large and geographically and temporally scattered. Additional improvements to the CNN architecture are suggested, such as expansions to three-dimensional (i.e., video) processing or the insertion of separable filters.

According to (Naranjo-Torres, et al., 2020), Given the high demand from every home, fruit production is particularly crucial. As a result, the agri-food industry depends heavily on the use of cutting-edge technologies. Nowadays, one crucial technical instrument that is frequently utilized in contemporary culture is artificial intelligence. In particular, because Deep Learning (DL) can build robust representations from pictures, it has many uses. The primary DL architecture for image categorization is Convolutional Neural Networks (CNN). Given the increased interest in CNNs in recent years, we provide an overview of their application to various automatic processing duties for fruit photos: detection, quality assurance, and categorization. We note that the usage of CNN for fruit recognition has significantly risen over the past two years (2019–2020), with both new models and pre-trained networks for transfer learning yielding good results. It is important to note that different kinds of pictures are utilized in datasets depending on the work at hand. Additionally, this page discusses the principles, resources, and two applications of CNNs for quality control and fruit sorting.

Based on the findings of (Y. Sun, et al., 2020), For twenty years, evolutionary principles have been successfully included into neural network architectures. Unfortunately, because of the intricate structures and massive amounts of connection weights, these techniques are not able to scale effectively to modern deep neural networks. In order to solve picture classification challenges, we present a novel approach in this study that uses evolutionary algorithms to evolve the topologies and connection weight initialization values of a deep convolutional neural network. Convolutional neural networks' many building pieces and possibly ideal depth are represented by an effective variable-length gene encoding technique in the suggested algorithm. Furthermore, a novel representation technique is created to efficiently initialize the link weights of deep convolutional neural networks. This is anticipated to prevent networks from becoming trapped in a local minimum, which is usually a significant problem in backward gradient-based optimization. Additionally, a new fitness evaluation technique is suggested to accelerate the heuristic search while using significantly fewer computer resources. On nine popular image classification tasks, including the most advanced techniques, the suggested approach is analyzed and contrasted with 22 current algorithms. The experimental results show that the suggested method is significantly better than the state-of-the-art approaches in terms of both the number of parameters (weights) and the classification error rate.

As stated by (Y. Tian, 2020), An algorithm with strong performance capability, the convolutional neural network has been applied extensively in the image processing area and had achieved desirable results only by

leveraging the local receptive fields, weight-sharing, pooling, and sparse connections. To enhance the speed and accuracy of the convolutional neural network algorithm in convergence, a novel convolutional neural network algorithm is proposed in this paper. The recurrent neural network is first applied by the convolutional neural network, and both the recurrent neural network and the convolutional neural network concurrently learn the deep picture features in parallel. Second, the unique residual module ShortCut3-ResNet is created based on the ResNet concept's skip convolution layer. Thirdly, this research offers a dual optimization methodology that simultaneously optimize the complete connection and overall convolution. Finally, in order to ultimately In order to determine the best parameters for the convolutional neural network, this research uses a simulated experiment to examine how different parameters affect the network's performance. The convolutional neural network method described in this study may learn the different elements of the image and improve the convolutional neural network's performance in feature extraction and image recognition, according to experimental data.

Synthesis and Relevance of the Study

With the growing reliance on technology for surveillance and access control, a web-based facial recognition system that uses Convolutional Neural Networks (CNNs) provides an innovative and useful method of identification verification. CNNs' capacity to effectively handle complicated visual data and automatically extract features has made them extremely effective in image identification. CNNs are especially well-suited for facial recognition tasks where precision and instantaneous detection are crucial

because of these features. Without requiring complicated hardware configurations, the system's design guarantees simplicity of integration by identifying and classifying people using a normal webcam.

This study addresses weaknesses of conventional techniques, including manual inspection, human error, and delayed answers, which are contributing to the expanding field of intelligent surveillance systems. Security staff can respond to possible threats or unauthorized accesses more rapidly and efficiently with automatic logging and monitoring. Additionally, this study is relevant because it may be used as a basis for other businesses wishing to improve their security architecture through the use of face recognition empowered by machine learning. For academics, developers, and security experts looking to create more effective, precise, and scalable security systems, the study's conclusions and outcomes should offer insightful information.

Literature Matrix

Table 1. Literature Matrix

Author	Year	Title	Findings
Browne, M., & Ghidary, S.S.	2023	Convolutional Neural Networks for Image Processing: An Application in Robot Vision.	Convolutional Neural Networks (CNNs) have proven to be effective for adaptive image processing tasks, especially in scenarios involving large and spatially distributed data. Their application in practical systems, such as autonomous inspection robots, highlights their versatility and potential for further development in real-world use cases.

Y. Tian	2020	Artificial Intelligence Image Recognition Method Based on Convolutional Neural Network Algorithm	The proposed convolutional neural network algorithm demonstrates significant improvements in image recognition accuracy and feature extraction. By integrating a recurrent neural network and a new residual module, the system enhances learning efficiency and performance. Simulation results confirm the effectiveness of the optimized network in processing complex image data.
R. Chauhan, K. K. Ghanshala & R. C. Joshi,	2018	Convolutional Neural Network (CNN) for Image Detection and Recognition	Deep learning algorithms, particularly convolutional neural networks (CNNs), are designed to function similarly to the human brain by processing complex patterns and features. CNNs excel at handling large datasets by extracting multiple levels of abstraction through their layered architecture. Unlike shallow networks, CNNs are capable of working with millions of parameters while maintaining computational efficiency. This makes them highly effective for image processing tasks, enabling accurate and scalable performance in various real-world applications.
A. A. M. Al-Saffar, H. Tao & M. A. Talab	2017	Review of deep convolution neural network in image classification	Convolutional Neural Networks (CNNs) have evolved to handle complex data structures more effectively than traditional machine learning methods. Their deeper architectures allow for stronger feature learning and representation capabilities. As a result, CNNs have achieved outstanding performance in large-scale image identification tasks

			within the field of computer vision.
T. Ko	2018	A survey on behavior analysis in video surveillance for homeland security applications	Automated visual surveillance aims to transform these systems from passive recorders into active tools capable of detecting and responding to real-time events. With advancements in computer vision, video surveillance has become a promising area for enhancing public safety and security. However, technical challenges still remain, requiring further research and development to achieve fully reliable and intelligent surveillance systems.
M. T. Awad, S. M. Aldaw, S. M. Aldaw & B. A. r. Osman,	2019	Video Security System for Intrusion Detection	The increasing need for security in institutions, businesses, and residential areas highlights the importance of more efficient surveillance systems. Traditional surveillance methods face limitations such as high storage demands, constant manual monitoring, and intensive data processing. The study finds that using motion-detection algorithms can significantly improve surveillance efficiency by only recording relevant activity. This approach not only reduces storage requirements but also ensures timely and accurate detection of intrusion events.
Xu, Z., Hu, C. & Mei, L.	2016	Video structured description technology-based intelligence analysis of surveillance videos for	Video surveillance provides accurate and timely information for public safety, managing and interpreting large volumes of video data remains a major challenge. Traditional methods fall short in bridging the gap between raw footage and meaningful insights, reducing the

		public security applications.	efficiency of crime detection and security management. The introduction of the Video Structural Description (VSD) model offers a solution by organizing video content using intelligent analysis and spatiotemporal segmentation, greatly enhancing the effectiveness of surveillance systems.
Vetter, V., Zielke, T., von Seelen, W.	2017	Integrating face recognition into security systems.	Automated face image processing has become a vital component in the growing market of facial recognition technology. Its success depends not only on accurate recognition but also on the ease of capturing facial images and handling detection and segmentation. By combining model-based and neural network methods, the system can perform reliably even in real-time and resource-limited environments.
Trinanjana Bagchi, Ankita Mahapatra, Dipesh Yadav, Daiwik Mishra, Anish Pandey, P.Chandrasekhar, & Ashwani Kumar,	2022	Intelligent security system based on face recognition and IoT	The increasing need to protect data, property, and human life has led to the rapid use of advanced technologies like facial recognition, valued for its accuracy and ease of use. With ongoing advancements in digital technology, facial recognition systems have become more capable of processing complex visual data, making them a practical and efficient solution for modern security needs.
P. B. Balla and K. T. Jadhao,	2018	IoT Based Facial Recognition	Computers have played a vital role in advancing facial recognition technology due to

Security System	their high accuracy, processing power, and ability to handle large databases. Unlike humans, computers can store and recognize countless faces, making them highly effective for security applications. Facial recognition systems are now used in various fields such as private security, crowd surveillance, smart homes, and workplace access control. Among all biometric methods, facial recognition stands out for its user-friendliness and versatility in real-world implementations.

Conceptual Framework

The system uses InsightFace with a pre-trained model bundle (buffalo_l) to handle the entire recognition flow. Live video from the webcam is processed to detect and align faces, then embeddings are generated and compared with the Catalogue Module. If a match is found, the person is labeled Employee or Person of Interest based on their stored record; if no match is found, the person is tagged as Civilian. A simple similarity threshold and short-term smoothing are used to keep labels stable during live monitoring.

The recognition results are integrated into a web application that streams the camera feed, logs time-in/time-out for Employees and POIs, and creates notifications (with snapshots) for Civilians. These entries update the dashboard counts and recent activity. The finished system was then checked against ISO/IEC 25010 criteria (functionality, reliability, performance, and security) to confirm it meets the needs of Camp General Macario Sakay

without custom model training, since it relies on pre- trained InsightFace models.

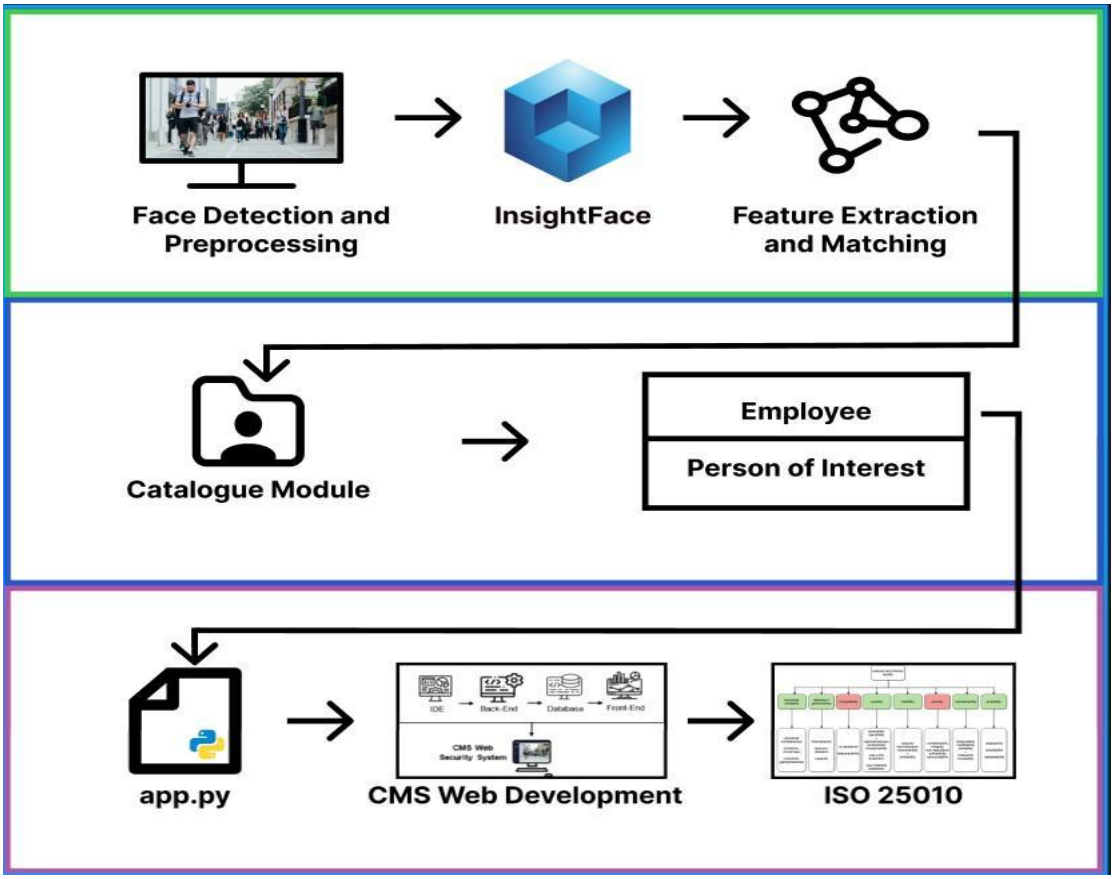


Figure 1. Conceptual Framework

CHAPTER III

METHODOLOGY

This chapter presented a systematic analysis of the methods applied throughout the study. Various methodologies were utilized as guides to ensure a structured and methodical approach in developing the application. It covered the project design, development process, as well as the testing and evaluation procedures involved in the system's implementation.

Project Design

Figure 2 presents the architecture of the Camp General Macario Sakay Web Security System. The pipeline starts with Face Detection and Recognition, where the live camera feed is processed by a lightweight Python service using InsightFace (pre- trained buffalo_l bundle) to detect, align, and embed faces, then perform similarity matching against the Inventory. Rather than training a custom model, the system relies on these pre-trained models and a fixed cosine-similarity threshold with brief smoothing to stabilize labels.

In the Web System Development phase, the recognition service is integrated with the CMS web app: a Python (Flask) module streams detections and snapshots, while the main web system runs on PHP with a MySQL database and a front end built with HTML, CSS, and JavaScript. The Monitoring Module captures the real-time feed and passes faces to InsightFace; embeddings are compared with the Catalogue Module (registered Employees and Persons of Interest). If a match is found, the identity is shown and logged; if no match exists, the individual is labeled Civilian.

Recognized events automatically populate the Logs Module (time-in/time-out for Employees and POIs) and trigger the Notification Module to alert staff whenever a Civilian, Employee, or Person of Interest is detected. The Dashboard Module summarizes key counts (e.g., total detections, unique employees, POI alerts, civilian sightings) and highlights recent activity with snapshots. The Reports Module provides date-range summaries and formatted exports (PDF) of detections per day and recent POI encounters.

The Catalogue Module allows the Super Administrator to register or update records by uploading photos. Role-based access (Administrator and Super Administrator) organizes permissions across modules, ensuring a secure, real-time facial identification workflow for Camp General Macario Sakay.

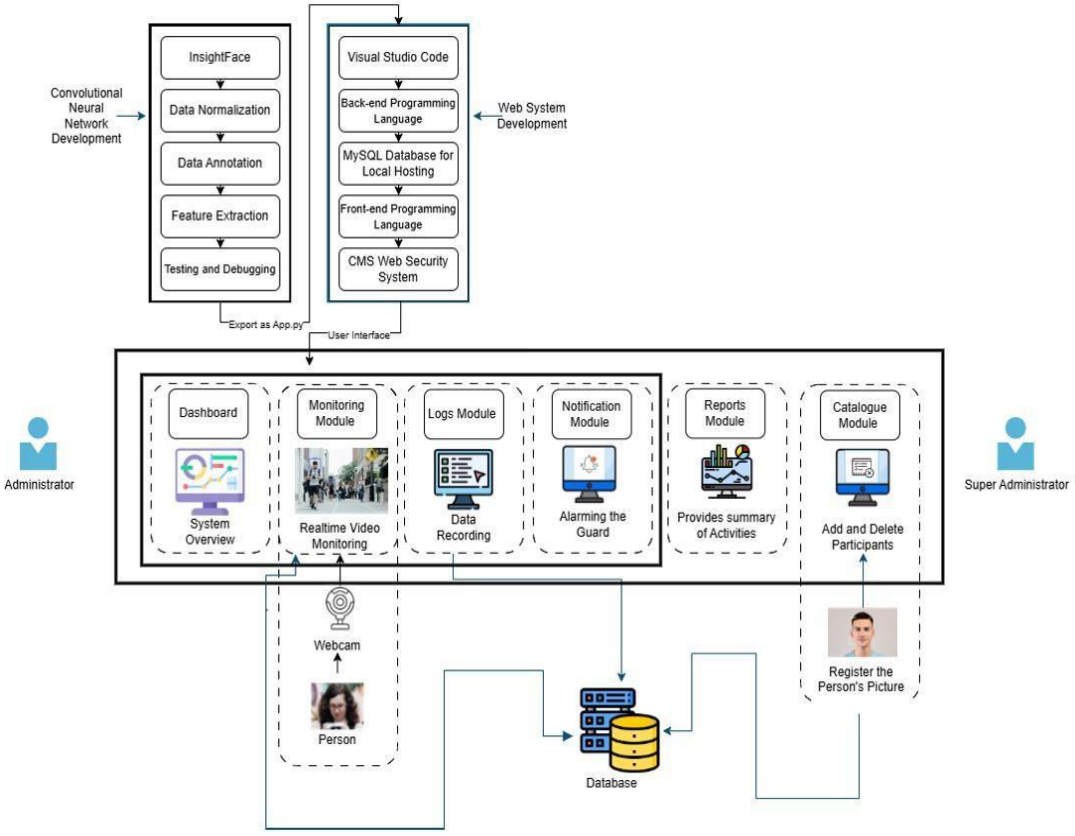


Figure 2. System Architecture

Project Locale

Figure 3. shows Camp General Macario Sakay, located in Los Baños, Laguna, which is the exclusive deployment site for this project. The system aims to enhance the current security measures by providing an automated solution for identifying and classifying individuals at the gate using facial recognition technology. This project is developed and tested within the boundaries of Camp General Macario Sakay and is specifically intended to address the security needs of this facility. It assists security personnel in monitoring real-time entries and generating accurate logs for every recognized individual. The deployment of the system is expected to streamline access control and strengthen the overall safety and operational efficiency of the camp.



Figure 3. Location of Camp General Macario Sakay

Project Development

Figure 4 presents the Agile Scrum model, which was the methodology applied in this study for the development of the web-based facial recognition security system. This approach allowed the project to progress iteratively, enabling the researchers to build and improve the system in manageable phases. The development cycle followed key Scrum phases, including the creation of a product backlog that outlined all required system features, followed by sprint planning meetings to determine task priorities and assignments. During each sprint, specific features such as facial recognition, monitoring, logging, and notifications were developed and tested. These tasks were managed through the sprint backlog, ensuring a focused and time-bound workflow. After completing each sprint, the updated system was deployed and reviewed for evaluation and improvements, ensuring continuous refinement and alignment with user needs.

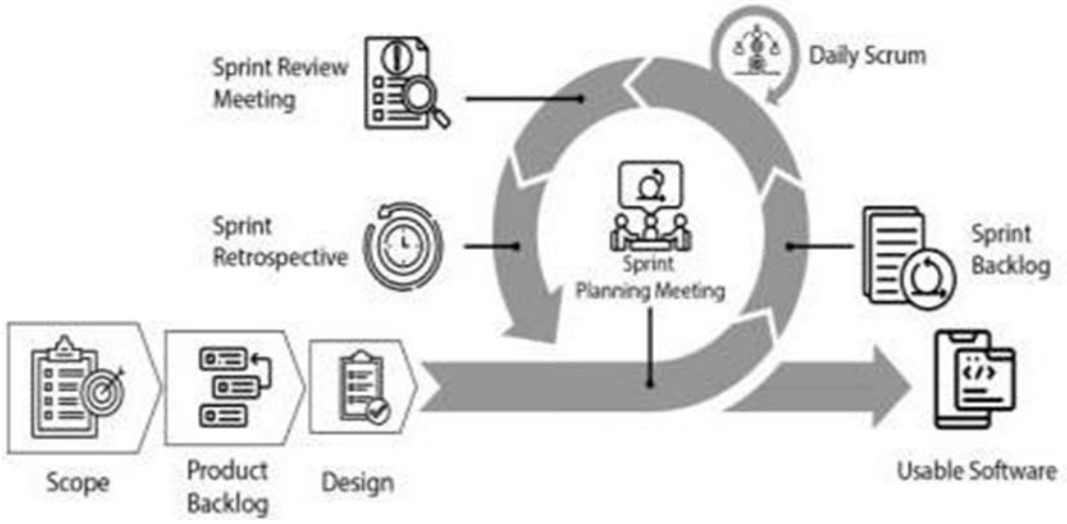


Figure 4. Agile Scrum Mode

Product Backlog

The product backlog illustrated in table 2 and 3 outlines the development of the Camp General Macario Sakay Web Security System by prioritizing essential features for the Super Administrator and Administrator roles. Tasks for the Super Administrator include adding and viewing employee and criminal records, accessing logs, receiving notifications, and managing login access, while the Administrator focuses on monitoring users, viewing logs, receiving alerts, and accessing the system. Each task is assigned estimated working hours to ensure efficient sprint planning and timely delivery. This prioritization allows the development team to focus on critical security features first, ensuring that user needs are addressed throughout each sprint. Effective backlog management supports a smooth development process, leading to a more reliable and responsive facial recognition security system.

Table 2. Product Backlog for Super Administrator

Priority	Features	Estimated hours per day
1	As a Super administrator, I want to add and view the employees and criminals to the system.	5 hours
2	As a Super administrator, I want to see all the logs.	4 hours
3	As a Super administrator, I want to be informed when there's a civilian or criminal entering within the premises.	2 hours
4	As a Super administrator, I want to access the login form.	1 hour

Table 3. Product Backlog for Administrator

Priority	Features	Estimated hours per day
1	As an Administrator, I want to monitor all the civilians, employees and criminals in the system.	7 hours
2	As an Administrator, I want to see all the logs.	4 hours
3	As an Administrator, I want to be informed when there's a civilian or criminal entering within the premises.	2 hours
4	As an Administrator, I want to access the login form.	1 hour

Sprint Planning

Table 4 presents the sprint planning for the Camp Macario Sakay Web Security System, highlighting the prioritization of essential tasks to be completed first within the sprint cycle. The most critical task was the development of the Monitoring System powered by CNN, assigned to the Administrator for 48 hours. Next, the Super Administrator focused on adding employees and criminals to the system, scheduled for 32 hours. Both the Super Administrator and Administrator were responsible for viewing system logs and handling notifications, each with 20 hours allocated. Additional tasks such as creating the Login Form and User Interface Design were planned with estimated times of 10 and 20 hours respectively. Tasks were selected based on their contribution to the system's core functionalities and feasibility within the sprint timeframe. This structured approach ensured that the most

important system modules were developed early, keeping the project on track.

Table 4. Sprint Planning

Priority	Features	Role	Estimated Hours
1	Monitoring system, powered by CNN	Administrator	48 Hours
2	Adding employee and criminals to the system	Super Administrator	32 Hours
3	View all logs	Super Administrator and Administrator	20 Hours
3	Notification for entering	Super Administrator and Administrator	20 Hours
4	Login form	Super Administrator and Administrator	10 Hours
4	User interface designing	System wide	20 Hours

System Flowchart

Figure 5 portrayed the flow of the Camp General Macario Sakay Web Security System. It starts with a login page to verify authorized personnel. Once logged in, users gain access to four main modules of the system. The Notification Module displays alerts triggered by facial detection events. The Monitoring Module provides real-time surveillance using a webcam and

performs facial recognition through a Convolutional Neural Network (CNN). The Logs Module records and stores data of every individual detected by the system. The Catalogue Module allows super administrators to add or remove participants such as employees and criminals from the database. When a face is detected, the system checks if the individual matches an employee record. If not, and the person is classified as a civilian or criminal, the system triggers an alert and logs the event; otherwise, it silently records the entry for employees.

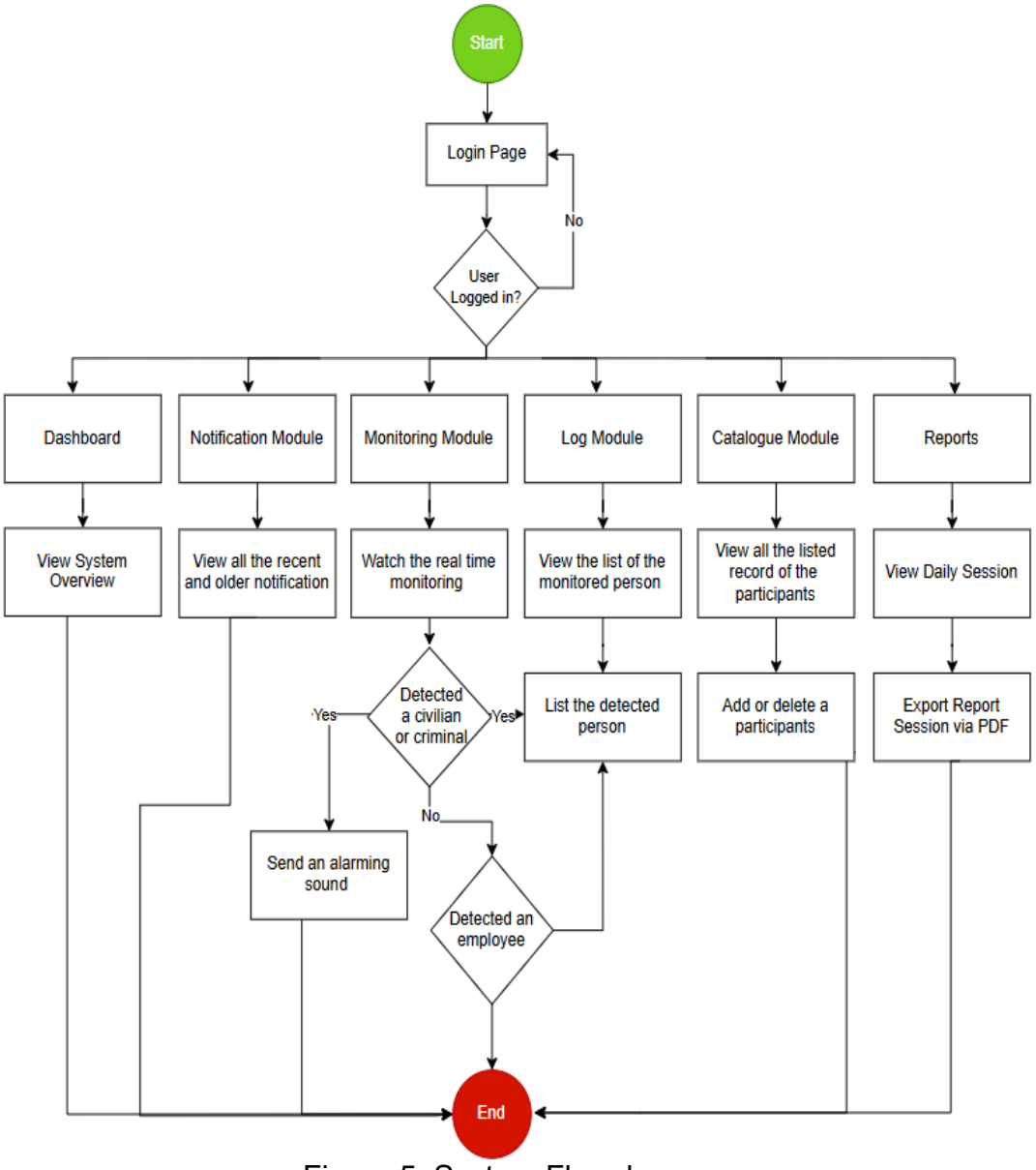


Figure 5. System Flowchar

Use Case Diagram

Figure 6 depicts the Camp General Macario Sakay Web Security System. The diagram defines clear interactions between users through its use case design. Both the Super Administrator and Administrator can log in, monitor real-time video feeds, view system logs, and access notifications generated by facial detection events. The Super Administrator holds additional privileges, such as adding or deleting employee and criminal profiles within the system database. These user roles work together to maintain secure access control, ensure continuous monitoring, and provide quick responses to any detected threats. Each functionality is structured to allow seamless user interaction with the facial recognition system. This division of responsibilities supports efficient system management and improves overall security operations. Through these roles, the system ensures that all identification, monitoring, and alert processes are properly handled. The use case framework simplifies user access while safeguarding the camp's premises.

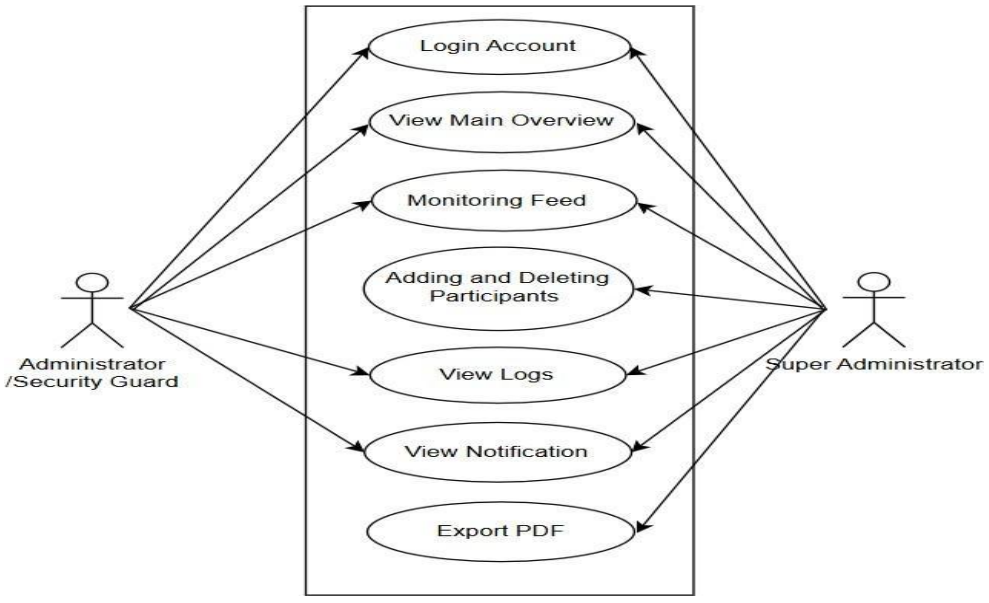


Figure 6. Use Case Diagram

Daily Scrum

The researchers conducted daily scrum meetings, each lasting two hours, to track the development of the web-based facial recognition security system for Camp Macario Sakay. These meetings focused on reviewing the sprint goal, evaluating current progress, and identifying any necessary changes to the sprint or product backlog to keep the project aligned with the system's objectives.

Sprint Execution

During the sprint execution phase, the researchers collaborated with users such as IT specialists, security personnel, and camp administrators to determine deliverables and outline specific tasks required to meet the sprint goals. This engagement ensured that the development process remained focused on the real- world needs of the security system and provided clarity on key features such as facial recognition accuracy, monitoring, and logging.

Sprint Review Meetings

Sprint review meetings were held regularly to evaluate completed features of the system. During these sessions, the team demonstrated working modules such as the monitoring interface, login system, and notification alerts. Feedback was gathered to identify any missing functionalities or enhancements needed to meet the expectations of the camp's security operations.

Sprint Retrospective

At the end of each sprint, the researchers held a sprint retrospective to assess team performance and improve development practices.

Discussions focused on key areas such as communication, tools, process efficiency, and collaboration with users. The goal was to continuously enhance both the development workflow and user experience, aiming to complete over 60% of the project's core features within the first month.

Materials

This section displays the software and hardware used to develop and finish the Camp Macario Sakay Web Security System. It consists of the tools and platforms employed during development, such as programming language, frameworks, and databases.

InsightFace

InsightFace is a high-performance face analysis toolkit built on pre-trained models (e.g., ArcFace backbones bundled in the buffalo_l pipeline) for detection, alignment, embedding, and recognition. In our system, InsightFace processes the live camera feed to detect faces, generate normalized embeddings, and compare them against the Catalogue Module. When a match is found, the individual is identified as an Employee or Person of Interest based on the stored record; if no match exists, the person is labeled a Civilian. This approach enables accurate, real-time recognition without manual dataset collection or custom model training.

Software

The software required to create the application, including its features and descriptions, are essential for building, training, deploying, and managing the facial recognition system. These tools ensure smooth development workflows, system functionality, and integration of the CNN model into the web-based security platform.

Visual Studio Code. A lightweight and powerful code editor developed by Microsoft, widely used for web and software development. In the development of this system, Visual Studio Code served as the primary environment for writing and organizing the project's source code, including the frontend (HTML, CSS, JavaScript) and backend (PHP, SQL) components. Its integration features and support for extensions allowed the researchers to efficiently manage code, debug, and collaborate throughout the system development lifecycle.

XAMPP. A free and open-source web server solution that includes Apache, MySQL, and PHP, providing a complete local environment for developing, hosting, and storing data for web applications. In this project, XAMPP was used to locally host the backend of the facial recognition security system and to store all system data, including employee records, criminal profiles, facial recognition logs, and notifications. It enabled the researchers to run PHP scripts, manage the MySQL database, and ensure that all information captured by the system was securely stored on a local server. This setup allowed smooth interaction between the facial recognition model and system modules such as user registration, monitoring, logs, and notifications during development and deployment phases.

Python. It is used for implementing the core facial recognition features and machine learning processes. It supports powerful libraries such as InsightFace and OpenCV, which handle face detection, embedding generation, and recognition. Python also allows fast development and easy integration with both frontend and backend components.

JavaScript. It is used to add dynamic and interactive features to the web interface. It enables real-time updates, camera integration for live feeds, and responsive interaction between the user and the system. It also handles client-side validations and improves the overall user experience. In the context of this project, JavaScript plays a crucial role in displaying facial recognition results instantly after processing. It also supports the seamless communication between the frontend and backend, ensuring smooth user interaction with the system modules.

HTML. It provides the structure and layout of the web pages for the system. It defines how content such as login forms, dashboards, and user interfaces are organized and displayed in a browser. HTML works in combination with CSS and JavaScript to build a complete and functional frontend.

CSS. It is used to style the HTML elements and make the user interface visually appealing. It controls the layout, color schemes, typography, and responsiveness of the pages. With CSS, the system can maintain a clean and modern design that works well across different devices.

PHP. It is used to handle the server-side operations of the system. It processes requests from the frontend, manages user sessions, and communicates with the database to fetch or store data. PHP is essential for handling the logic behind login authentication, user access, and secure data flow.

SQL (Structured Query Language). It is used to manage and manipulate the system's database. It stores critical data such as employee and visitor profiles, facial recognition results, and access logs. SQL enables efficient

queries for inserting, updating, deleting, and retrieving data needed by the system. It ensures that all stored information is organized, accessible, and secure for real-time system operations.

Hardware

The minimal requirements depicted in Table 5 are essential for a computer to run the system smoothly and without errors. An internet connection and a web browser are also required to ensure full access to the web-based features of the system. These specifications help guarantee optimal performance and compatibility during operation.

This allows the system to process real-time video input, perform facial recognition efficiently, and store logs without interruptions. Ensuring that devices meet these requirements contributes to the system's overall reliability and user satisfaction.

Table 5. Minimum Requirements for Computer

Hardware	Specification
Operating System	Windows 7
Central Processing Unit	Intel Core i3
SSD	30GB
RAM	4GB

Sprint Backlog

Table 6 and 7 shows the sprint backlog served as a structured guide for developing the Camp General Macario Sakay Web Security System. The

researchers prioritized essential features like real-time monitoring, facial recognition, and notifications, ensuring that tasks for the Super Administrator and Administrator roles were clearly defined. This organization helped the team focus on critical activities, track progress, and adjust plans during each sprint. By regularly reviewing and updating the backlog, the researchers maintained consistent development flow and met project goals efficiently.

Table 6. Sprint Backlog for Super Administrator

Priority	Features	Task	Estimated hours per day
1	As a Super administrator, I want to add and view the employees and criminals to the system.	Create a table in the database for employees and criminals. Develop and design a module for adding and viewing of the listed employees and criminals.	5 hours
2	As a Super administrator, I want to see all the logs.	Design and develop a module for data recording, once an individual is classified a civilian or criminal	4 hours
3	As a Super administrator, I want to be informed when there's a civilian or criminal entering within the premises.	Design and develop a module for notification when there's a civilian or criminal entering within the premises	2 hours
4	As a Super administrator, I want to access the login form.	Design and develop a login form for super administrator	1 hour

Table 7. Sprint Backlog for Administrator

Priority	Features	Task	Estimated hours per day
1	As an Administrator, I want to monitor all the civilians, employees and criminals in the system.	Design and develop a monitoring module powered by CNN for identifying any person entering the premises with high accuracy.	7 hours
2	As an Administrator, I want to see all the logs.	Design and develop a module for data recording, once an individual is classified a civilian or criminal	4 hours
3	As an Administrator, I want to be informed when there's a civilian or criminal entering within the premises.	Design and develop a module for notification when there's a civilian or criminal entering within the premises	2 hours
4	As an Administrator, I want to access the login form.	Design and develop a login form for super administrator	1 hour

Procedure and Methods

The following procedures and methods are used to develop and finish the Camp Macario Sakay Web Security System in an orderly and systematic manner. These steps ensure that each phase of development from planning

to deployment follows a structured workflow that meets the project's goals and user requirements.

Algorithm

Figure 7 illustrates the Convolutional Neural Network (CNN), The system captures facial images in real-time using a webcam placed at the entrance of Camp General Macario Sakay. These images are then processed by the Convolutional Neural Network (CNN), where the first step involves applying a convolution operation to scan the image with filters that generate feature maps highlighting important facial characteristics like edges, contours, and textures. The feature maps are passed through pooling layers, which reduce the dimensionality of the data while preserving essential information, thus minimizing computational load and preventing overfitting. As the data passes through multiple layers of convolution and pooling, the CNN gradually learns to detect more complex and unique features specific to each face. The processed data then reaches the fully connected layers, where all extracted features are consolidated, and the classification process begins. The final output layer uses the softmax function to produce a probability distribution, determining the likelihood that the detected face matches one of the individuals registered in the system. This step-by-step process allows the CNN to transform raw facial images into accurate, real-time identity classifications, identifying individuals as employees, visitors, or recorded criminals, and supporting secure access control at the facility. Stated by W. Rawat and Z. Wang (2017) in "Deep Convolutional Neural Networks for Image Classification: A Comprehensive Review," in Neural Computation. Through this intelligent, step-by-step processing, the CNN transforms raw facial images into accurate, real-time identity classifications

that support secure access control within the facility.

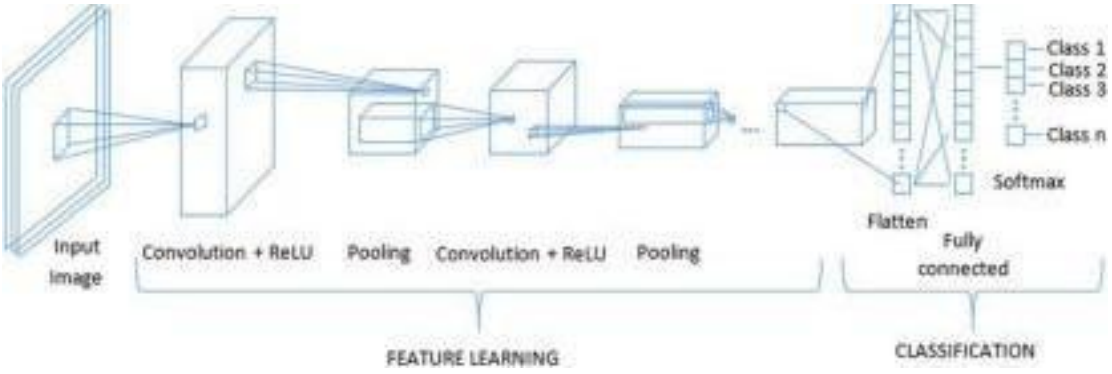


Figure 7. Convolutional Neural Network

Model Development

Figure Instead of training a new neural network, the study implements its recognition “model” as the end-to-end pipeline coded in app.py using the InsightFace framework. At runtime, the application loads the pre-trained buffalo_l package for face detection, alignment, and embedding extraction, using GPU when available and falling back to CPU. A local gallery (inventory) is constructed from images stored under /faces/{role}/{id}.jpg for employees and persons of interest. On startup and whenever the folder changes the system detects the face in each image, computes an L2- normalized embedding, and caches these vectors with their role and ID. During monitoring, frames from the webcam are processed in real time: faces are detected on a down-scaled copy for efficiency, embeddings are averaged over a short sliding window to stabilize results, and identity is decided by cosine-similarity matching against the gallery using a fixed acceptance threshold. Matched subjects are labeled according to the catalogue (Employee or Person of Interest), while unmatched faces are treated as Civilians, with debounced snapshot saving and notification. The pipeline also

manages session state (time-in/time-out) with grace and re-log timers, writes snapshots to /snapshots, posts JSON events to the PHP backend, and streams the annotated video via a Flask endpoint. In this design, InsightFace provides the learned components, while the project's contribution lies in the recognition policy, gallery management, thresholding and smoothing strategy, session logic, and integration with the web system eliminating the need for Google Colab training or model export.

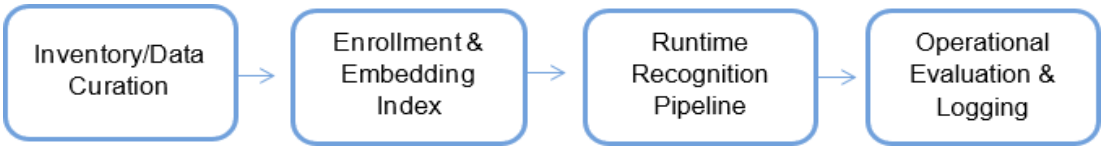


Figure 8. Model Development

Development Process

Figure 9 illustrates the development process of the Camp General Macario Sakay Web Security System. Instead of training a custom Convolutional Neural Network (CNN), the system leverages the InsightFace library, which provides pre- trained models for accurate face recognition. The process begins by setting up the data of employee and person of interest profiles, which are stored in the system database along with their corresponding images. InsightFace generates embeddings for each image and uses similarity measurements to compare detected faces against the stored records.

Data Gathering

Instead of collecting a large public dataset, the system builds a small internal gallery of faces used only for recognition at Camp General Macario Sakay. Images of employees and persons of interest are captured (or uploaded) and saved in a simple folder structure faces/employee/{id}.jpg and

faces/person_of_interest/{id}.jpg. These images are automatically standardized by InsightFace (face detection, alignment, embedding) so no manual preprocessing or dataset labeling is required. Civilians are not enrolled; their faces are never stored in the gallery only brief encounter snapshots are logged for notifications when needed. The gallery can be updated at any time by adding or replacing pictures, keeping the system current while minimizing data collection and preserving privacy.

Export Model for Deployment

The system deploys the pre-trained InsightFace pipeline directly. At runtime, the application loads the FaceAnalysis model pack (buffalo_l) which InsightFace downloads and caches automatically so there is no .h5/checkpoint to maintain. For integration, we simply bundle the Python environment and model cache with the server code (Flask). Recognition runs in real time by (1) using CUDA with CPU fallback, (2) setting a compact detector size (e.g., det_size=(640,640)), (3) frame skipping to reduce compute, and (4) storing only L2-normalized embeddings for enrolled users (employees/POIs). This approach preserves model architecture and weights via InsightFace's package, minimizes latency, and avoids a separate export/convert step.

Integration of InsightFace into the Web-Based System

The backend integrates InsightFace to handle real-time face detection and identification. Instead of a custom-trained model, InsightFace's pre-trained FaceAnalysis (buffalo_l) pipeline generates embeddings and matches them against stored profiles in the database (employees and persons of interest). Security personnel monitor a live webcam feed in the

Monitoring module; when a face is detected, the system classifies it as Employee, Person of Interest, or Civilian (no match). Results are logged automatically, notifications are issued for each detection, and summary counts appear on the Dashboard and Reports modules for review and export.

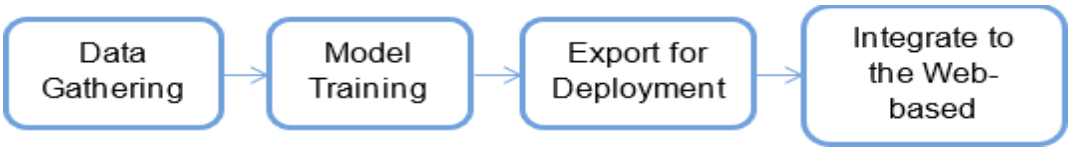


Figure 9. Development Process

Project Testing and Evaluation

Figure 10 displays the 4 different levels of software testing. To ensure the reliability, accuracy, and overall quality of the developed web-based facial recognition security system, four levels of testing were conducted: Unit Testing, Integration Testing, System Testing, and Acceptance Testing. Each level played a vital role in identifying issues and verifying that the system functions correctly as per its intended design.



Figure 10. Levels of Testing

Unit Testing

Unit testing focused on verifying the smallest components or functions of the system individually. This included testing each module such as the facial detection algorithm, CNN-based classification function, login authentication, and log recording features. By testing these components separately, the developers ensured that each function performed accurately before combining it with others. Distinct attention was given to the facial recognition module to ensure it could consistently detect and extract features from various face angles and lighting conditions. This level of testing helped prevent errors during later stages and ensured the system's core functionalities were stable and reliable.

Integration Testing

Integration testing was conducted to evaluate how different modules work together. This included testing the connection between the webcam input and the CNN recognition model, the interaction between user input and the database, and the flow of data from the monitoring module to the logging and notification modules. This step ensured seamless data communication and module coordination within the system.

System Testing

System testing involved testing the complete and fully integrated system to verify that all functionalities meet the original project requirements. This included end- to-end testing of real-time facial recognition, user login, classification of individuals (employee, visitor, or criminal), data logging, and system alerts. The entire workflow was tested in a simulated live environment to confirm that the system behaves as expected.

Acceptance Testing

Acceptance testing was carried out by selected end users including IT experts, administrators, and security personnel. This final phase was conducted to ensure the system is user-friendly, functions as intended and meets the needs of the actual users in the camp setting. Feedback from this phase was used to finalize the system and confirm its readiness for deployment.

Evaluation Metrics

Face matching verification metrics was used to assess the effectiveness of face matching in the CNN-based facial recognition system, vector-based similarity metrics were used. These metrics compare the extracted facial feature vectors to determine whether two images belong to the same individual. The two main metrics used in this study are Cosine Similarity and Angular Distance.

Cosine Similarity

Cosine Similarity measures the cosine of the angle between two non zero vectors. It evaluates how similar two vectors are regardless of their magnitude, making it ideal for comparing facial embeddings. A higher cosine similarity (close to 1) indicates greater similarity between faces.

$$\text{similarity } (A,B) = \cos(\theta) = \frac{A \cdot B}{||A||\,||B||}$$

The (A · B) is the dot product of vectors A and B, and ||A|| and ||B|| are their magnitudes. High cosine similarity means the vectors are pointing in nearly the same direction, suggesting the faces are of the same person.

Angular Distance

Angular Distance is derived from the angle between two vectors and is a variation of cosine similarity. It represents how far apart two vectors are in terms of angular separation, with smaller angles indicating more similar vectors.

$$\text{Angular Distance} = \arccos = \frac{A \cdot B}{||A||\ ||B||}$$

The (A · B) / (||A|| × ||B||) is the cosine of the angle between two vectors A and B, calculated using their dot product and magnitudes. The arccos (arc cosine) function returns the angle (in radians or degrees) between these vectors based on their similarity. A smaller angular distance means the vectors are closer in direction, suggesting that the two faces being compared are highly similar or likely belong to the same person.

Evaluation Procedure

Figure 11 illustrates the ISO 25010 standard that was used by IT experts to evaluate the technical quality of the developed web-based facial recognition security system. This evaluation focused on key quality characteristics including Functional Suitability, Reliability, Performance Efficiency, Usability, Security, Compatibility, Maintainability, and Portability. ISO 25010 is widely recognized for assessing both the functional and non-functional aspects of software systems, including internal and external system behaviors and architecture. The IT experts assessed whether the system components met these standards to ensure the system's stability and effectiveness before it was tested by end users. They verified that each module such as monitoring, logs, notifications, and participant management functioned as intended and delivered accurate information. A 4-point Likert

scale was used to rate each criterion, providing measurable results that reflect the overall quality and readiness of the system for deployment.

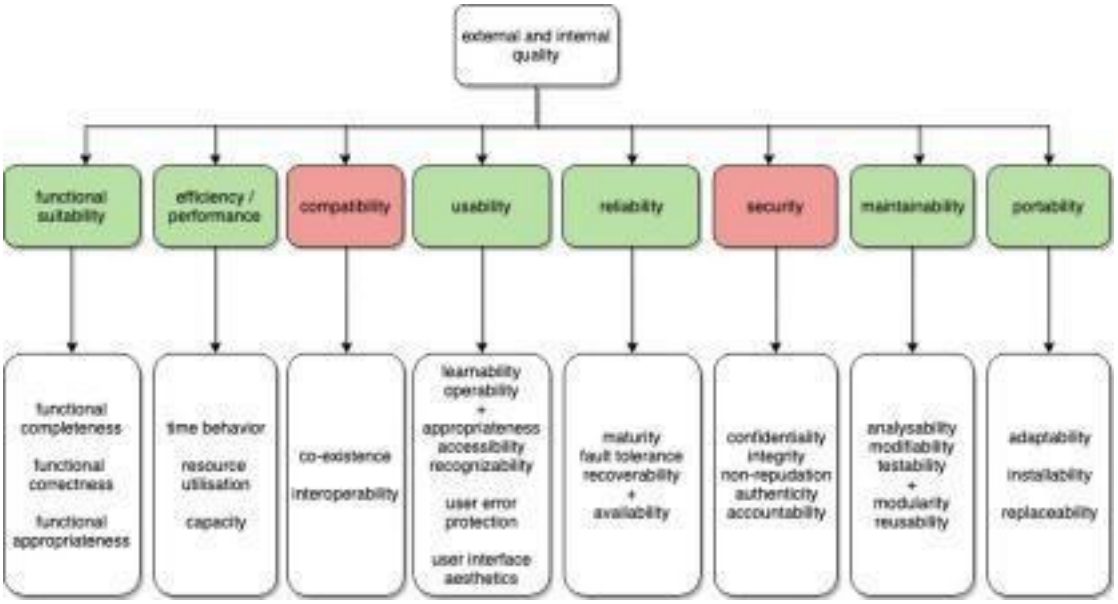


Figure 11. ISO 25010 Quality Characteristics

Standards such as ISO 25010 are applied to ensure the quality of the Camp Macario Sakay Web-System. The project team used a Likert scale to gather feedback from Camp Macario Sakay and IT specialists, evaluating important factors such as usability, efficiency, and reliability. The assessment followed the ISO 25010 quality model, a recognized standard for software quality measurement, focusing on critical aspects including functionality, security, maintainability, and portability.

In addition, the evaluation also considered other quality characteristics from the ISO 25010 model such as performance efficiency, compatibility, interaction capability (usability), reliability, flexibility, and safety. These include sub-factors like functional completeness, time behavior, interoperability, learnability, confidentiality, scalability, and risk identification, ensuring a holistic assessment of the system’s quality.

The feedback collected was crucial in identifying areas for improvement,

which were then refined to enhance the system's usability and overall effectiveness. By adhering to ISO 25010 guidelines, the project ensures that the system not only meets recognized quality standards but also effectively supports the efficient and reliable processing of Camp Macario Sakay.

Table 8. 4-point Verbal Interpretation Scale for ISO 25010

Rating	Verbal Interpretation
5	Excellent
4	Good
3	Fair
2	Need Improvement
1	Poor

IT experts used ISO 25010 to evaluate all technical components of the system. Before end users test each module of the system, the experts ensures that the system functions as intended and that all information is accurate. The 4-point Likert scale is used to score each item in order to obtain the results of the evaluation conducted by IT experts. This evaluation focuses on the critical quality attributes defined in ISO 25010, such as Functionality, Reliability, Usability, Efficiency, Maintainability, and Portability, to ensure that the system meets the required standards of software quality before deployment.

Technology Acceptance Model

The researchers used the method to determine the context to which the system is easy to use and how each module is functionable and reliable, and if the system is useful. This research aims to collect date and responses to conducted survey. The Camp Macario Sakay personnel of Los Banos Laguna evaluated the system. As a result, it is important to evaluate the effectiveness

of the system as well as how user requirements like interface design and the system benefits are addressed.

The 30 Respondents evaluates the 5 categories of TAM; Perceived of Usefulness, Information Awareness, Perceived Ease of Use, Attitude Towards Use, and Behavioral Intention to use to determine the outcomes of the respondent's evaluation, each item is scored on a 4-point Likert scale. The results helps to identify key factors influencing user adaption and satisfaction. Figure 12 below illustrates the Technology Acceptance Model (TAM).

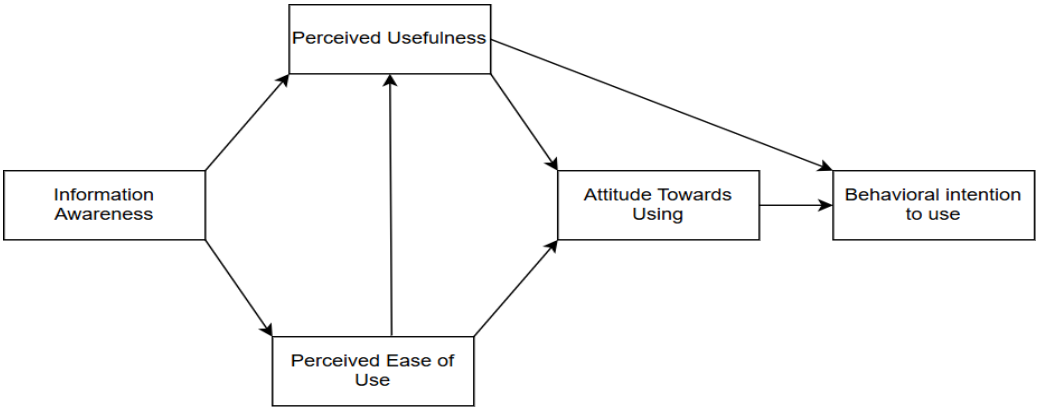


Figure 12. Technology Acceptance Model (TAM)

The model also considers Behavioral Intention to Use (BI), which predicts whether users are likely to continue using the system based on their perceptions of its usefulness and ease of use. Additionally, and System Quality are assessed to ensure the system's reliability, performance, and support. By evaluating these key factors, the TAM provides valuable insights Camp Macario Sakay Web-System, enhancing user experience, and promoting higher adoption and sustained usage among residents.

Table 9. 4-point Verbal Interpretation Scale for TAM

Rating	Verbal Interpretation
4	Strongly Agree
3	Agree
2	Disagree
1	Strongly Disagree

The table 10 below present the 4-point Likert scale Technology Acceptance Model (TAM) This scale serves as a guide to interpret the level of user acceptance and their perception of the system's usefulness, ease of use, attitude, and behavioral intention. The composite mean was computed from the responses to determine the overall interpretation of each construct.

Table 10. 4-point Likert Scale for TAM

Rating	Verbal Interpretation
4	Strongly Agree
3	Agree
2	Disagree
1	Strongly Disagree

Population of the Study

The study involves 33 participants who contributed to examining and assessing the viability of creating a web-based security system. This includes 3 I.T. specialists who provided technical insights on system development, integration, and assessment, 30 Personnel of Camp Macario Sakay who evaluated the system's relevance to facility management. Their combined feedback played a vital role in ensuring the system's functionality, practicality, and alignment with the camp's security needs. A specialist in

information technology with experience in online appointment systems also supports the evaluation process by providing insights on system performance. Additionally, they offer recommendations to enhance and optimize the system further.

Table11. Respondents of the Research Project (Population of the Study)

Category of Respondents	Sample
IT Expert	3
Camp Personnel	30
Total	33

This study shows the combination of IT Experts and Camp Macario Sakay Personnel provided a well-rounded view of the system's strengths and areas for improvement. Their feedback played a big role in shaping the final version of the platform and making sure it works for the Camp Macario Sakay Security.

Sampling Design

The researchers intentionally selects participants based on specific criteria and relevance to the study. This non-probability method was chosen because it enables the collection of data from respondents who are readily accessible within a limited timeframe. A total of 33 individuals participated in this study, including thirty (30) Employee and three (3) IT experts or professionals with knowledge of the web-based system.

Data Collection Instrument

The conducted interview is a tool used to collect data for this study. The interview questions focused on understanding the current security challenges, expectations, and usability concerns of the intended users, such as security personnel, administrators, and employees. Questions were directed toward the system's design, real-time monitoring capabilities, facial recognition accuracy, and its potential impact on daily operations within the camp. Evaluation of the system was conducted by distributing two types of questionnaires to the respondents. The Set 1 questionnaire consisted of 10 questions focused on the system’s functionality, ease of use, responsiveness, and user experience. It assessed aspects such as interface design, accessibility, information accuracy, and system performance. The Set 2 questionnaire was based on the ISO 25010 software quality model, covering criteria such as reliability, security, maintainability, and usability. Printed questionnaires were used to ensure convenient and consistent data collection from all participants.

Ethical Consideration

To ensure the privacy and protection of all participants, the researchers handles all facial images and personal data collected during the study with strict confidentiality. The data gathered, including facial recognition inputs and identification details, is used solely for the development and evaluation of the capstone project and is not shared or repurposed for any unrelated activities. In compliance with the Data Privacy Act of 2012, no identifiable information such as names or personal records is disclosed or published without the participant’s consent. All participants were informed about the

purpose of the data collection, and their involvement was voluntary, with assurances that the information is accessed by the researchers for academic purposes.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the result and discussion to support the methodology utilized in the achievement of the objectives. This section presents the system outputs, interface designs, and evaluation results based on the specific objectives of the study. Each objective is restated and followed by the corresponding findings, screenshots, and explanations. The results demonstrate how each system feature was developed, implemented, and assessed according to the goals of the project.

Camp Macario Sakay Web Security System

Figure 13 shows the dashboard also the main interface of Camp Macario Sakay Web Security System (CMS), highlighting its core features designed to navigate all the important information easily. The interface shows the overall total of employee and person of interest, total logs, detections daily, and graphs. The design emphasizes accessibility and ease of navigation for both admin and super admin.

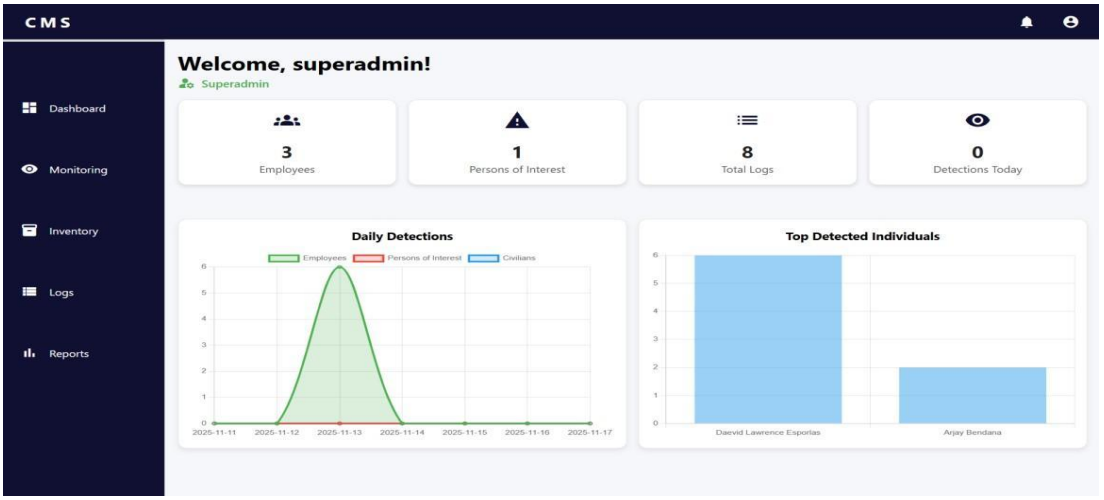
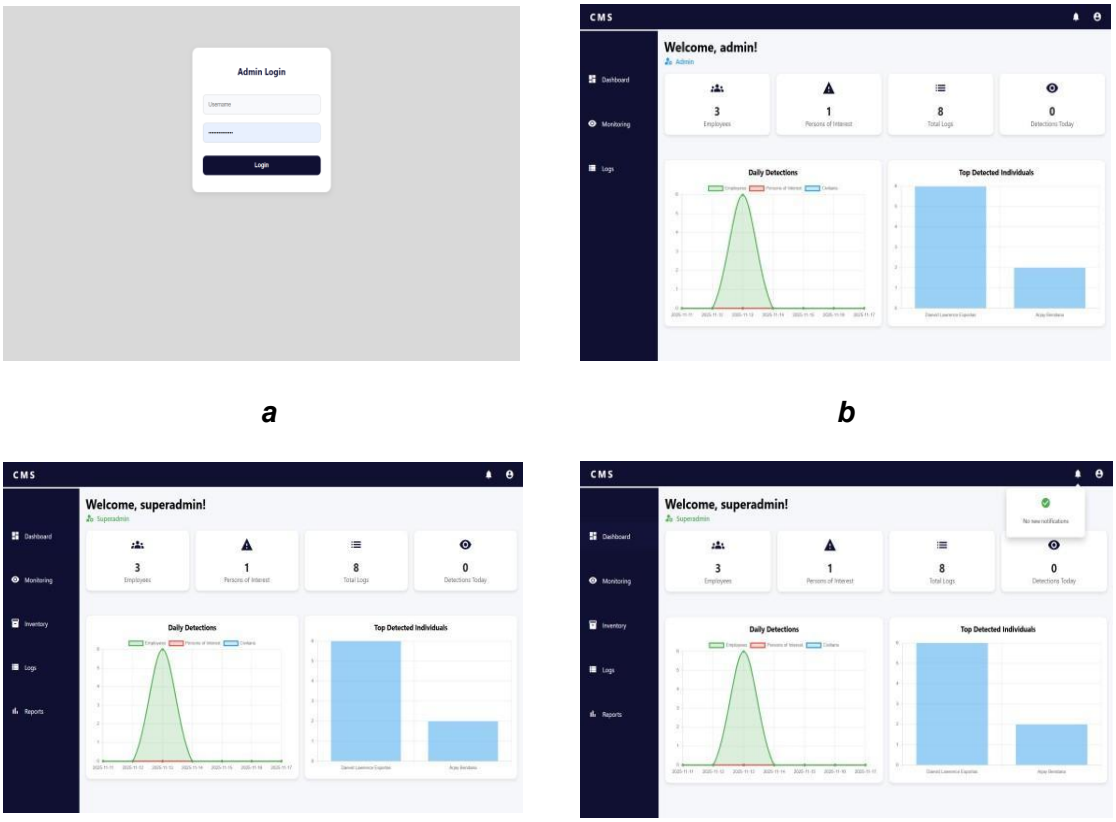


Figure 13. User Interface for dashboard

Figure 14 section (a) shows the Login Module, where admin and super admin are able to log in and access the system. The system allows the admin and super admin to access features depends on their role. Section (b) System View of Admin displays the accessible features which is limited for this role. It can only access Dashboard Module, Monitoring Module, Logs Module and Notification Module. Section (c) System View of Super Admin displays all the modules that Admin can access, and addition to it, Catalogue Module and Reports Module is now accessible for Super admin role. Section (d) Notification Module provides notification to notify the admin whenever there's detected individual. It comes with an alarm where it rang a sound depends on what role that individual

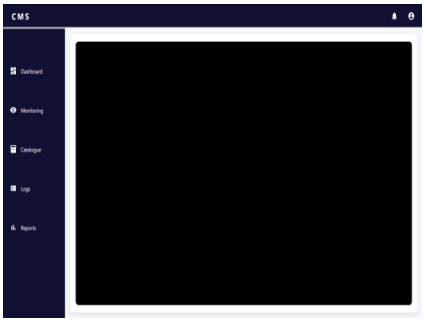


a ***b***

c ***d***

Figure 14. (a) Login Module, (b) System View of Admin, (c) System View of Super Admin, and (d) Notification Module

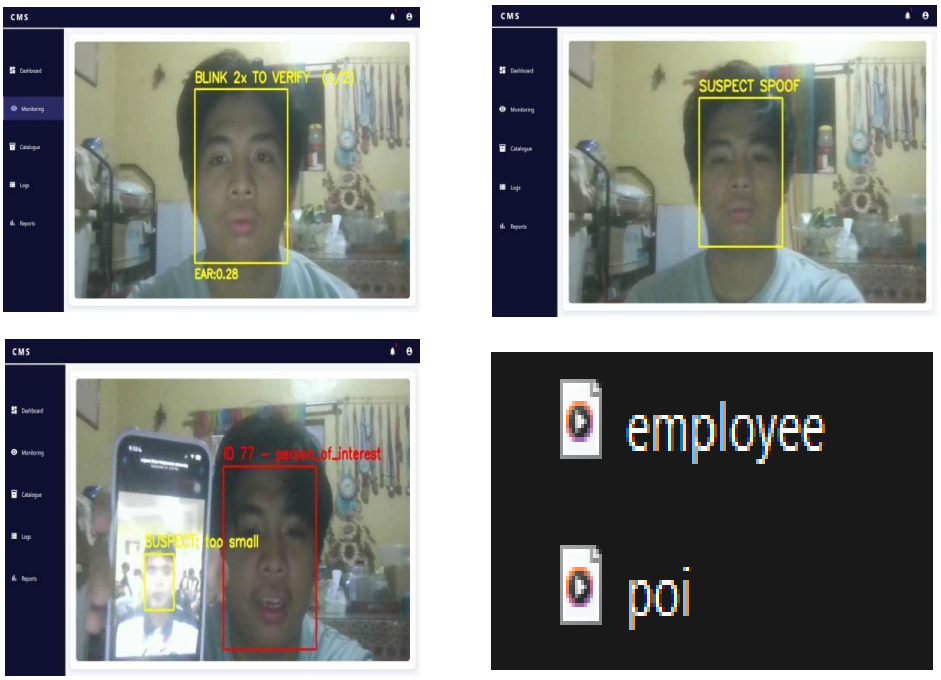
Figure 15 section (a) shows Monitoring Module, where admin monitors every individual that enters the premises. Section (b) InsightFace Detection, where it detects faces and compare if that individual is registered in Catalogue Module, and if so, it shows a colored line box depends on what role that individual has. The detection comes with a feature called “Anti Spoofing Security” that needed to blink twice to know if that person facing on the camera is a real person or not. If an individual didn’t blink and showed an image from phone or printed image, it categorized as “Suspect Spoof” or “Suspect: too small”. Section (c) Role Detection Sound, whenever an Employee or Person of Interest got detected, it rangs an alarm depends on what role that individual has when it got detected.



a



b



c

Figure 15. (a) Monitoring Module, (b) InsightFace Detection, (c) Role Detection Sound

Figure 16 section (a) 4 Pin Lock Security displays a security feature for Catalogue Module and Reports Module. It makes the system more secured even if the user is the super admin where it can access everything in the system. It also has customed 4 numerical pin to use for accessing Catalogue Module and Reports Module. Section (b) Catalogue Module shows two different in tables, Employee Catalogue and Person of Interest Catalogue. Both tables show Users ID, Image, Name, Position/Record, Action (Edit and Delete). Section (c) Add Person/Edit Person Form shows a form to add/edit person. The form contains role selection, full name text box, position text box and image picker. Once submitted, person is added to its desired catalogue or its credentials is edited. It also comes with a popup message for confirmation for add/edit and deleting credential of individual registered in the system.

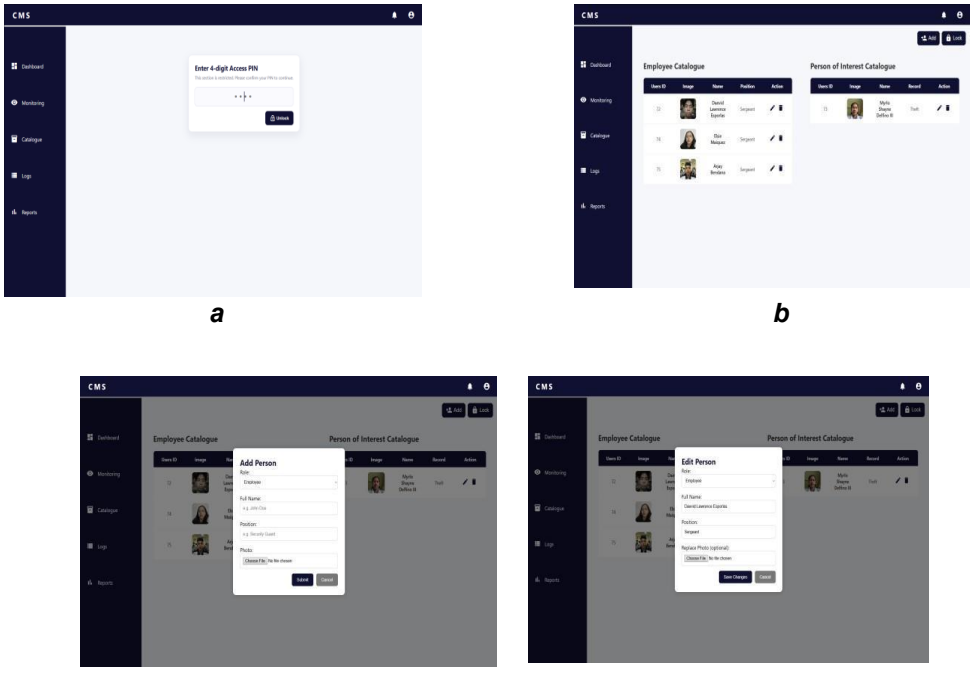
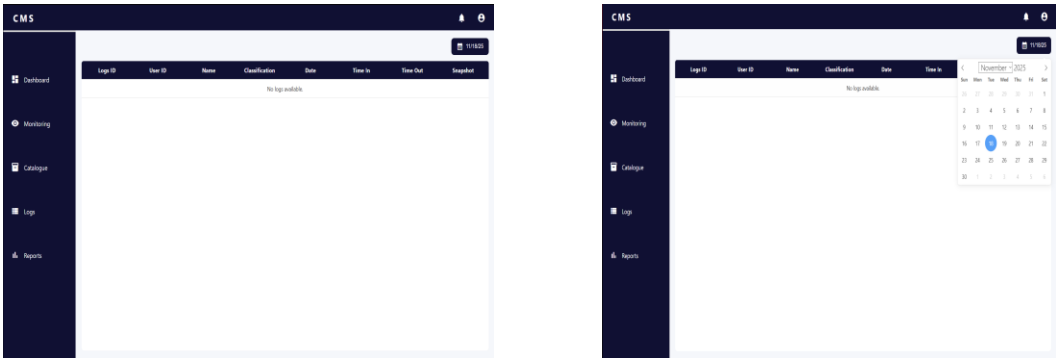


Figure 16. (a) 4 Pin Lock Security, (b) Catalogue Module, (c) Add Person/Edit Person Form

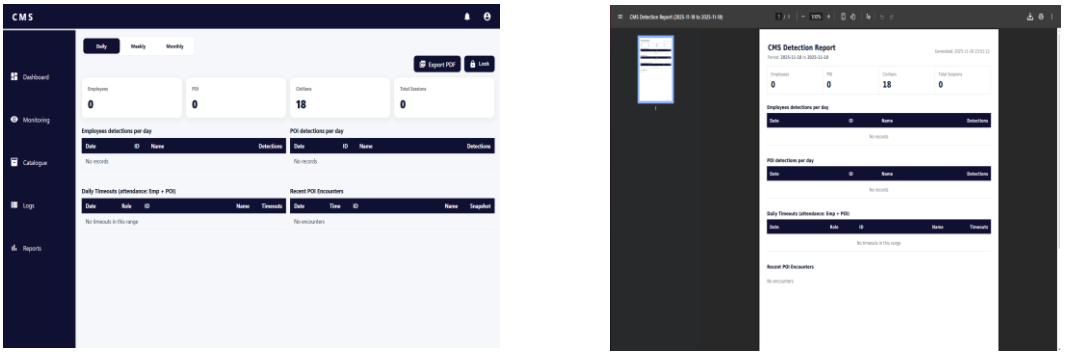
Figure 17 section (a) Logs Module displays the table where every detected individual with role either Employee or Person of Interest is logged. The table contains Logs Id, Users Id, Name, Classification/Role, Date, Time In, Time Out, and Snapshot. Section (b) Date Picker shows a mini calendar to choose what date is desired. By clicking on the date, it can choose what month and year and upon choosing, it shows all the existing logs on that date, if there's no log on that date it shows a plain table.



a ***b***

Figure 17. (a) Logs Module, (b) Date Picker

Figure 18 section (a) Reports Module, it displays the counters for Employee, Civilian, Person of Interest and overall session. It also have table for Employee Detection per day, Person of Interest Detection per day, Daily Timeouts Attendance for Employee and Person of Interest, and lastly Recent Person of Interest Encounters. Section (b) Report to Pdf, upon clicking the Export Pdf button, it makes a pdf file that shows all the counters, all the tables, and for the Person of Interest Encounters, it shows all the snapshots that it got from the Logs Module. The exporting can be toggled daily, weekly and monthly.



a ***b***

Figure 18. (a) Reports Module, (b) Report to Pdf

System Development and Functional Performance

In this section, the results of development and functionality assessment of web-based facial recognition security system using Convolutional Neural Network (CNN) are presented. The goal of this assessment is centered on examining how well the proposed system can automatically detect, recognize, and classify people entering Camp Macario Sakay. Different success factors were employed in ascertaining the effectiveness of the proposed system in relation to accuracy, speed, ease of use, and overall functionality.

Table 12. System Development and Functional Performance

Performance Indicator	**Description**	**Result**	**Interpretation**
Face Detection Accuracy	Ability of the system to detects faces in real-time using CNN	High accuracy	The system successfully detects faces at the camo entrance under normal lightning conditions
Face Identification Rate	Correct identification of registered individuals	High recognition rate	CNN effectively classifies authorized personnel
Classification Speed	Time taken to detect and identify a face	Fast response time	System meets real-time security requirements
System Accessibility	Web-based access across devices	Accessible via browser	Allows easy deployment without additional software

Surveillance, Tracking, and Logging Capabilities

This section addresses the outcome of incorporating additional security surveillance functionalities within the facial recognition system. The aim of the objective is to assess its ability to offer a real-time monitoring system, facial tracking services, and an automatic log of identified persons. These services are crucial for improving situational awareness and maintaining an accurate security audit trail within the camp environment. The assessment addresses the efficiency of the system in recording and storing facial information for security surveillance and future identification.

Table 13. Surveillance, Tracking, and Logging Capabilities

Feature	**Functionality Description**	**System Output**	**Remarks**
Real-Time Monitoring	Displays live camera feed at the entrance	Live video stream	Enable Continuous Surveillance
Facial Tracking	Tracks detected faces in the frame	Active face bounding boxes	Improves monitoring accuracy
Entry logging	Records date, time, and identity	Automatically stored logs	Provides audit trail for security
Database Storage	Stires captured facial data	Secure database	Ensures data integrity and retrieval

Test and evaluate using the ISO 25010 and TAM.

This section presents the evaluation findings of the Web-Based Facial Recognition Security System for Camp Macario Sakay using two recognized assessment models**:** ISO 25010 and the Technology Acceptance Model

(TAM). ISO 25010 served as the basis for examining the system's technical quality, including aspects such as functionality, reliability, usability, efficiency, maintainability, security, and safety. In parallel, TAM was utilized to determine user acceptance by evaluating perceived usefulness, ease of use, and willingness to adopt the system. Together, these results offer an extensive overview of both the system's operational performance and the users' level of acceptance.

The TAM below shows the reliability analysis of different categories related to our system, measured using user. Each category, including Perceived Usefulness, Ease of Use, Attitude, and Behavior, received a SD score above 0.05, which indicates good internal consistency. All categories are interpreted as "GOOD," highlighting overall positive reliability in measuring these aspects. These findings support the validity of the survey instrument, ensuring that the results accurately reflect user perceptions and can be used confidently for further analysis and decision-making.

Table 14. Technology Acceptance Model

CATEGORIES	USER	VERBAL INTERPRETATION
Perceived Usefulness	0.12	GOOD
Perceive Ease of Use	0.36	GOOD
Attitude	0.20	GOOD
Behavior	0.40	GOOD
Composite Score	**0.55**	**GOOD**

Statistical Treatment

To evaluate the effectiveness and user satisfaction of the developed web-based facial recognition security system, the researchers employed purposive sampling by selecting respondents directly involved in the

system’s usage and evaluation. A four- point Likert scale was used to gather feedback, with the following response options: Highly Acceptable – four (4), Moderately Acceptable – three (3), Slightly Acceptable – two (2), and Not Acceptable – one (1). This method was applied to assess key areas such as system functionality, usability, reliability, and interface design. The use of the Likert scale enabled the collection of quantifiable feedback, allowing the researchers to analyze the system’s overall acceptability and performance from the users’ perspective.

Table 15. Likert Scale Method

Rating	Interpretation	Range
4	Highly Acceptable	3.25 – 4.00
3	Moderately Acceptable	2.50 – 3.24
2	Slightly Acceptable	1.75 – 2.49
1	Not Acceptable	1.0 – 1.74

Table 14 shows the Likert Scale that was used as a survey method to gauge how satisfied users were with the application. The respondents' evaluations are measured using a percentage technique. The formula is as follows:

Highest Score – Lowest Score

4 – 1 = 3 Highest Score 3 / 4

= 0.75

ISO 25010

Table 16 shows the evaluated system by the IT experts using the ISO 25010 software quality model. The results show outstanding performance

across most quality attributes. Functional suitability, reliability, performance efficiency, and portability all received perfect mean scores, indicating excellent operation and system responsiveness. Compatibility and maintainability were also rated highly, reflecting good adaptability and ease of updates. Although usability and security still performed well, they were noted as areas requiring further improvement to better support user experience and data protection. Overall, the composite mean score of 4.88 signifies that the system demonstrates very good quality and meets the required standards effectively.

Table 16. Results of survey answered by the IT expert

Categories	**IT 1**	**IT 2**	**IT 3**	**Mean**	**SD**	**Verbal interpretation**
Functional Suitability	5	4	5	5.00	0.00	EXCELLENT
Reliability	5	5	5	5.00	0.00	EXCELLENT
Performance Efficiency	5	5	5	5.00	0.00	EXCELLENT
Usability	5	5	5	5.00	0.00	FAIR
Security	5	5	5	5.00	0.00	NEEDS IMPROVEMENT
Compatibility	4	4	5	4.33	0.58	EXCELLENT
Maintainability	5	4	4	4.80	0.31	GOOD
Portability	5	5	5	5.00	0.00	EXELLENT
Composite Mean				**4.88**	**0.22**	**GOOD**

The IT experts rated the system with a high overall quality, reflected in a composite mean of 4.88 (SD = 0.22) and a verbal interpretation of Good. The system achieved excellent scores in several key areas, including functional suitability, reliability, performance efficiency, and portability, indicating that it performs its intended functions accurately, operates

consistently, and can run smoothly across different environments.

Technology Acceptance Model (TAM)

The findings show's that respondents strongly agree that the system is useful in enhancing camp security operations, with a composite mean of 3.66. Users particularly value its ability to efficiently track and log individuals (3.80), while also recognizing that it accurately detects faces, speeds up identity verification, and improves reliability in security tasks (3.63–3.67). Overall, responses are consistent, indicating a positive perception of the system's benefits in security management.

Table 17. Results of Perceived Usefulness

ITEM N	MEA	STANDARD DEVIATION	INTERPRETATION
The system accurately detects faces.	3.67	±0.48	Strongly Agree
The system enhances camp security.	3.63	0.56	Strongly Agree
The system makes identity verification faster.	3.67	±0.55	Strongly Agree
The system tracks and logs individual efficiency.	3.80	±0.41	Strongly Agree
The system improves reliability in security operation.	3.67	±0.48	Strongly Agree
COMPOSITE MEAN	**3.66**	**±0.49**	**Strongly Agree**

The results shows that respondents strongly agree that the system is easy to use, with a composite mean of 3.68 and a low standard deviation, indicating consistent feedback. Users particularly appreciate the interface design and setup process, both receiving perfect mean scores of 4.00. Learning to use the system and understanding its instructions received slightly lower, yet still strongly positive ratings (3.60), suggesting minor improvements

may further enhance usability. Overall, the findings confirm that the system is user-friendly and manageable even for first-time users.

Table 18. Results of Perceived Ease of Use

ITEM	MEAN	STANDARD DEVIATION	INTERPRETATION
The interface of the system.	4	0.00	Strongly Agree
The set-up process	4	0.00	Strongly Agree
Learning to use the system.	3.60	±0.54	Strongly Agree
Clarity of instructions.	3.60	±0.54	Strongly Agree
System independence.	3.80	±0.44	Strongly Agree
COMPOSITE MEAN	**3.68**	**±0.24**	**Strongly Agree**

The results indicate a strong willingness among users to adopt the system, with a composite mean of 3.72 and a low standard deviation, showing consistent positive responses. Users highly prefer the system over manual verification (4.00), reflecting strong trust in its efficiency. They also show a strong intention to continue using the system (3.80) and are willing to recommend it to others (3.60). Additionally, respondents agree that the system should be integrated into daily operations (3.60). Overall, the findings demonstrate that users are confident in the system and are motivated to use it regularly in camp security processes.

Table 19. Behavioral Intention to use

ITEM	MEAN	STANDARD DEVIATION	INTERPRETATION
Intention to use the system regularly.	3.60	±0.54	Strongly Agree
Willingness to recommended.	3.60	±0.54	Strongly Agree
Preference over manual Verification.	4	0.00	Strongly Agree
Tracking and managing Intention for continuous use.	3.80	±0.44	Strongly Agree
Integration in daily operations.	3.60	±0.54	Strongly Agree
COMPOSITE MEAN	**3.72**	**±0.23**	**Strongly Agree**

The results show a very positive attitude toward the system, with a composite mean of 3.92 indicating strong agreement among users. They find the system enjoyable and modern, believe it improves the efficiency of document processing, and feel confident in its performance. Overall, the system enhances their trust in the barangay's services.

Table 20. Attitude Toward Use

ITEM	MEAN	STANDARD DEVIATION	INTERPRETATION
Enjoyment while using the system.	4	0.00	Strongly Agree
Confidence in system performance.	3.80	±0.44	Strongly Agree
Impression of modernization.	4	0.00	Strongly Agree
Feeling that document acquisition more modern effective because of the system.	4	0.00	Strongly Agree
The system builds confidence in the barangay's service	3.80	±0.44	Strongly Agree
COMPOSITE MEAN	**3.92**	**±0.24**	**Strongly Agree**

CHAPTER V

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

SUMMARY

This project developed a web-based facial recognition security system using CNN to improve access control and security at Camp Macario Sakay. It addressed issues such as lack of centralized identification, limited manpower, and human error in manual verification. The system includes modules for Login, Dashboard, Monitoring, Face Recognition, Logs, Catalogue, Notifications, and Reports, enabling real-time identification of individuals and automated logging to support security personnel.

CNN-driven recognition process enhanced safety and supports SDG goals on innovation, sustainable communities, and strong institutions. However, the system has limitations, including sensitivity to lighting, difficulty recognizing masked or obstructed faces, a limited dataset, manual registration requirements, and lack of mobile or multi-channel notifications.

The reviewed literature shows that modern security increasingly relies on advanced technologies like facial recognition and intelligent video surveillance, which offer greater accuracy and reliability than traditional human-dependent methods. Studies highlight the effectiveness of CNNs in real-time facial identification due to their ability to process complex visual data. While challenges such as lighting and privacy concerns remain, current systems demonstrate strong benefits in automated monitoring and faster response.

Synthesis confirms that a web-based facial recognition system using pre- trained CNN models can address common security flaws, reduce human

error, and improve decision-making. The conceptual framework integrates InsightFace, live webcam detection, and a web application for logging, notifications, and classification, with automated dashboard updates for real-time monitoring. Evaluation using ISO/IEC 25010 shows that the system meets Camp Macario Sakay's functional and performance needs.

System combines a PHP-based web application with MySQL, a Python service for live camera processing, and InsightFace for facial detection and recognition. It identifies individuals by matching embeddings to the Catalogue and classifies them as employees, persons of interest, or civilians. Detected entries automatically update the Logs, Notifications, Dashboard, and Reports, while role- based access ensures secure management.

Developed and tested exclusively at Camp General Macario Sakay, the system enhances gate security through real-time automated facial identification. Using the Agile Scrum methodology, development progressed through iterative sprints, prioritizing key features such as monitoring, facial recognition, catalogue management, logs, notifications, and user interface design.

Camp General Macario Sakay Web Security System's entire development process was described in this chapter. It explained how InsightFace with CNN recognition identifies people in real time and how authorized users can access monitoring, notifications, logs, and catalogue modules. The Administrator and Super Administrator's responsibilities for safe and effective monitoring were outlined in a use case diagram.

The project prioritized important features including facial recognition, monitoring, logging, and notifications utilizing backlogs, daily scrums, sprints,

and reviews in accordance with the Agile Scrum methodology. The hardware and software utilized, including InsightFace, Python, PHP, MySQL, JavaScript, HTML, CSS, and XAMPP, were described in length in the materials section.

The CNN workflow, InsightFace pipeline, internal face gallery setup, and online system integration of the recognition service were all covered in this chapter. Lastly, unit, integration, system, and acceptance tests were used to validate the quality of the system.

CONCLUSION

The Camp General Macario Sakay Web Security System has successfully established a web-based facial recognition technology that employs CNN and InsightFace to improve security, automate identification, and eliminate human error. To guarantee the safe and effective administration of workers, people of interest, and citizens, the system incorporates real-time monitoring, logging, notifications, and role-based access. The project, which was developed using Agile Scrum, used reliable hardware and software tools, such as Python, PHP, MySQL, and XAMPP, and prioritized important features through iterative sprints. Multiple levels of testing verified the operation, performance, and dependability of the system.

Overall, the experiment shows that in a regulated setting like Camp Macario Sakay, a pre-trained CNN-based facial recognition system may efficiently enhance security operations, expedite access management, and support decision- making.

RECOMMENDATION

Based on the evaluation results and the identified limitations of the Web-Based Facial Recognition Security System for Camp General Macario Sakay, the researchers recommend several improvements to enhance its performance and reliability. First, the system's usability may be further improved by refining the interface and providing clearer instructions or guided tutorials to support new users. Strengthening the system's security features is also necessary, particularly through the implementation of stronger data encryption, multi-factor authentication, and regular security audits to protect sensitive facial data.

To address challenges in recognizing faces under poor lighting or partial obstructions, the researchers recommend expanding the dataset and enhancing the facial recognition model with more advanced techniques to improve accuracy. Automating the registration process, such as through bulk uploads or ID scanning, may also help streamline data entry and reduce manual workload.

Moreover, the system would benefit from the integration of mobile access or multi-channel notifications to ensure real-time monitoring and faster response from security personnel. Future work should also consider testing the system in other locations to validate its performance in different environments and to gather broader user feedback. Finally, compliance with data privacy regulations must be maintained through clear policies on consent, storage, and data protection to ensure ethical and responsible system use.

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APPENDIX A.

Plagiarism Results


Match Overview
15%
1 Submitted to Laguna S... 1%
Student Paper
2 link.springer.com 1%
Internet Source
3 www.researchgate.net 1%
Internet Source
4 www.mdpi.com <1%
Internet Source
5 www.sws.org.ph <1%
Internet Source
6 www.coursehero.com <1%
Internet Source
7 V. Sharmila, S. Kannad... <1%
Publication
8 wiredspace.wits.ac.za <1%
Internet Source
9 mcc.edu.ph <1%
Internet Source
10 worldwidescience.org <1%
Internet Source

APPENDIX B.

Survey Questionnaires

Technology Acceptance Questionnaire

Dear Respondent,

Good day! We are 4th-year students pursuing a BS in Information Technology specializing in Web and Mobile Application Development conducting a research study and developing a system entitled, **"Web-Based Facial Recognition Security System for Camp Macario Sakay."** We seek to determine the acceptability of the technology developed by asking for your opinion using the indicators below. Please use the description for each score so that you will be guided accordingly. Rest assured that any information you provide will be treated with the utmost confidentiality in accordance with RA 10173 (Data Privacy Act) and will be used solely for research purposes. Thank you for your cooperation.

Sincerely,

Arjay I. Bendaña, Daevid Lawrence Esporlas, Myrlo Delfino III
LSPU Los Baños, College of Computer Studies

Technology Acceptance Model Rubric

Please encircle the score associated with each category

1. Perceived Usefulness. Measures how well the system improves security, accuracy, and overall usefulness in identifying individuals entering the camp.

Indicator	4 – excellent	3 – Good	2- Fair	1 - Poor
1. The system accurately detects faces.	Accurately recognizes individuals instantly with high reliability.	Detects faces correctly most of the time with few delays.	Detects faces sometimes but misses a few.	Often fails to detect or identify faces.
2. The system enhances camp security.	Greatly strengthens monitoring and safety within the premises.	Improves security but still needs refinement.	Slightly helps improve security.	Has no effect on camp security.
3. The system makes identity verification faster.	Significantly reduces time and effort during entry.	Moderately improves speed and reduces delays.	Slightly faster than manual verification.	No improvement in verification speed.
4. The system tracks and logs individuals efficiently.	Accurately records and logs all entries in real-time.	Logs most faces correctly but sometimes misses data.	Logs are incomplete or inaccurate at times.	Fails to log or track faces properly.
5. The system improves reliability in security operations.	Highly dependable and consistent in detecting faces.	Reliable with few errors or interruptions.	Occasionally unreliable during use.	Often fails or produces errors.

2. **Perceived Ease of Use.** Measures simplicity, usability, and learning comfort when using the system.

Indicator	4 – Excellent	3 – Good	2 – Fair	1 - Poor
1. The interface of the system.	Very clear, intuitive, and easy to navigate.	Simple to use with a few minor confusions.	Sometimes confusing but manageable.	Hard to understand and operate.
2. The setup process.	Quick and easy to install without help.	Easy setup but requires small guidance.	Takes time to set up correctly.	Complicated and needs technical help.
3. Learning to use the system.	Very easy to learn and remember.	Easy after a few tries.	Takes effort to recall how to use features.	Difficult to learn or remember.
4. Clarity of instructions.	Instructions are clear and easy to follow.	Mostly understandable with few unclear parts.	Some instructions are confusing.	Instructions are hard to follow.
5. System independence.	Can be used easily without any help.	Needs minimal help sometimes.	Needs frequent assistance.	Cannot be used without help.

3. **Behavioral Intention to Actual Use.** Measures the user's intention to continue using and adopt the system for daily camp operations.

Indicator	4 – excellent	3 – Good	2 – Fair	1 – Poor
1. Intention to use the system regularly.	Strongly intends to use it every day.	Likely to use it often.	May use it occasionally.	No intention to use again.
2. Willingness to recommend.	Actively recommends the system to others.	Sometimes recommends it.	Rarely recommends it.	Does not recommend.
3. Preference over manual verification.	Fully prefers it over manual methods.	Prefers it but still uses manual sometimes.	Neutral between both.	Prefers traditional checking.
4. Intention for continuous use.	Plans to continue using it long-term.	Will use it until new updates are needed.	May continue using if improved.	Does not plan to use again.
5. Integration in daily operations.	Fully integrates the system in camp procedures.	Partly integrated with other systems.	Used occasionally in security.	Not integrated at all.

4. **Attitude Toward Using the System.** Measures satisfaction, comfort, and emotional response when using the system.

Indicator	4- Excellent	3 – Good	2 – Fair	1 - Poor
1. Enjoyment while using the system.	Greatly enjoys using it and finds it engaging.	Enjoys using it most of the time.	Rarely enjoys using it.	Does not enjoy using it at all.
2. Confidence in system performance.	Feels very confident in its accuracy and reliability.	Confident but sees small issues.	Some doubts about performance.	Not confident in system results.
3. Impression of modernization.	Strongly believes it modernizes camp security.	Considers it a good technological improvement.	Somewhat modern but needs updates.	Does not seem modern.
4. Comfort level in usage.	Feels very comfortable and secure using it.	Feels comfortable most of the time.	Sometimes uncomfortable.	Feels unsafe or uneasy.
5. Overall satisfaction.	Highly satisfied with system experience.	Satisfied but sees areas for improvement.	Slightly satisfied.	Not satisfied at all.

_____ _____

Signature over printed name Respondent — Date Signed

ISO Questionnaire

Dear IT Expert,

Good day! We are 4th-year students conducting a research study and developing a system entitled,

“Web-Based Facial Recognition Security System for Camp Macario Sakay.”

We are currently pursuing a BS in Information Technology specializing in Web

and Mobile App Development. We now seek to determine if the said system adheres to ISO 25010 standards. To assess the product quality, we have devised a rubric aligned with our project objectives. Your responses helps us evaluate various aspects of the system based on the said international standard.

Sincerely,
Arjay I. Bendaña, Daevid Lawrence Esporlas, Myrlo Shane Delfino III
LSPU Los Baños, College of Computer Studies

ISO 25010 RUBRIC

Please encircle the score associated with each category

FUNCTIONAL SUITABILITY

Indicator	5 - Excellent	4 - Very Good	3 - Good	2 - Fair	1 - Poor
The system provides all necessary facial recognition functions for accurate identification at the camp entrance.	All facial recognition features work accurately and efficiently.	Most features work well with rare detection errors.	Most features function properly but may miss a few identifications.	Several recognition issues occur.	Many recognition functions fail.
The system detects and classifies faces correctly using CNN.	Always accurate detection and classification.	Minor misidentifications.	Occasionally misclassifies faces.	Often inaccurate detections.	Fails to classify or detect faces properly.
The system fulfills user requirements for camp entry validation.	Fully supports security validation needs.	Supports most validation functions.	Adequately relevant.	Limited accuracy.	Not suitable for validation.

PERFORMANCE EFFICIENCY

Indicator	5 - Excellent	4 - Very Good	3 - Good	2 - Fair	1 - Poor
The system processes facial recognition quickly and efficiently.	Very fast and responsive.	Usually fast; minor delay.	Acceptable response time	Often slow or lagging.	Very slow and unresponsive.
The system efficiently handles multiple recognition requests.	Highly optimized for multiple inputs.	Efficient most of the time.	Acceptable performance.	Struggles with high loads.	Poor optimization.

USABILITY

Indicator	5 - Excellent	4 Very Good	3 - Good	2 - Fair	1 - Poor
The interface is clear and easy for security personnel to use.	Highly intuitive and user-friendly.	Easy with minimal training.	Understandable with effort.	Somewhat confusing.	Hard to use.

Users can easily view, track, and log facial data.	Smooth and simple tracking and logging.	Minor navigation issues.	Usable but needs improvement.	Difficult to track data.	Not functional.

Provides clear feedback and alerts during recognition.	Real-time alerts and logs.	Reliable feedback.	Some delay in alerts.	Occasional alert failure.	No feedback or alerts.

COMPATIBILITY

Indicator	5 - Excellent	4 - Very Good	3 - Good	2 - Fair	1 - Poor
The system works well with other existing camp security systems or platforms.	Fully compatible.	Mostly compatible.	Some integration issues.	Limited compatibility.	Not compatible at all.
The system exchanges and synchronizes data effectively with related applications or devices.	Excellent interoperability.	Good integration.	Partial data exchange.	Limited integration.	Cannot exchange data.

REALIABILITY

Indicator	5 - Excellent	4 - Very Good	3 - Good	2 - Fair	1 - Poor
The system operates continuously without crashes or errors.	Always stable.	Rare minor bugs.	Occasionally unstable.	Often unstable.	Crashes frequently.
The system consistently recognizes faces under different conditions..	Always reliable.	Mostly reliable.	Occasionally fails.	Often inconsistent.	Unreliable recognition.
Data logs are stored and retrievable after interruptions.	Always recoverable	Recovers most data.	Some data loss.	Frequent losses.	No recovery possible.

SECURITY

Indicator	5 - Excellent	4 - Very Good	3 - Good	2 - Fair	1 - Poor
Protects facial data from unauthorized access.	Strong data encryption and access control.	Secure with few concerns.	Minor vulnerabilities.	Risky protection.	Not secure.
Ensures integrity and accuracy of logged data.	Fully maintains accuracy.	Mostly consistent.	Minor inconsistencies.	Often inaccurate.	Data easily altered..
Tracks and audits user access and activity.	Complete audit trail.	Reliable tracking.	Partial tracking.	Limited logs.	No tracking.

MAINTANABILITY

Indicator	5 - Excellent	4 - Very Good	3 - Good	2 - Fair	1 - Poor

The system's modules are well-organized and easy to update.	Highly modular.	Mostly modular.	Some dependency issues.	Hard to update.	Poorly structured.
Code is reusable and easy to modify for improvements.	Highly reusable.	Reusable with adjustments.	Moderately reusable.	Hard to reuse.	Not reusable.

Issues and bugs are easy to identify and fix.	Easy to diagnose and resolve.	Manageable debugging.	Moderate effort needed.	Hard to fix.	Very difficult.

PORTABILITY

Indicator	5 - Excellent	4 - Very Good	3 - Good	2 - Fair	1 - Poor
The system works on multiple devices (desktop, laptop, tablet).	Fully adaptive.	Works on most devices.	Limited responsiveness.	Minor device issues.	Not adaptive.
The system can handle increased user traffic and data volume.	Scales efficiently.	Handles moderate load well.	Limited scalability.	Struggles under load.	Cannot scale.

SAFETY

Indicator	5 - Excellent	4 - Very Good	3 - Good	2 - Fair	1 – Poor
The system operates safely within expected parameters.	No risks at all.	Very few minor risks.	Occasionally risky.	Often risky.	Unsafe operation.
The system identifies and prevents potential risks or data loss.	Fully detects and Prevents risk.	Detects most issue.	Sometimes misses.	Often fails to detect.	No detection.
Provides alerts or warnings for system errors or threats.	Always provides timely alerts.	Usually alerts users.	Occasionally alerts.	Rare alerts.	No warning system.

_____

Signature over printed name Date
Signed
Survey Questionnaire Validator

APPENDIX C.

Relevant Source Code

login.php

```php
<?php
session_start();
require 'includes/conn.php';

if ($_SERVER['REQUEST_METHOD'] == 'POST') {
    $username = trim($_POST['username']);
    $password = trim($_POST['password']);

    // Prepared statement to prevent SQL injection
    $stmt = $conn->prepare("SELECT * FROM admin WHERE username = ?");
    $stmt->bind_param("s", $username);
    $stmt->execute();
    $result = $stmt->get_result();
    $admin = $result->fetch_assoc();

    // Check SHA-256 password
    if ($admin && hash('sha256', $password) === $admin['password']) {
        $_SESSION['username'] = $admin['username']; // store username
        $_SESSION['role'] = $admin['role'];         // store role

        // Success popup and redirect
        echo "<script>
                alert('Login successful! Redirecting to main page...');
                window.location.href='dashboard.php';
              </script>";
        exit();
    } else {
        // Failure popup
        echo "<script>
                alert('Invalid username or password.');
                window.location.href='login.php';
              </script>";
        exit();
    }
}
?>
```

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Admin Login</title>
    <link rel="stylesheet" href="css/style.css">
</head>
<body>

<form method="POST" class="login-form">
    <h2>Admin Login</h2>
    <input type="text" name="username" required placeholder="Username"><br>
    <input type="password" name="password" required placeholder="Password"><br>
    <button type="submit">Login</button>
</form>

</body>
</html>
```

logout.php

```php
<?php
session_start();
session_unset();
session_destroy();
header("Location: login.php");
exit();
```

dashboard.php

```php
<?php
session_start();
require 'includes/conn.php';

// Redirect if not logged in
if (!isset($_SESSION['username'])) {
    header("Location: login.php");
    exit();
}
$role  = $_SESSION['role'] ?? 'user';
$today = date('Y-m-d');

/* ------------------------------------------------
   Quick Stats
------------------------------------------------ */
$totalEmployees = (int)($conn->query("SELECT COUNT(*) AS cnt FROM users WHERE role='employee'")->fetch_assoc()['cnt'] ?? 0);
$totalPOI       = (int)($conn->query("SELECT COUNT(*) AS cnt FROM users WHERE role='person_of_interest'")->fetch_assoc()['cnt'] ?? 0);
$totalLogs      = (int)($conn->query("SELECT COUNT(*) AS cnt FROM logs")->fetch_assoc()['cnt'] ?? 0);

/* ------------------------------------------------
   Detections Today (logs + visitors.json civilians)
------------------------------------------------ */
$todayLogs = (int)($conn->query("SELECT COUNT(*) AS cnt FROM logs WHERE date = '{$conn->real_escape_string($today)}'")->fetch_assoc()['cnt'] ?? 0);

$visitors_file  = __DIR__ . '/snapshots/visitors.json';
$todayVisitors  = 0;
if (is_file($visitors_file)) {
    $vdata = json_decode(@file_get_contents($visitors_file), true);
    if (is_array($vdata)) {
        foreach ($vdata as $meta) {
            // we only treat civilians as 'today' if last_seen is exactly today
            if (!empty($meta['last_seen']) && substr($meta['last_seen'], 0, 10) === $today) {
                $todayVisitors++;
            }
        }
    }
}
$recentDetected = $todayLogs + $todayVisitors;

/* ------------------------------------------------
   Chart Data (last 7 days, zero-filled)
------------------------------------------------ */
$days = [];
for ($i = 6; $i >= 0; $i--) {
    $days[] = date('Y-m-d', strtotime("-{$i} days", strtotime($today)));
}
```

```php
// fetch counts for last 7 days from logs
$start = $conn->real_escape_string($days[0]);
$end   = $conn->real_escape_string($days[count($days)-1]);

$sql = "
  SELECT date, role, COUNT(*) AS cnt
  FROM logs
  WHERE date BETWEEN '$start' AND '$end'
  GROUP BY date, role
";
$res = $conn->query($sql);
$byDayRole = []; // date => ['employee'=>n, 'person_of_interest'=>n, 'civilian'=>n]
while ($row = $res->fetch_assoc()) {
    $d = $row['date'];
    $r = $row['role'];
    $c = (int)$row['cnt'];
    if (!isset($byDayRole[$d])) $byDayRole[$d] = ['employee'=>0, 'person_of_interest'=>0, 'civilian'=>0];
    $byDayRole[$d][$r] = $c;
}

// build aligned arrays
$labels          = $days;
$employeeCounts  = [];
$poiCounts       = [];
$civilianCounts  = [];

foreach ($days as $d) {
    $employeeCounts[] = (int)($byDayRole[$d]['employee'] ?? 0);
    $poiCounts[]      = (int)($byDayRole[$d]['person_of_interest'] ?? 0);
    $civilianCounts[] = (int)($byDayRole[$d]['civilian'] ?? 0);
}

// add today's civilians from visitors.json to today's bucket
$todayIndex = array_search($today, $labels, true);
if ($todayIndex !== false) {
    $civilianCounts[$todayIndex] += (int)$todayVisitors;
}
```

```php
/* ------------------------------------------------
   Top detected individuals (top 5, last 30 days for relevance)
------------------------------------------------ */
$since = $conn->real_escape_string(date('Y-m-d', strtotime('-30 days', strtotime($today))));
$topSql = "
  SELECT COALESCE(u.name, CONCAT('ID ', l.user_id)) AS name, COUNT(l.id) AS cnt
  FROM logs l
  LEFT JOIN users u ON u.id = l.user_id
  WHERE l.date >= '$since'
  GROUP BY name
  ORDER BY cnt DESC
  LIMIT 5
";
$topRes = $conn->query($topSql);
$topNames = [];
$topCounts = [];
while ($row = $topRes->fetch_assoc()) {
    $topNames[]  = $row['name'];
    $topCounts[] = (int)$row['cnt'];
}
?>
```

```php
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1, viewport-fit=cover">
  <title>Dashboard</title>
  <link href="https://fonts.googleapis.com/icon?family=Material+Icons" rel="stylesheet">
  <link rel="stylesheet" href="css/style.css">
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
  <style>
    /* ensure charts have height */
    .charts-container { display:grid; grid-template-columns: 1fr 1fr; gap:16px; }
    .chart-card { background:#fff; border:1px solid #e5e7eb; border-radius:12px; padding:16px; }
    .chart-card canvas { width:100%; height:320px; }
    .stats-cards { display:grid; grid-template-columns: repeat(4,1fr); gap:16px; margin:16px 0; }
    .stats-cards .card { background:#fff; border:1px solid #e5e7eb; border-radius:12px; padding:16px; text-align:center; }
    .stats-cards .card h2 { margin:6px 0 0; font-size:28px; }
    .role-label.role-superadmin{ color:#4CAF50; font-weight:600;}
    .role-label.role-admin{ color:#2196F3; font-weight:600;}
  </style>
</head>
<body>
<?php include('includes/header.php'); ?>
<div class="layout">
  <?php include('includes/sidebar.php'); ?>
  <main class="main-content">
    <h1>Welcome, <?= htmlspecialchars($_SESSION['username']); ?>!</h1>

    <!-- Role display -->
    <p>
      <span class="material-icons role-icon"
            style="color: <?= $role === 'superadmin' ? '#4CAF50' : ($role === 'admin' ? '#2196F3' : '#101033'); ?>;">
        <?= ($role === 'superadmin' || $role === 'admin') ? 'manage_accounts' : 'person'; ?>
      </span>
      <span class="role-label <?= $role === 'superadmin' ? 'role-superadmin' : ($role === 'admin' ? 'role-admin' : 'role-other'); ?>">
        <?= htmlspecialchars(ucfirst($role)); ?>
      </span>
    </p>

    <!-- Quick Stats -->
    <div class="stats-cards">
      <div class="card">
        <span class="material-icons">groups</span>
        <h2><?= $totalEmployees; ?></h2>
        <p>Employees</p>
      </div>
      <div class="card">
        <span class="material-icons">warning</span>
        <h2><?= $totalPOI; ?></h2>
        <p>Persons of Interest</p>
      </div>
```

```
<div class="layout">
  <main class="main-content">
    <div class="stats-cards">
      <div class="card">
        <span class="material-icons">list</span>
        <h2><?= $totalLogs; ?></h2>
        <p>Total Logs</p>
      </div>
      <div class="card">
        <span class="material-icons">visibility</span>
        <h2 id="detections-today" data-db="<?= $recentDetected; ?>"><?= $recentDetected; ?></h2>
        <p>Detections Today</p>
      </div>
    </div>

    <!-- Graphs Section -->
    <div class="charts-container">
      <div class="chart-card">
        <h3>Daily Detections</h3>
        <canvas id="dailyChart"></canvas>
      </div>
      <div class="chart-card">
        <h3>Top Detected Individuals</h3>
        <canvas id="topChart"></canvas>
      </div>
    </div>
  </main>
</div>

<!-- JSON blobs for charts -->
<script id="daily-data" type="application/json">
<?= json_encode([
  'labels'    => $labels,
  'employees' => $employeeCounts,
  'poi'       => $poiCounts,
  'civilians' => $civilianCounts
], JSON_UNESCAPED_SLASHES); ?>
</script>

<script id="top-data" type="application/json">
<?= json_encode([
  'names'  => $topNames,
  'counts' => $topCounts
], JSON_UNESCAPED_SLASHES); ?>
</script>

<script src="js/script.js"></script>
</body>
</html>
```

monitoring.php

```php
<?php
session_start();
require 'includes/conn.php';

// Redirect if not logged in
if (!isset($_SESSION['username'])) {
    header("Location: login.php");
    exit();
}
$role = $_SESSION['role'];
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1, viewport-fit=cover">
    <title>Monitoring</title>
    <link href="https://fonts.googleapis.com/icon?family=Material+Icons" rel="stylesheet">
    <link rel="stylesheet" href="css/style.css">
    <?php include('includes/conn.php'); ?>
</head>
<body>
    <?php include('includes/header.php'); ?>
    <div class="layout">
        <?php include('includes/sidebar.php'); ?>
        <main class="main-content">
            <!-- Large Monitoring Section -->
            <section class="monitoring-section">
                <div class="monitoring-placeholder monitoring-active" id="monitoringBox">
                    <img src="http://127.0.0.1:5000/video_feed" class="monitoring-stream" alt="Live Feed">
                </div>
            </section>
        </main>
    </div>
    <script src="js/script.js"></script>
</body>
</html>
```

inventory.php

```php
<?php
session_start();
require 'includes/conn.php';
require 'includes/code.php'; // defines: const INVENTORY_PIN = '1234';

if (!defined('INVENTORY_PIN')) {
    die('PIN not configured: INVENTORY_PIN is not defined. Check includes/code.php.');
}

// Restrict access to superadmin only
if (!isset($_SESSION['role']) || $_SESSION['role'] !== 'superadmin') {
    header("Location: index.php?msg=Access Denied");
    exit();
}

/** ---- 4-digit PIN Gate ---- */
$PIN_SESSION_FLAG = 'inventory_pin_ok';
$PIN_ATTEMPTS     = 'inventory_pin_attempts';
$PIN_LOCK_UNTIL   = 'inventory_pin_lock_until';
$MAX_ATTEMPTS     = 5;
$LOCK_SECONDS     = 5 * 60;

// initialize counters
$_SESSION[$PIN_ATTEMPTS] = $_SESSION[$PIN_ATTEMPTS] ?? 0;

// lockout check
if (!empty($_SESSION[$PIN_LOCK_UNTIL]) && time() < $_SESSION[$PIN_LOCK_UNTIL]) {
    $remain = $_SESSION[$PIN_LOCK_UNTIL] - time();
    $mins = floor($remain/60); $secs = $remain%60;
    $lock_msg = "Too many incorrect attempts. Try again in {$mins}m {$secs}s.";
}

// handle PIN submit
if (($_SERVER['REQUEST_METHOD'] === 'POST') && isset($_POST['pin_submit']) && !isset($lock_msg)) {
    // CSRF
    if (!isset($_POST['csrf']) || !hash_equals($_SESSION['csrf'] ?? '', $_POST['csrf'])) {
        $pin_error = "Invalid request. Please try again.";
    } else {
        $pin = $_POST['pin'] ?? '';
        if (!preg_match('/^\d{4}$/', $pin)) {
            $pin_error = "PIN must be 4 digits.";
        } elseif (hash_equals(INVENTORY_PIN, $pin)) {
            session_regenerate_id(true);
            $_SESSION[$PIN_SESSION_FLAG] = true;
            $_SESSION[$PIN_ATTEMPTS] = 0;
            unset($_SESSION[$PIN_LOCK_UNTIL]);

            // PRG: clear old ?msg=... from URL
            header("Location: inventory.php");
            exit();
        } else {
            $_SESSION[$PIN_ATTEMPTS]++;
            if ($_SESSION[$PIN_ATTEMPTS] >= $MAX_ATTEMPTS) {
                $_SESSION[$PIN_LOCK_UNTIL] = time() + $LOCK_SECONDS;
                $lock_msg = "Too many incorrect attempts. Try again in 5 minutes.";
            } else {
                $left = $MAX_ATTEMPTS - $_SESSION[$PIN_ATTEMPTS];
                $pin_error = "Incorrect PIN. {$left} attempt(s) left.";
            }
        }
    }
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1, viewport-fit=cover">
  <title>Inventory</title>
  <link href="https://fonts.googleapis.com/icon?family=Material+Icons" rel="stylesheet" />
  <link rel="stylesheet" href="css/style.css" />
</head>
<body class="<?php echo empty($_SESSION[$PIN_SESSION_FLAG]) ? 'locked' : ''; ?>">
  <?php include('includes/header.php'); ?>
  <div class="layout">
    <?php include('includes/sidebar.php'); ?>
    <main class="main-content">

      <?php if (empty($_SESSION[$PIN_SESSION_FLAG])): ?>
        <?php $_SESSION['csrf'] = bin2hex(random_bytes(16)); ?>
        <!-- ======= LOCKED: PIN card ======= -->
        <div class="pin-card">
          <h1>Enter 4-digit Access PIN</h1>
          <p class="pin-sub">This section is restricted. Please confirm your PIN to continue.</p>

          <form method="POST" autocomplete="off">
            <input type="hidden" name="csrf" value="<?= htmlspecialchars($_SESSION['csrf']) ?>">
            <!-- Anti-autofill dummies -->
            <input type="text" style="position:absolute; left:-9999px; opacity:0;" autocomplete="username">
            <input type="password" style="position:absolute; left:-9999px; opacity:0;" autocomplete="current-password">

            <div class="pin-field">
              <input class="pin-input"
                     type="password"
                     name="pin"
                     inputmode="numeric"
                     pattern="\d{4}"
                     maxlength="4"
```

```
                    placeholder="••••"
                    value=""
                    autocomplete="new-password"
                    autocapitalize="off"
                    spellcheck="false"
                    oninput="this.value=this.value.replace(/[^0-9]/g,'').slice(0,4)"
                    required
                    autofocus />
            </div>

            <div class="pin-actions">
              <button class="btn-primary" type="submit" name="pin_submit">
                <span class="material-icons" style="vertical-align:-6px;font-size:20px;">lock_open</span>
                Unlock
              </button>
            </div>

            <?php if (!empty($lock_msg)): ?>
              <div class="pin-msg"><?= htmlspecialchars($lock_msg) ?></div>
            <?php elseif (!empty($pin_error)): ?>
              <div class="pin-err"><?= htmlspecialchars($pin_error) ?></div>
            <?php endif; ?>
          </form>
        </div>

      <?php else: ?>
        <!-- ======= UNLOCKED: normal page content ======= -->

        <?php if (isset($_GET['msg'])): ?>
          <script>
            // lightweight feedback like your old version
            window.addEventListener('DOMContentLoaded', function () {
              alert("<?= htmlspecialchars($_GET['msg']) ?>");
              <?php if (isset($_GET['error']) && $_GET['error'] == 1): ?>
                if (typeof openAddForm === 'function') openAddForm();
              <?php endif; ?>
            });
          </script>
        <?php endif; ?>

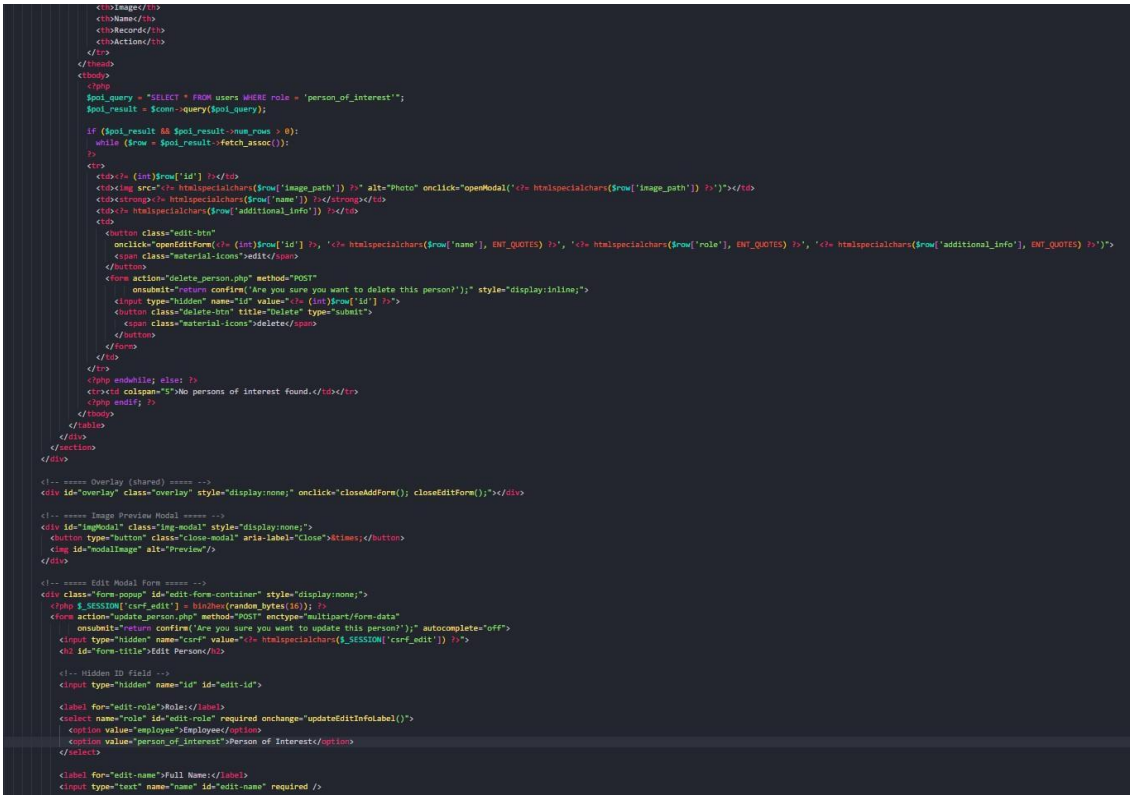
        <div class="top-actions">
          <button class="add-person-btn" onclick="openAddForm()">
            <span class="material-icons">person_add</span> Add
          </button>
```

```
    <!-- Quick lock button -->
    <form method="post" action="pin_lock.php" style="display:inline;margin-left:8px;">
      <input type="hidden" name="page" value="inventory">
      <button class="add-person-btn btn-lock" type="submit" title="Lock PIN gate">
        <span class="material-icons">lock</span> Lock
      </button>
    </form>
  </div>

  <div class="inventory-wrapper">
    <!-- Employee Inventory -->
    <section class="inventory-section">
      <h1>Employee Catalogue</h1>
      <div class="inventory-table-wrapper">
        <table class="inventory-table">
          <thead>
            <tr>
              <th>Users ID</th>
              <th>Image</th>
              <th>Name</th>
              <th>Position</th>
              <th>Action</th>
            </tr>
          </thead>
          <tbody>
            <?php
            $employee_query = "SELECT * FROM users WHERE role = 'employee'";
            $employee_result = $conn->query($employee_query);

            if ($employee_result && $employee_result->num_rows > 0):
              while ($row = $employee_result->fetch_assoc()):
            ?>
            <tr>
              <td><?= (int)$row['id'] ?></td>
              <td><img src="<?= htmlspecialchars($row['image_path']) ?>" alt="Photo" onclick="openModal('<?= htmlspecialchars($row['image_path']) ?>')"></td>
              <td><strong><?= htmlspecialchars($row['name']) ?></strong></td>
              <td><?= htmlspecialchars($row['additional_info']) ?></td>
              <td>
                <button class="edit-btn"
                  onclick="openEditForm(<?= (int)$row['id'] ?>, '<?= htmlspecialchars($row['name'], ENT_QUOTES) ?>', '<?= htmlspecialchars($row['role'], ENT_QUOTES) ?>', '<?= htmlspecialchars($row['additional_info'], ENT_QUOTES) ?>')">
                  <span class="material-icons">edit</span>
                </button>
                <form action="delete_person.php" method="POST"
                      onsubmit="return confirm('Are you sure you want to delete this person?');" style="display:inline;">
                  <input type="hidden" name="id" value="<?= (int)$row['id'] ?>">
                  <button class="delete-btn" title="Delete" type="submit">
                    <span class="material-icons">delete</span>
                  </button>
                </form>
              </td>
            </tr>
            <?php endwhile; else: ?>
            <tr><td colspan="5">No employees found.</td></tr>
            <?php endif; ?>
          </tbody>
        </table>
      </div>
    </section>

    <!-- Person of Interest Inventory -->
    <section class="inventory-section">
      <h1>Person of Interest Catalogue</h1>
      <div class="inventory-table-wrapper">
        <table class="inventory-table">
          <thead>
            <tr>
              <th>Users ID</th>
```



```
            <th>Image</th>
            <th>Name</th>
            <th>Record</th>
            <th>Action</th>
          </tr>
        </thead>
        <tbody>
          <?php
          $poi_query = "SELECT * FROM users WHERE role = 'person_of_interest'";
          $poi_result = $conn->query($poi_query);

          if ($poi_result && $poi_result->num_rows > 0):
            while ($row = $poi_result->fetch_assoc()):
          ?>
          <tr>
            <td><?= (int)$row['id'] ?></td>
            <td><img src="<?= htmlspecialchars($row['image_path']) ?>" alt="Photo" onclick="openModal('<?= htmlspecialchars($row['image_path']) ?>')"></td>
            <td><strong><?= htmlspecialchars($row['name']) ?></strong></td>
            <td><?= htmlspecialchars($row['additional_info']) ?></td>
            <td>
              <button class="edit-btn"
                onclick="openEditForm(<?= (int)$row['id'] ?>, '<?= htmlspecialchars($row['name'], ENT_QUOTES) ?>', '<?= htmlspecialchars($row['role'], ENT_QUOTES) ?>', '<?= htmlspecialchars($row['additional_info'], ENT_QUOTES) ?>')">
                <span class="material-icons">edit</span>
              </button>
              <form action="delete_person.php" method="POST"
                  onsubmit="return confirm('Are you sure you want to delete this person?');" style="display:inline;">
                <input type="hidden" name="id" value="<?= (int)$row['id'] ?>">
                <button class="delete-btn" title="Delete" type="submit">
                  <span class="material-icons">delete</span>
                </button>
              </form>
            </td>
          </tr>
          <?php endwhile; else: ?>
          <tr><td colspan="5">No persons of interest found.</td></tr>
          <?php endif; ?>
        </tbody>
      </table>
    </div>
  </section>
</div>

<!-- ===== Overlay (shared) ===== -->
<div id="overlay" class="overlay" style="display:none;" onclick="closeAddForm(); closeEditForm();"></div>

<!-- ===== Image Preview Modal ===== -->
<div id="imgModal" class="img-modal" style="display:none;">
  <button type="button" class="close-modal" aria-label="Close">&times;</button>
  <img id="modalImage" alt="Preview"/>
</div>

<!-- ===== Edit Modal Form ===== -->
<div class="form-popup" id="edit-form-container" style="display:none;">
  <?php $_SESSION['csrf_edit'] = bin2hex(random_bytes(16)); ?>
  <form action="update_person.php" method="POST" enctype="multipart/form-data"
        onsubmit="return confirm('Are you sure you want to update this person?');" autocomplete="off">
    <input type="hidden" name="csrf" value="<?= htmlspecialchars($_SESSION['csrf_edit']) ?>">
    <h2 id="form-title">Edit Person</h2>

    <!-- Hidden ID field -->
    <input type="hidden" name="id" id="edit-id">

    <label for="edit-role">Role:</label>
    <select name="role" id="edit-role" required onchange="updateEditInfoLabel()">
      <option value="employee">Employee</option>
      <option value="person_of_interest">Person of Interest</option>
    </select>

    <label for="edit-name">Full Name:</label>
    <input type="text" name="name" id="edit-name" required />
```

```
        <label for="edit-additional-info" id="edit-info-label">Position:</label>
        <input type="text" name="additional_info" id="edit-additional-info" required />

        <label for="edit-image">Replace Photo (optional):</label>
        <input type="file" name="image" id="edit-image" accept="image/*" />

        <div class="form-buttons">
          <button type="submit" class="add-btn">Save Changes</button>
          <button type="button" class="add-btn cancel-btn" onclick="closeEditForm()">Cancel</button>
        </div>
      </form>
    </div>

    <!-- ===== Add Modal Form ===== -->
    <div class="form-popup" id="add-form-container" style="display:none;">
      <?php $_SESSION['csrf_add'] = bin2hex(random_bytes(16)); ?>
      <form action="add_person.php" method="POST" enctype="multipart/form-data" autocomplete="off">
        <input type="hidden" name="csrf" value="<?= htmlspecialchars($_SESSION['csrf_add']) ?>">
        <h2 id="form-title">Add Person</h2>

        <label for="role">Role:</label>
        <select name="role" id="role" required onchange="updateAdditionalInfoLabel()">
          <option value="employee">Employee</option>
          <option value="person_of_interest">Person of Interest</option>
          <option value="civilian">Civilian</option>
        </select>

        <label for="name">Full Name:</label>
        <input type="text" id="name" name="name" placeholder="e.g. John Doe" required />

        <label for="additional_info" id="info-label">Position:</label>
        <input type="text" name="additional_info" placeholder="e.g. Security Guard" required />

        <label for="image">Photo:</label>
        <input type="file" name="image" id="image" accept="image/*" />

        <div class="form-buttons">
          <button type="submit" class="add-btn">Submit</button>
          <button type="button" class="add-btn cancel-btn" onclick="closeAddForm()">Cancel</button>
        </div>
      </form>
    </div>

    <!-- Optional: Prevent audio .play() null refs in your notifications JS -->
    <audio id="employeeSound" preload="auto" src="sounds/employee.mp3"></audio>
    <audio id="personSound" preload="auto" src="sounds/poi.mp3"></audio>
    <audio id="civilianSound" preload="auto" src="sounds/civilian.mp3"></audio>

  <?php endif; // end unlocked ?>

  </main>
</div>

<script src="js/script.js"></script>
</body>
</html>
```

logs.php

```php
<?php
session_start();
require 'includes/conn.php';

// Redirect if not logged in
if (!isset($_SESSION['username'])) {
    header("Location: login.php");
    exit();
}

$role = $_SESSION['role'];
$today = date('Y-m-d');
$selectedDate = isset($_GET['date']) ? $_GET['date'] : $today;
?>

<?php if (isset($_GET['msg'])): ?>
  <script>
    alert("<?= htmlspecialchars($_GET['msg']) ?>");
  </script>
<?php endif; ?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1, viewport-fit=cover">
    <title>Logs</title>
    <link href="https://fonts.googleapis.com/icon?family=Material+Icons" rel="stylesheet">
    <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/flatpickr/dist/flatpickr.min.css">
    <link rel="stylesheet" href="css/style.css">
    <script src="https://cdn.jsdelivr.net/npm/flatpickr"></script>
</head>
<body>
    <?php include('includes/header.php'); ?>
    <div class="layout">
        <?php include('includes/sidebar.php'); ?>
        <main class="main-content">
            <!-- Calendar Section -->
            <div class="calendar-container">
                <button class="calendar-btn" type="button" id="calendar-btn">
                    <span class="material-icons">calendar_month</span>
                    <span id="calendar-date"><?= date('m/d/y', strtotime($selectedDate)) ?></span>
                </button>
                <input type="text" id="calendar" class="calendar-input">
            </div>

            <!-- Logs Table Section -->
            <section class="logs-section">
                <div class="logs-table-container">

                    <!-- Fixed header -->
                    <div class="logs-table-header">
                        <table class="logs-table">
                            <thead>
                                <tr>
                                    <th>Logs ID</th>
                                    <th>User ID</th>
                                    <th>Name</th>
                                    <th>Classification</th>
                                    <th>Date</th>
                                    <th>Time In</th>
                                    <th>Time Out</th>
                                    <th>Snapshot</th>
                                </tr>
                            </thead>
                        </table>
                    </div>

                    <!-- Scrollable body -->
                    <div class="logs-table-body">
                        <table class="logs-table">
                            <tbody>
                                <?php
                                $stmt = $conn->prepare("
```

```
                            SELECT logs.*, users.name AS user_name, users.id AS user_id
                            FROM logs
                            LEFT JOIN users ON logs.user_id = users.id
                            WHERE logs.date = ?
                            ORDER BY logs.id DESC
                        ");
                        $stmt->bind_param("s", $selectedDate);
                        $stmt->execute();
                        $result = $stmt->get_result();

                        if ($result && $result->num_rows > 0):
                            while ($row = $result->fetch_assoc()):
                                // Format role
                                if ($row['role'] === 'employee') {
                                    $roleDisplay = 'Employee';
                                } elseif ($row['role'] === 'person_of_interest') {
                                    $roleDisplay = 'Person of Interest';
                                } else {
                                    $roleDisplay = 'Civilian';
                                }

                                // Handle civilian name/id
                                $userIdDisplay = $row['user_id'] ? $row['user_id'] : '-';
                                $nameDisplay   = $row['user_name'] ? htmlspecialchars($row['user_name']) : ($row['role'] === 'civilian' ? 'Civilian' : 'Unknown');
                        ?>
                        <tr>
                            <td><?= $row['id'] ?></td>
                            <td><?= $userIdDisplay ?></td>
                            <td><?= $nameDisplay ?></td>
                            <td><?= htmlspecialchars($roleDisplay) ?></td>
                            <td><?= htmlspecialchars($row['date']) ?></td>
                            <td><?= date("g:i A", strtotime($row['time_in'])) ?></td>
                            <td><?= $row['time_out'] ? date("g:i A", strtotime($row['time_out'])) : '-' ?></td>
                            <td>
                                <?php if (!empty($row['image_capture'])): ?>
                                    <img src="<?= htmlspecialchars($row['image_capture']) ?>"
                                         alt="Snapshot"
                                         class="thumbnail"
                                         onclick="openModal(this.src)">
                                <?php else: ?>
                                    N/A
                                <?php endif; ?>
                            </td>
                        </tr>
                        <?php endwhile; else: ?>
                        <tr><td colspan="8">No logs available.</td></tr>
                        <?php endif; ?>
                    </tbody>
                </table>
            </div>

        </div>
    </section>
</main>
</div>

<!-- Image Modal -->
<div id="imgModal" class="img-modal">
    <span class="close-modal" onclick="closeModal()">&times;</span>
    <img class="img-modal-content" id="modalImage">
</div>

<script src="js/script.js"></script>
</body>
</html>
```

reports.php

```php
<?php
session_start();
require 'includes/conn.php';
require 'includes/code.php'; // defines: const INVENTORY_PIN = '1234';

if (!defined('INVENTORY_PIN')) { die('PIN not configured: INVENTORY_PIN is not defined. Check includes/code.php.'); }
if (!isset($_SESSION['username'])) { header("Location: login.php"); exit(); }

date_default_timezone_set('Asia/Manila');

/* ---------------- Period helpers ---------------- */
function isoWeekBounds(string $anchor): array {
  // Monday = start of week, Sunday = end (ISO-8601)
  $dt = new DateTime($anchor);
  $dt->modify('Monday this week');
  $start = $dt->format('Y-m-d');
  $end = (clone $dt)->modify('Sunday this week')->format('Y-m-d');
  return [$start, $end];
}

function monthBounds(string $anchor): array {
  $dt = new DateTime($anchor);
  $start = $dt->modify('first day of this month')->format('Y-m-d');
  $end   = (new DateTime($start))->modify('last day of this month')->format('Y-m-d');
  return [$start, $end];
}

/* ---------------- PIN gate (same as inventory) ---------------- */
$PIN_SESSION_FLAG = 'reports_pin_ok';
$PIN_ATTEMPTS     = 'reports_pin_attempts';
$PIN_LOCK_UNTIL   = 'reports_pin_lock_until';
$MAX_ATTEMPTS = 5; $LOCK_SECONDS = 5 * 60;
$_SESSION[$PIN_ATTEMPTS] = $_SESSION[$PIN_ATTEMPTS] ?? 0;

if (!empty($_SESSION[$PIN_LOCK_UNTIL]) && time() < $_SESSION[$PIN_LOCK_UNTIL]) {
  $remain = $_SESSION[$PIN_LOCK_UNTIL] - time();
  $lock_msg = "Too many incorrect attempts. Try again in " . floor($remain/60) . "m " . ($remain%60) . "s.";
}

if (($_SERVER['REQUEST_METHOD'] === 'POST') && isset($_POST['pin_submit']) && !isset($lock_msg)) {
  if (!isset($_POST['csrf']) || !hash_equals($_SESSION['csrf'] ?? '', $_POST['csrf'])) {
    $pin_error = "Invalid request. Please try again.";
  } else {
    $pin = $_POST['pin'] ?? '';
    if (!preg_match('/^\d{4}$/', $pin)) {
      $pin_error = "PIN must be 4 digits.";
    } elseif (hash_equals(INVENTORY_PIN, $pin)) {
      session_regenerate_id(true);
      $_SESSION[$PIN_SESSION_FLAG] = true;
      $_SESSION[$PIN_ATTEMPTS] = 0;
      unset($_SESSION[$PIN_LOCK_UNTIL]);
      header("Location: reports.php"); exit();
    } else {
      $_SESSION[$PIN_ATTEMPTS]++;
      if ($_SESSION[$PIN_ATTEMPTS] >= $MAX_ATTEMPTS) {
        $_SESSION[$PIN_LOCK_UNTIL] = time() + $LOCK_SECONDS;
        $lock_msg = "Too many incorrect attempts. Try again in 5 minutes.";
      } else {
        $left = $MAX_ATTEMPTS - $_SESSION[$PIN_ATTEMPTS];
        $pin_error = "Incorrect PIN. {$left} attempt(s) left.";
      }
    }
  }
}
```

```php
/* ---------------- Params ---------------- */
function h($v){ return htmlspecialchars((string)$v, ENT_QUOTES, 'UTF-8'); }

$today  = date('Y-m-d');
$range  = strtolower(preg_replace('/[^a-z]/','', $_GET['range'] ?? 'daily')); // daily|weekly|monthly
$anchor = preg_replace('/[^0-9\-]/','', $_GET['anchor'] ?? $today);

// Optional manual overrides still supported:
$startQ = array_key_exists('start', $_GET) ? preg_replace('/[^0-9\-]/','', $_GET['start']) : null;
$endQ   = array_key_exists('end',   $_GET) ? preg_replace('/[^0-9\-]/','', $_GET['end'])   : null;

if (!empty($startQ) && !empty($endQ)) {
  // Honor explicit range if both provided
  $start = $startQ;
  $end   = $endQ;
} else {
  // Compute from range + anchor
  if ($range === 'weekly') {
    [$start, $end] = isoWeekBounds($anchor);
  } elseif ($range === 'monthly') {
    [$start, $end] = monthBounds($anchor);
  } else {
    // daily
    $start = $anchor;
    $end   = $anchor;
  }
}

$endPlus1 = (new DateTime($end))->modify('+1 day')->format('Y-m-d');

/* ---------------- KPIs ---------------- */
$kpiEmployees = 0; $kpiPoi = 0; $kpiSessions = 0; $kpiCivilians = 0;

/* unique employees */
if ($stmt = $conn->prepare("
  SELECT COUNT(DISTINCT user_id) AS c
  FROM logs
  WHERE role='employee' AND date >= ? AND date < ? AND user_id IS NOT NULL
")) {
  $stmt->bind_param('ss',$start,$endPlus1); $stmt->execute();
  $r = $stmt->get_result()->fetch_assoc(); $kpiEmployees = (int)($r['c'] ?? 0); $stmt->close();
}

/* total POI events */
if ($stmt = $conn->prepare("
  SELECT COUNT(*) AS c
  FROM logs
  WHERE role='person_of_interest' AND date >= ? AND date < ?
")) {
  $stmt->bind_param('ss',$start,$endPlus1); $stmt->execute();
  $r = $stmt->get_result()->fetch_assoc(); $kpiPoi = (int)($r['c'] ?? 0); $stmt->close();
}

/* total sessions (emp + poi) */
if ($stmt = $conn->prepare("
  SELECT COUNT(*) AS c
  FROM logs
  WHERE role IN ('employee','person_of_interest') AND date >= ? AND date < ?
")) {
  $stmt->bind_param('ss',$start,$endPlus1); $stmt->execute();
  $r = $stmt->get_result()->fetch_assoc(); $kpiSessions = (int)($r['c'] ?? 0); $stmt->close();
}

/* civilians from visitors.json (no timeouts) */
$visitorsFile = __DIR__ . '/snapshots/visitors.json';
if (is_file($visitorsFile)) {
  $data = json_decode(@file_get_contents($visitorsFile), true);
  if (is_array($data)) {
    foreach ($data as $meta) {
      $d = substr((string)($meta['created_at'] ?? ''), 0, 10);
      if ($d && $d >= $start && $d <= $end) { $kpiCivilians++; }
    }
  }
}
```

```php
/* ---------------- Tables (detections) ---------------- */
$employeesAgg = [];
if ($stmt = $conn->prepare("
  SELECT l.date, l.user_id, COALESCE(u.name,'') AS name, COUNT(*) AS detections
  FROM logs l
  LEFT JOIN users u ON u.id = l.user_id
  WHERE l.role='employee' AND l.date >= ? AND l.date < ? AND l.user_id IS NOT NULL
  GROUP BY l.date, l.user_id
  ORDER BY l.date DESC, detections DESC
")) {
  $stmt->bind_param('ss',$start,$endPlus1); $stmt->execute();
  $employeesAgg = $stmt->get_result()->fetch_all(MYSQLI_ASSOC); $stmt->close();
}

$poiAgg = [];
if ($stmt = $conn->prepare("
  SELECT l.date, l.user_id, COALESCE(u.name,'') AS name, COUNT(*) AS detections
  FROM logs l
  LEFT JOIN users u ON u.id = l.user_id
  WHERE l.role='person_of_interest' AND l.date >= ? AND l.date < ?
  GROUP BY l.date, l.user_id
  ORDER BY l.date DESC, detections DESC
")) {
  $stmt->bind_param('ss',$start,$endPlus1); $stmt->execute();
  $poiAgg = $stmt->get_result()->fetch_all(MYSQLI_ASSOC); $stmt->close();
}

/* ---------------- Extra tables (attendance-style) ---------------- */
/* Recent POI encounters (keep) */
$poiEncounters = [];
if ($stmt = $conn->prepare("
  SELECT l.date, l.time_in, l.image_capture, l.user_id, COALESCE(u.name,'') AS name
  FROM logs l
  LEFT JOIN users u ON u.id = l.user_id
  WHERE l.role='person_of_interest' AND l.date >= ? AND l.date < ?
  ORDER BY l.date DESC, l.time_in DESC
  LIMIT 25
")) {
  $stmt->bind_param('ss',$start,$endPlus1); $stmt->execute();
  $poiEncounters = $stmt->get_result()->fetch_all(MYSQLI_ASSOC); $stmt->close();
}

/* NEW: Daily timeouts per person (attendance) */
$dailyTimeouts = [];
if ($stmt = $conn->prepare("
  SELECT l.date,
         l.role,
         l.user_id,
         COALESCE(u.name,'') AS name,
         COUNT(*) AS timeouts
  FROM logs l
  LEFT JOIN users u ON u.id = l.user_id
  WHERE l.time_out IS NOT NULL
    AND l.role IN ('employee','person_of_interest')
    AND l.date >= ? AND l.date < ?
  GROUP BY l.date, l.role, l.user_id
  ORDER BY l.date DESC, timeouts DESC, l.role ASC
")) {
  $stmt->bind_param('ss',$start,$endPlus1); $stmt->execute();
  $dailyTimeouts = $stmt->get_result()->fetch_all(MYSQLI_ASSOC); $stmt->close();
}

?>
```

```php
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1, viewport-fit=cover">
  <title>Reports</title>
  <link href="https://fonts.googleapis.com/icon?family=Material+Icons" rel="stylesheet">
  <link rel="stylesheet" href="css/style.css">
  <link rel="stylesheet" href="css/reports.css">
</head>
<body class="<?php echo empty($_SESSION[$PIN_SESSION_FLAG]) ? 'locked' : ''; ?>">
<?php include('includes/header.php'); ?>
<div class="layout">
  <?php include('includes/sidebar.php'); ?>

  <main class="main-content">
    <?php if (empty($_SESSION[$PIN_SESSION_FLAG])): ?>
      <?php $_SESSION['csrf'] = bin2hex(random_bytes(16)); ?>
      <div class="pin-card">
        <h1>Enter 4-digit Access PIN</h1>
        <p class="pin-sub">This section is restricted. Please confirm your PIN to continue.</p>
        <form method="POST" autocomplete="off">
          <input type="hidden" name="csrf" value="<?php echo htmlspecialchars($_SESSION['csrf']); ?>">
          <input type="text" style="position:absolute; left:-9999px; opacity:0;" autocomplete="username">
          <input type="password" style="position:absolute; left:-9999px; opacity:0;" autocomplete="current-password">
          <div class="pin-field">
            <input class="pin-input" type="password" name="pin" inputmode="numeric" pattern="\d{4}" maxlength="4"
                   placeholder="••••" autocomplete="new-password"
                   oninput="this.value=this.value.replace(/[^0-9]/g,'').slice(0,4)" required autofocus />
          </div>
          <div class="pin-actions">
            <button class="btn-primary" type="submit" name="pin_submit">
              <span class="material-icons" style="vertical-align:-6px;font-size:20px;">lock_open</span>
              Unlock
            </button>
          </div>
          <?php if (!empty($lock_msg)): ?>
            <div class="pin-msg"><?php echo htmlspecialchars($lock_msg); ?></div>
          <?php elseif (!empty($pin_error)): ?>
            <div class="pin-err"><?php echo htmlspecialchars($pin_error); ?></div>
          <?php endif; ?>
        </form>
      </div>
    <?php else: ?>

    <!-- ------------------- Period selector (PRO) ------------------- -->
    <form method="get" class="periodbar">
      <div class="period-shell">
        <!-- Segmented control -->
        <fieldset class="segmented" role="tablist" aria-label="Report Range">
          <input type="radio" id="seg-daily"   name="range" value="daily"   <?php echo ($range==='daily'  ?'checked':''); ?> onchange="this.form.submit()">
          <label for="seg-daily"   role="tab" aria-selected="<?php echo ($range==='daily'?'true':'false'); ?>">Daily</label>

          <input type="radio" id="seg-weekly"  name="range" value="weekly"  <?php echo ($range==='weekly' ?'checked':''); ?> onchange="this.form.submit()">
          <label for="seg-weekly"  role="tab" aria-selected="<?php echo ($range==='weekly'?'true':'false'); ?>">Weekly</label>

          <input type="radio" id="seg-monthly" name="range" value="monthly" <?php echo ($range==='monthly'?'checked':''); ?> onchange="this.form.submit()">
          <label for="seg-monthly" role="tab" aria-selected="<?php echo ($range==='monthly'?'true':'false'); ?>">Monthly</label>
        </fieldset>
      </div>
    </form>
```

```
<!-- ------------------- Normal Reports Content ------------------- -->

<div class="export-bar">
  <a class="btn" href="export_pdf.php?start=<?php echo h($start); ?>&end=<?php echo h($end); ?>&range=<?php echo h($range); ?>&anchor=<?php echo h($anchor); ?>">
    <span class="material-icons mi">picture_as_pdf</span>Export PDF
  </a>
  <form method="post" action="pin_lock.php" class="inline-form">
    <input type="hidden" name="page" value="reports">
    <button type="submit" class="btn btn-lock">
      <span class="material-icons mi">lock</span>Lock
    </button>
  </form>
</div>

<!-- Row 1: 4 KPI cards -->
<section class="kpis">
  <div class="card kpi"><h3>Employees</h3><p><?php echo $kpiEmployees; ?></p></div>
  <div class="card kpi"><h3>POI</h3><p><?php echo $kpiPoi; ?></p></div>
  <div class="card kpi"><h3>Civilians</h3><p><?php echo $kpiCivilians; ?></p></div>
  <div class="card kpi"><h3>Total Sessions</h3><p><?php echo $kpiSessions; ?></p></div>
</section>

<!-- Row 2: 2 wide tables -->
<div class="grid grid-2 tables-row">
  <section class="card">
    <h3>Employees detections per day</h3>
    <div class="table-responsive">
      <table class="table">
        <thead><tr><th>Date</th><th>ID</th><th>Name</th><th>Detections</th></tr></thead>
        <tbody>
          <?php if (!$employeesAgg): ?>
            <tr><td colspan="4" class="text-center">No records</td></tr>
          <?php else: foreach ($employeesAgg as $r): ?>
            <tr>
              <td><?php echo h($r['date']); ?></td>
              <td><?php echo (int)$r['user_id']; ?></td>
              <td><?php echo h($r['name']); ?></td>
              <td><?php echo (int)$r['detections']; ?></td>
            </tr>
          <?php endforeach; endif; ?>
        </tbody>
      </table>
    </div>
  </section>

  <section class="card">
    <h3>POI detections per day</h3>
    <div class="table-responsive">
      <table class="table">
        <thead><tr><th>Date</th><th>ID</th><th>Name</th><th>Detections</th></tr></thead>
        <tbody>
          <?php if (!$poiAgg): ?>
            <tr><td colspan="4" class="text-center">No records</td></tr>
          <?php else: foreach ($poiAgg as $r): ?>
            <tr>
              <td><?php echo h($r['date']); ?></td>
              <td><?php echo h($r['user_id']); ?></td>
              <td><?php echo h($r['name']); ?></td>
              <td><?php echo (int)$r['detections']; ?></td>
            </tr>
          <?php endforeach; endif; ?>
        </tbody>
      </table>
    </div>
  </section>
</div>
```

```
  <!-- Row 3: 2 wide tables -->
  <div class="grid grid-2 tables-row">
    <section class="card">
      <h3>Daily Timeouts (attendance: Emp + POI)</h3>
      <div class="table-responsive">
        <table class="table">
          <thead>
            <tr><th>Date</th><th>Role</th><th>ID</th><th>Name</th><th>Timeouts</th></tr>
          </thead>
          <tbody>
            <?php if (!$dailyTimeouts): ?>
              <tr><td colspan="5" class="text-center">No timeouts in this range</td></tr>
            <?php else: foreach ($dailyTimeouts as $t): ?>
              <tr>
                <td><?php echo h($t['date']); ?></td>
                <td><?php echo h($t['role']); ?></td>
                <td><?php echo (int)$t['user_id']; ?></td>
                <td><?php echo h($t['name']); ?></td>
                <td style="text-align:right; font-variant-numeric:tabular-nums;"><?php echo (int)$t['timeouts']; ?></td>
              </tr>
            <?php endforeach; endif; ?>
          </tbody>
        </table>
      </div>
    </section>

    <section class="card">
      <h3>Recent POI Encounters</h3>
      <div class="table-responsive">
        <table class="table">
          <thead><tr><th>Date</th><th>Time</th><th>ID</th><th>Name</th><th>Snapshot</th></tr></thead>
          <tbody>
            <?php if (!$poiEncounters): ?>
              <tr><td colspan="5" class="text-center">No encounters</td></tr>
            <?php else: foreach ($poiEncounters as $e): ?>
              <tr>
                <td><?php echo h($e['date']); ?></td>
                <td><?php echo h($e['time_in']); ?></td>
                <td><?php echo h($e['user_id']); ?></td>
                <td><?php echo h($e['name']); ?></td>
                <td>
                  <?php if (!empty($e['image_capture'])): ?>
                    <img src="<?php echo h($e['image_capture']); ?>" alt="snap">
                  <?php endif; ?>
                </td>
              </tr>
            <?php endforeach; endif; ?>
          </tbody>
        </table>
      </div>
    </section>
  </div>

  <!-- blobs for optional PDF export -->
  <script type="application/json" id="employees-data"><?php echo json_encode($employeesAgg, JSON_UNESCAPED_SLASHES); ?></script>
  <script type="application/json" id="poi-data"><?php echo json_encode($poiAgg, JSON_UNESCAPED_SLASHES); ?></script>
  <script type="application/json" id="poi-encounters-data"><?php echo json_encode($poiEncounters, JSON_UNESCAPED_SLASHES); ?></script>
  <script type="application/json" id="timeouts-daily-data"><?php echo json_encode($dailyTimeouts, JSON_UNESCAPED_SLASHES); ?></script>

  <?php endif; ?>
 </main>
</div>

<script src="js/script.js"></script>
</body>
</html>
```

notifications.php

```
// robust sort by created_at desc, prefer DB items on tie
usort($merged, function ($a, $b) {
    if ($a['created_at'] === $b['created_at']) {
        return strcmp($a['_source'], $b['_source']); // 'db' before 'json'
    }
    return strcmp($b['created_at'], $a['created_at']);
});

$merged = array_slice($merged, 0, $LIMIT);

// ---------- Unread badge: DB-only ----------
$unreadCount = 0;
$res2 = $conn->query("SELECT COUNT(*) AS c FROM notifications WHERE is_read = 0");
if ($res2) {
    $row = $res2->fetch_assoc();
    $unreadCount = (int)($row['c'] ?? 0);
}

echo json_encode([
    'unreadCount'   => $unreadCount,
    'notifications' => $merged
]);
```

app.py

```python
# app.py — GPU Face Recognition + Blink Liveness (MediaPipe) + Sounds + Sticky Verification
#          + Face-only snapshots + Anti-hijack + Crash-safe GPU wrapper

import os, cv2, numpy as np, threading, requests, json, time, platform, traceback
from flask import Flask, Response
from collections import deque
from datetime import datetime, date

# ---------- Third-party inference ----------
import onnxruntime as ort
from insightface.app import FaceAnalysis
import mediapipe as mp

# ---------- Watch files for DB reload ----------
from watchdog.observers import Observer
from watchdog.events import FileSystemEventHandler

# ---------- WAV sounds (Windows only) ----------
IS_WINDOWS = platform.system().lower().startswith("win")
if IS_WINDOWS:
    import winsound  # built-in

# ---------- Paths / Dirs ----------
BASE_DIR     = os.path.dirname(os.path.abspath(__file__))
SOUNDS_DIR   = os.path.join(BASE_DIR, "sounds")
SNAPSHOT_DIR = os.path.join(BASE_DIR, "snapshots")
os.makedirs(SNAPSHOT_DIR, exist_ok=True)

EMPLOYEE_WAV = os.path.join(SOUNDS_DIR, "employee.wav")
POI_WAV      = os.path.join(SOUNDS_DIR, "poi.wav")

# Silence Albumentations banner
os.environ.setdefault("NO_ALBUMENTATIONS_UPDATE", "1")

# ---------- Sounds ----------
SOUND_COOLDOWN = 5.0
_last_sound = {"employee": 0.0, "person_of_interest": 0.0}

def _play_wav_async(path: str):
    if not IS_WINDOWS:
        return
    if os.path.isfile(path):
        winsound.PlaySound(path, winsound.SND_FILENAME | winsound.SND_ASYNC | winsound.SND_NODEFAULT)
    else:
        winsound.Beep(820, 120)

def maybe_play_role_sound(role: str):
    now = time.time()
    if role not in _last_sound:
        return
    if now - _last_sound[role] < SOUND_COOLDOWN:
        return
    _last_sound[role] = now
    if role == "employee":
        _play_wav_async(EMPLOYEE_WAV)
    elif role == "person_of_interest":
        _play_wav_async(POI_WAV)

# ---------- Anti-spoof / Liveness ----------
LIVENESS_WINDOW_SEC     = 6.0
VERIFY_REQUIRED_BLINKS  = 2
SUSPECT_TIMEOUT_SEC     = 4.0
MIN_FACE_SIZE_PX        = 120
EAR_BLINK_THRESH        = 0.23
EAR_RELEASE             = 0.25
LOST_RESET_SEC          = 2.5

STATE_CANDIDATE = "candidate"
STATE_VERIFIED  = "verified"
STATE_SUSPECT   = "suspect"

# ---------- Tracking / Matching ----------
IOU_MIN = 0.15
CENTER_DIST_MAX = 70.0
CONSISTENCY_MIN = 0.65
COSINE_THRESHOLD = 0.25
```

```python
# ---------- Session / Logging ----------
active_sessions = {}   # user_id -> {log_id, last_seen, absent_since, date}
last_timeout    = {}   # user_id -> datetime marking start of cooldown
SESSION_GRACE   = 30   # seconds unseen before ending session
RELOG_DELAY     = 10   # << COOL-DOWN FIX: 10s wait AFTER timeout before re-login allowed
LOG_URL         = "http://localhost/cms/log_event.php"

def async_request(payload):
    def worker():
        try:
            requests.post(LOG_URL, json=payload, timeout=0.6)
        except Exception as e:
            print("[WARN] async_request failed:", e)
    threading.Thread(target=worker, daemon=True).start()

def start_log(user_id, role, snapshot):
    payload = {"action": "start", "user_id": user_id, "role": role, "snapshot": snapshot}
    try:
        r = requests.post(LOG_URL, json=payload, timeout=2)
        if r.status_code == 200:
            text = r.text.strip()
            if text.isdigit():
                print(f"[LOG START] user {user_id}, log {text}")
                return int(text)
    except Exception as e:
        print("[WARN] start_log error:", e)
    return None

def end_log(log_id, user_id):
    async_request({"action": "end", "log_id": log_id, "user_id": user_id})
    print(f"[LOG END] user {user_id}, log {log_id}")

def seconds_since(past, now):  # past, now are datetimes
    return (now - past).total_seconds()

# ---------- Flask / Camera ----------
app = Flask(__name__)

camera = cv2.VideoCapture(0, cv2.CAP_DSHOW)
camera.set(cv2.CAP_PROP_FRAME_WIDTH,  640)
camera.set(cv2.CAP_PROP_FRAME_HEIGHT, 480)
camera.set(cv2.CAP_PROP_FPS, 24)  # lighter than 30
camera.set(cv2.CAP_PROP_FOURCC, cv2.VideoWriter_fourcc(*'MJPG'))
camera.set(cv2.CAP_PROP_BUFFERSIZE, 1)
cv2.setNumThreads(1)

# ---------- Colors ----------
ROLE_COLORS = {
    "employee": (0, 255, 0),
    "person_of_interest": (0, 0, 255),
    "civilian": (255, 0, 0),
    "suspect": (0, 255, 255),
}

# ---------- Face Analysis (GPU guarded) ----------
FACE_LOCK  = threading.Lock()
GPU_ACTIVE = True

def build_face_app(use_gpu: bool):
    providers = ["CUDAExecutionProvider", "CPUExecutionProvider"] if use_gpu else ["CPUExecutionProvider"]
    fa = FaceAnalysis(name="buffalo_l", providers=providers)
    fa.prepare(ctx_id=0, det_size=(384, 384))  # slightly lighter than 512
    return fa

face_app = build_face_app(use_gpu=True)
print("[INFO] ORT available providers:", ort.get_available_providers())

def safe_face_get(img_rgb):
    """
    Call face_app.get() under a lock; if CUDA/ONNX crashes, rebuild on CPU and continue.
    """
    global face_app, GPU_ACTIVE
    with FACE_LOCK:
        try:
            return face_app.get(img_rgb)
```

```
def safe_face_get(img_rgb):
        except Exception:
            print("[ERROR] face_app.get() failed. Falling back to CPU...")
            traceback.print_exc()
            GPU_ACTIVE = False
            try:
                face_app = build_face_app(use_gpu=False)
                print("[INFO] Rebuilt FaceAnalysis on CPU. Continuing stream.")
                return face_app.get(img_rgb)
            except Exception as e2:
                print("[FATAL] CPU fallback failed:", e2)
                return []

# ---------- MediaPipe FaceMesh ----------
mp_face_mesh = mp.solutions.face_mesh.FaceMesh(
    static_image_mode=False,
    refine_landmarks=True,
    max_num_faces=5,
    min_detection_confidence=0.5,
    min_tracking_confidence=0.5
)
LEFT_EYE  = (33, 160, 158, 133, 153, 144)
RIGHT_EYE = (362, 385, 387, 263, 373, 380)

# ---------- Embedding DB (CPU-only to avoid GPU contention) ----------
DB_PATH = os.path.join(BASE_DIR, "faces")
faces_db, roles_db = {}, {}

def l2_normalize(v: np.ndarray) -> np.ndarray:
    n = np.linalg.norm(v)
    return v / n if n > 0 else v

def load_db():
    global faces_db, roles_db
    faces_db, roles_db = {}, {}

    if not os.path.isdir(DB_PATH):
        print(f"[WARN] DB_PATH '{DB_PATH}' does not exist.")
        return

    try:
        cpu_app = FaceAnalysis(name="buffalo_l", providers=["CPUExecutionProvider"])
        cpu_app.prepare(ctx_id=0, det_size=(384, 384))
    except Exception as e:
        print("[WARN] Could not init CPU FaceAnalysis for DB load:", e)
        return

    loaded = 0
    for role in os.listdir(DB_PATH):
        role_dir = os.path.join(DB_PATH, role)
        if not os.path.isdir(role_dir):
            continue
        for img_name in os.listdir(role_dir):
            img_path = os.path.join(role_dir, img_name)
            img = cv2.imread(img_path)
            if img is None:
                continue
            faces = cpu_app.get(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
            if not faces:
                continue
            emb = l2_normalize(faces[0].normed_embedding)
            faces_db[img_name] = emb
            roles_db[img_name] = role
            loaded += 1
    print(f"[INFO] Loaded {loaded} embeddings.")

load_db()

class DBHandler(FileSystemEventHandler):
    def on_any_event(self, e):
        if not e.is_directory:
            print("[INFO] Faces DB changed, reloading...")
            load_db()

observer = Observer()
```

```
observer.schedule(DBHandler(), DB_PATH, recursive=True)
observer.start()

# ---------- Tracking structures ----------
tracked_faces = {}   # fid -> track dict
next_face_id = 0

def _iou(b1, b2):
    x1, y1, x2, y2 = b1; X1, Y1, X2, Y2 = b2
    ix1, iy1 = max(x1, X1), max(y1, Y1)
    ix2, iy2 = min(x2, X2), min(y2, Y2)
    iw, ih = max(0, ix2 - ix1), max(0, iy2 - iy1)
    inter = iw * ih
    if inter <= 0: return 0.0
    a1 = max(0, x2 - x1) * max(0, y2 - y1)
    a2 = max(0, X2 - X1) * max(0, Y2 - Y1)
    return inter / float(a1 + a2 - inter + 1e-6)

def _center(b):
    x1, y1, x2, y2 = b
    return ((x1 + x2) * 0.5, (y1 + y2) * 0.5)

def _cos(a, b):
    if a is None or b is None: return 0.0
    na = np.linalg.norm(a); nb = np.linalg.norm(b)
    if na == 0 or nb == 0: return 0.0
    return float(np.dot(a/na, b/nb))

def get_centroid(b):
    x1, y1, x2, y2 = b
    return ((x1 + x2) // 2, (y1 + y2) // 2)

def ensure_track(fid, centroid, bbox=None):
    if fid not in tracked_faces:
        tracked_faces[fid] = {
            'centroid': centroid,
            'bbox': bbox if bbox is not None else (0,0,1,1),
            'embeddings': deque(maxlen=5),
            'role': 'civilian',
            'blink_times': deque(),
            'blink_state': False,
            'state': STATE_CANDIDATE,
            'state_since': time.time(),
            'candidate_started': time.time(),
            'blink_count_window': 0,
            'last_ear': 0.0,
            'lost_since': None,
            'verified_embedding': None
        }

def match_tracked(bbox):
    best_fid, best_score = None, -1.0
    bx, by = _center(bbox)
    for fid, d in tracked_faces.items():
        tb = d.get('bbox', None)
        if tb is None:
            continue
        iou = _iou(bbox, tb)
        cx, cy = _center(tb)
        dist = ((bx - cx)**2 + (by - cy)**2) ** 0.5
        if iou >= IOU_MIN and dist <= CENTER_DIST_MAX:
            score = iou - 0.0005 * dist
            if score > best_score:
                best_score, best_fid = score, fid
    return best_fid

def recognize_face_smooth(emb, bbox):
    global next_face_id
    fid = match_tracked(bbox)
    if fid is None:
        fid = next_face_id
        next_face_id += 1
        ensure_track(fid, get_centroid(bbox), bbox=bbox)
    else:
        ensure_track(fid, get_centroid(bbox), bbox=bbox)
```

```
    tr = tracked_faces[fid]
    tr['centroid'] = get_centroid(bbox)
    tr['bbox'] = bbox
    tr['embeddings'].append(l2_normalize(emb))

    avg = l2_normalize(np.mean(tr['embeddings'], axis=0))
    best, score = None, -1.0
    for name, db_emb in faces_db.items():
        s = float(np.dot(avg, db_emb))
        if s > score:
            score, best = s, name

    if score >= COSINE_THRESHOLD:
        role = roles_db.get(best, 'civilian')
        tr['role'] = role
        return fid, best, role
    return fid, None, "civilian"

def _dist(a, b): return np.linalg.norm(a - b)

def _ear(pts):
    p1, p2, p3, p4, p5, p6 = pts
    return (_dist(p2, p6) + _dist(p3, p5)) / (2.0 * _dist(p1, p4) + 1e-6)

def update_blink(fid, crop_bgr, eligible_for_verification=True):
    """Always track EAR/blinks; promotion to VERIFIED only if eligible."""
    if crop_bgr.size == 0:
        return

    h, w = crop_bgr.shape[:2]
    res = mp_face_mesh.process(cv2.cvtColor(crop_bgr, cv2.COLOR_BGR2RGB))
    if not res.multi_face_landmarks:
        return

    lms = res.multi_face_landmarks[0].landmark

    def pts(indexes):
        return np.array([[lms[i].x * w, lms[i].y * h] for i in indexes], dtype=np.float32)

    left_ear  = _ear(pts(LEFT_EYE))
    right_ear = _ear(pts(RIGHT_EYE))
    ear = float((left_ear + right_ear) / 2.0)

    tr = tracked_faces.get(fid)
    if tr is None:
        return

    now_ts = time.time()
    blinking = tr['blink_state']

    # hysteresis for blink
    if not blinking and ear < EAR_BLINK_THRESH:
        tr['blink_state'] = True
    elif blinking and ear >= EAR_RELEASE:
        tr['blink_state'] = False
        tr['blink_times'].append(now_ts)
        tr['blink_count_window'] += 1

    # keep blinks within window
    while tr['blink_times'] and now_ts - tr['blink_times'][0] > LIVENESS_WINDOW_SEC:
        tr['blink_times'].popleft()

    tr['last_ear'] = ear

    # sticky: never downgrade here
    if tr['state'] == STATE_VERIFIED:
        return

    if tr['state'] == STATE_SUSPECT:
        if now_ts - tr['state_since'] > LIVENESS_WINDOW_SEC:
            tr['state'] = STATE_CANDIDATE
            tr['state_since'] = now_ts
            tr['candidate_started'] = now_ts
            tr['blink_count_window'] = 0
```

```
def update_blink(fid, crop_bgr, eligible_for_verification=True):
    # keep blinks within window
    while tr['blink_times'] and now_ts - tr['blink_times'][0] > LIVENESS_WINDOW_SEC:
        tr['blink_times'].popleft()

    tr['last_ear'] = ear

    # sticky: never downgrade here
    if tr['state'] == STATE_VERIFIED:
        return

    if tr['state'] == STATE_SUSPECT:
        if now_ts - tr['state_since'] > LIVENESS_WINDOW_SEC:
            tr['state'] = STATE_CANDIDATE
            tr['state_since'] = now_ts
            tr['candidate_started'] = now_ts
            tr['blink_count_window'] = 0
        return

    # candidate -> verified only if allowed
    if tr['blink_count_window'] >= VERIFY_REQUIRED_BLINKS:
        if eligible_for_verification:
            tr['state'] = STATE_VERIFIED
            tr['state_since'] = now_ts
            tr['blink_count_window'] = 0
            tr['verified_embedding'] = l2_normalize(np.mean(tr['embeddings'], axis=0)) if len(tr['embeddings']) else None
        return

    # suspect if no blinks for too long
    if (now_ts - tr['candidate_started'] >= SUSPECT_TIMEOUT_SEC) and (len(tr['blink_times']) == 0):
        tr['state'] = STATE_SUSPECT
        tr['state_since'] = now_ts

def is_verified(fid) -> bool:
    tr = tracked_faces.get(fid)
    return bool(tr and tr['state'] == STATE_VERIFIED)

def is_suspect(fid) -> bool:
    tr = tracked_faces.get(fid)
    return bool(tr and tr['state'] == STATE_SUSPECT)

def face_area(bbox):
    x1, y1, x2, y2 = bbox
    return max(0, x2 - x1) * max(0, y2 - y1)

# ---------- Snapshots ----------
VISITORS_JSON = os.path.join(SNAPSHOT_DIR, "visitors.json")
CIVILIAN_WRITE_COOLDOWN = 8
_last_civilian_write_ts = 0.0

def _expand_clamp_bbox(bbox, img_w, img_h, margin=0.18):
    x1, y1, x2, y2 = bbox
    w = max(1, x2 - x1); h = max(1, y2 - y1)
    cx = x1 + w / 2.0; cy = y1 + h / 2.0
    w2 = w * (1.0 + 2.0 * margin); h2 = h * (1.0 + 2.0 * margin)
    nx1 = int(max(0, round(cx - w2 / 2.0)))
    ny1 = int(max(0, round(cy - h2 / 2.0)))
    nx2 = int(min(img_w, round(cx + w2 / 2.0)))
    ny2 = int(min(img_h, round(cy + h2 / 2.0)))
    if nx2 <= nx1: nx2 = min(img_w, nx1 + 1)
    if ny2 <= ny1: ny2 = min(img_h, ny1 + 1)
    return nx1, ny1, nx2, ny2

def save_face_crop(frame_bgr, bbox, prefix, margin=0.20):
    H, W = frame_bgr.shape[:2]
    ex1, ey1, ex2, ey2 = _expand_clamp_bbox(bbox, W, H, margin=margin)
    face_crop = frame_bgr[ey1:ey2, ex1:ex2]
    fname = f"{datetime.now().strftime('%Y%m%d_%H%M%S')}_{prefix}.jpg"
    abs_path = os.path.join(SNAPSHOT_DIR, fname)
    cv2.imwrite(abs_path, face_crop, [int(cv2.IMWRITE_JPEG_QUALITY), 90])
    return f"snapshots/{fname}"

def notify_civilian(snapshot_relpath: str):
    try:
        requests.post(
```

```
def notify_civilian(snapshot_relpath: str):
            LOG_URL,
            json={"action": "notify_civilian", "role": "civilian", "snapshot": snapshot_relpath},
            timeout=0.7
        )
    except Exception as e:
        print("[WARN] notify_civilian failed:", e)

def write_civilian_event_and_notify(frame_bgr=None, bbox=None, face_crop_bgr=None):
    global _last_civilian_write_ts
    now_ts = time.time()
    if now_ts - _last_civilian_write_ts < CIVILIAN_WRITE_COOLDOWN:
        return
    _last_civilian_write_ts = now_ts

    if face_crop_bgr is None:
        if frame_bgr is None or bbox is None:
            return
        H, W = frame_bgr.shape[:2]
        ex1, ey1, ex2, ey2 = _expand_clamp_bbox(bbox, W, H, margin=0.20)
        face_crop_bgr = frame_bgr[ey1:ey2, ex1:ex2]

    fname = f"civilian_{datetime.now().strftime('%Y%m%d_%H%M%S')}.jpg"
    abs_path = os.path.join(SNAPSHOT_DIR, fname)
    cv2.imwrite(abs_path, face_crop_bgr, [int(cv2.IMWRITE_JPEG_QUALITY), 90])
    rel_path = f"snapshots/{fname}"

    data = {}
    if os.path.isfile(VISITORS_JSON):
        try:
            with open(VISITORS_JSON, "r", encoding="utf-8") as f:
                data = json.load(f)
            if not isinstance(data, dict):
                data = {}
        except Exception:
            data = {}

    civ_id = str(int(now_ts))
    data[civ_id] = {
        "created_at": datetime.now().strftime("%Y-%m-%d %H:%M:%S"),
        "last_seen": datetime.now().strftime("%Y-%m-%d"),
        "snapshot": rel_path
    }

    try:
        with open(VISITORS_JSON, "w", encoding="utf-8") as f:
            json.dump(data, f, ensure_ascii=False, indent=2)
        print(f"[CIVILIAN] wrote {civ_id} to visitors.json")
    except Exception as e:
        print("[WARN] Failed writing visitors.json:", e)

    notify_civilian(rel_path)

# ---------- Frame generator ----------
def gen_frames():
    frame_skip, frame_count = 3, 0  # lighter than 2; raise if stable
    recent_results = {}
    jpeg_params = [int(cv2.IMWRITE_JPEG_QUALITY), 70]

    while True:
        ok, frame = camera.read()
        if not ok:
            break

        try:
            frame_count += 1
            now = datetime.now()
            today = date.today()

            # When skipping compute frames, just redraw last results
            if frame_count % frame_skip != 0:
                for bbox, (name, role) in recent_results.items():
                    x1, y1, x2, y2 = bbox
                    color = ROLE_COLORS.get(role, (255, 255, 255))
                    lbl = "civilian" if role == "civilian" else f"ID {os.path.splitext(name)[0]} - {role}"
```

```
            cv2.rectangle(frame, (x1, y1), (x2, y2), color, 2)
            cv2.putText(frame, lbl, (x1, y1 - 10), cv2.FONT_HERSHEY_SIMPLEX, 0.7, color, 2)

        ret, buf = cv2.imencode('.jpg', frame, jpeg_params)
        if ret:
            yield (b'--frame\r\nContent-Type: image/jpeg\r\n\r\n' + buf.tobytes() + b'\r\n')
        continue

    # Detect on downscaled for speed
    scale = 0.5
    small = cv2.resize(frame, (0, 0), fx=scale, fy=scale)
    rgb = cv2.cvtColor(small, cv2.COLOR_BGR2RGB)

    # Crash-safe call
    faces = safe_face_get(rgb)

    current_results, seen_users = {}, set()
    seen_fids = set()

    # Build detections
    dets = []
    for f in faces:
        x1, y1, x2, y2 = (f.bbox / scale).astype(int)

        # Ignore tiny faces (likely phone screens)
        if face_area((x1, y1, x2, y2)) < (MIN_FACE_SIZE_PX * MIN_FACE_SIZE_PX):
            dets.append({
                'fid': None, 'bbox': (x1, y1, x2, y2), 'area': 0,
                'match_name': None, 'role_guess': 'civilian', 'tr': None, 'skip': 'tiny'
            })
            continue

        fid, match_name, role_guess = recognize_face_smooth(f.normed_embedding, (x1, y1, x2, y2))
        seen_fids.add(fid)
        tr = tracked_faces.get(fid)
        if tr:
            tr['lost_since'] = None  # visible

        dets.append({
            'fid': fid,
            'bbox': (x1, y1, x2, y2),
            'area': face_area((x1, y1, x2, y2)),
            'match_name': match_name,
            'role_guess': role_guess,
            'tr': tr,
            'skip': None
        })

    # Pick primary fid for liveness
    verified_candidates = [d for d in dets if d['fid'] is not None and d['tr'] and d['tr']['state'] == STATE_VERIFIED]
    if verified_candidates:
        primary = max(verified_candidates, key=lambda d: d['area'])
        primary_fid = primary['fid']
    else:
        valid = [d for d in dets if d['fid'] is not None]
        primary_fid = max(valid, key=lambda d: d['area'])['fid'] if valid else None

    # Process / draw
    for d in dets:
        x1, y1, x2, y2 = d['bbox']
        if d['skip'] == 'tiny':
            color = ROLE_COLORS["suspect"]
            cv2.rectangle(frame, (x1, y1), (x2, y2), color, 2)
            cv2.putText(frame, "SUSPECT: too small", (x1, y1 - 10),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.7, color, 2)
            continue

        fid = d['fid']
        tr  = d['tr']
        match_name  = d['match_name']
        role_guess  = d['role_guess']

        # Parse user_id once
        user_id = None
```

```
            try:
                if match_name:
                    user_id = int(os.path.splitext(match_name)[0])
            except Exception:
                user_id = None

            # Sticky verified while session active (prevent UI flicker/downgrade)
            if user_id and tr and active_sessions.get(user_id):
                if tr.get('state') != STATE_VERIFIED:
                    tr['state'] = STATE_VERIFIED
                    tr['state_since'] = time.time()

            # Only primary updates blink
            face_crop = frame[y1:y2, x1:x2]
            eligible_for_verification = True
            cooldown_msg = None
            if fid is not None and fid == primary_fid:
                # Gate promotion to VERIFIED (sessions & cooldown)
                if user_id and role_guess in ["employee", "person_of_interest"]:
                    sess = active_sessions.get(user_id)
                    if sess:
                        eligible_for_verification = False  # already logged
                        cooldown_msg = "ACTIVE SESSION — liveness off"
                    else:
                        if user_id in last_timeout:
                            # remaining seconds until cooldown over
                            elapsed = seconds_since(last_timeout[user_id], now)
                            if elapsed < RELOG_DELAY:
                                eligible_for_verification = False
                                cooldown_msg = f"COOLDOWN: wait {int(RELOG_DELAY - elapsed)}s"
                update_blink(fid, face_crop, eligible_for_verification)

            # Identity consistency check ONLY when no active session
            if fid is not None and is_verified(fid):
                active_for_user = bool(user_id and active_sessions.get(user_id))
                if not active_for_user:
                    tr_now = tracked_faces.get(fid)
                    if tr_now is not None:
                        cur_avg = l2_normalize(np.mean(tr_now['embeddings'], axis=0)) if len(tr_now['embeddings']) else None
                        sim = _cos(cur_avg, tr_now.get('verified_embedding'))
                        if sim < CONSISTENCY_MIN:
                            tr_now['state'] = STATE_CANDIDATE
                            tr_now['state_since'] = time.time()
                            tr_now['candidate_started'] = tr_now['state_since']
                            tr_now['blink_count_window'] = 0

            # Draw / log
            if fid is not None and is_verified(fid):
                color = ROLE_COLORS.get(role_guess, (255, 255, 255))
                label = (f"ID {user_id} - {role_guess}"
                         if (role_guess != "civilian" and user_id)
                         else (role_guess if role_guess != "civilian" else "civilian"))
                cv2.rectangle(frame, (x1, y1), (x2, y2), color, 2)
                cv2.putText(frame, label, (x1, y1 - 10),
                            cv2.FONT_HERSHEY_SIMPLEX, 0.7, color, 2)
                current_results[(x1, y1, x2, y2)] = (match_name, role_guess)

                # Sessions (only when verified)
                if user_id and role_guess in ["employee", "person_of_interest"]:
                    seen_users.add(user_id)
                    sess = active_sessions.get(user_id)
                    if sess and sess.get("absent_since"):
                        if seconds_since(sess["absent_since"], now) >= SESSION_GRACE:
                            end_log(sess["log_id"], user_id)
                            # << COOL-DOWN FIX: cooldown starts AT TIMEOUT (now), not at absent_since
                            last_timeout[user_id] = now
                            del active_sessions[user_id]
                            sess = None
                    if not sess:
                        # only allow new log if cooldown done
                        if (user_id not in last_timeout) or (seconds_since(last_timeout[user_id], now) >= RELOG_DELAY):
                            rel_path = save_face_crop(frame, (x1, y1, x2, y2), role_guess, margin=0.20)
                            new_id = start_log(user_id, role_guess, rel_path)
                            if new_id:
```

```
                        active_sessions[user_id] = {
                            "log_id": new_id, "last_seen": now, "absent_since": None, "date": today
                        }
                        maybe_play_role_sound(role_guess)
                else:
                    sess["last_seen"] = now
                    sess["absent_since"] = None
                    sess["date"] = today

            elif role_guess == "civilian":
                # Log civilian detection (with cooldown)
                write_civilian_event_and_notify(frame_bgr=frame, bbox=(x1, y1, x2, y2))


        elif fid is not None and is_suspect(fid):
            color = ROLE_COLORS["suspect"]
            cv2.rectangle(frame, (x1, y1), (x2, y2), color, 2)
            cv2.putText(frame, "SUSPECT SPOOF", (x1, y1 - 10),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.7, color, 2)

        else:
            # Candidate UI (cap the shown counter; show cooldown reason if any)
            color = (0, 255, 255)
            blink_cnt = tr.get('blink_count_window', 0) if tr else 0
            ear_dbg   = tr.get('last_ear', 0.0) if tr else 0.0
            shown_cnt = min(blink_cnt, VERIFY_REQUIRED_BLINKS)
            cv2.rectangle(frame, (x1, y1), (x2, y2), color, 2)
            if fid is not None and fid != primary_fid:
                msg = "SECONDARY FACE (no liveness)"
            else:
                msg = f"BLINK {VERIFY_REQUIRED_BLINKS}x TO VERIFY  ({shown_cnt}/{VERIFY_REQUIRED_BLINKS})"
                # Extra info if blocked
                if user_id and role_guess in ["employee", "person_of_interest"] and (user_id in last_timeout):
                    elapsed = seconds_since(last_timeout[user_id], now)
                    if elapsed < RELOG_DELAY:
                        msg = f"{msg} — COOLDOWN {int(RELOG_DELAY - elapsed)}s"
            # If we computed a cooldown message earlier for primary fid, append it
            if fid is not None and fid == primary_fid and cooldown_msg:
                msg = f"{msg} | {cooldown_msg}"

            cv2.putText(frame, msg, (x1, max(24, y1 - 10)),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.7, color, 2)
            cv2.putText(frame, f"EAR:{ear_dbg:.2f}", (x1, y2 + 22),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.6, color, 2)

    # mark absent sessions (start absent_since when not seen)
    for user_id, sess in list(active_sessions.items()):
        if sess["date"] != today:
            end_log(sess["log_id"], user_id)
            # << COOL-DOWN FIX: on date rollover, start cooldown now
            last_timeout[user_id] = now
            del active_sessions[user_id]
            continue
        if user_id not in seen_users and sess["absent_since"] is None:
            sess["absent_since"] = now
            print(f"[ABSENT START] user {user_id}")
        # end the session after grace
        if sess["absent_since"] is not None:
            if seconds_since(sess["absent_since"], now) >= SESSION_GRACE:
                end_log(sess["log_id"], user_id)
                # << COOL-DOWN FIX: cooldown starts at timeout moment (now)
                last_timeout[user_id] = now
                del active_sessions[user_id]
                print(f"[SESSION ENDED] user {user_id}")

    # reset tracks not seen for LOST_RESET_SEC
    now_ts = time.time()
    for fid, tr in list(tracked_faces.items()):
        if fid in seen_fids:
            continue
        if tr['lost_since'] is None:
            tr['lost_since'] = now_ts
        elif (now_ts - tr['lost_since']) > LOST_RESET_SEC:
```

```
                    del tracked_faces[fid]

            # update last results for skipped frames display
            recent_results = current_results

            ret, buf = cv2.imencode('.jpg', frame, jpeg_params)
            if ret:
                yield (b'--frame\r\nContent-Type: image/jpeg\r\n\r\n' + buf.tobytes() + b'\r\n')

        except Exception:
            # contain per-frame errors and keep going
            print("[WARN] Exception in gen_frames loop (continuing):")
            traceback.print_exc()
            ret, buf = cv2.imencode('.jpg', frame, [int(cv2.IMWRITE_JPEG_QUALITY), 60])
            if ret:
                yield (b'--frame\r\nContent-Type: image/jpeg\r\n\r\n' + buf.tobytes() + b'\r\n')

# ---------- Flask route ----------
@app.route('/video_feed')
def video_feed():
    return Response(gen_frames(), mimetype='multipart/x-mixed-replace; boundary=frame')

# ---------- Main ----------
if __name__ == "__main__":
    try:
        app.run(debug=False, host="0.0.0.0", threaded=True)
    finally:
        observer.stop()
        observer.join()
```

APPENDIX D.

Evaluation Tool

Cosine Similarity

$$similarity\,(A,B) = \cos(\theta) = \frac{A \cdot B}{\|A\|\,\|B\|}$$

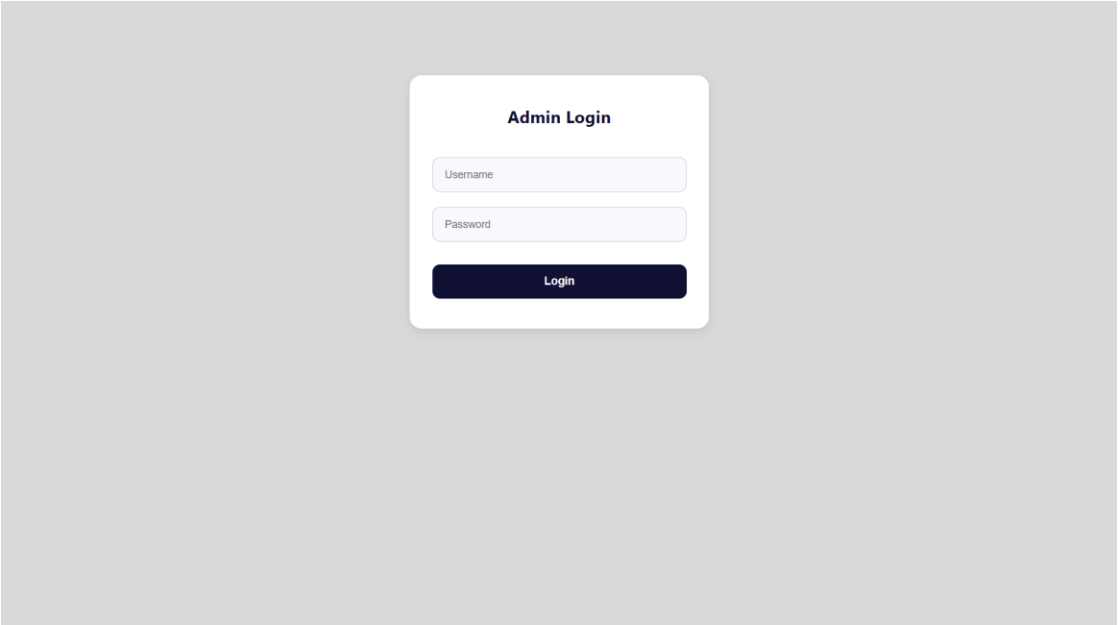
Angular Distance

$$\text{Angular Distance} = \arccos = \frac{A \cdot B}{\|A\|\,\|B\|}$$

APPENDIX E.

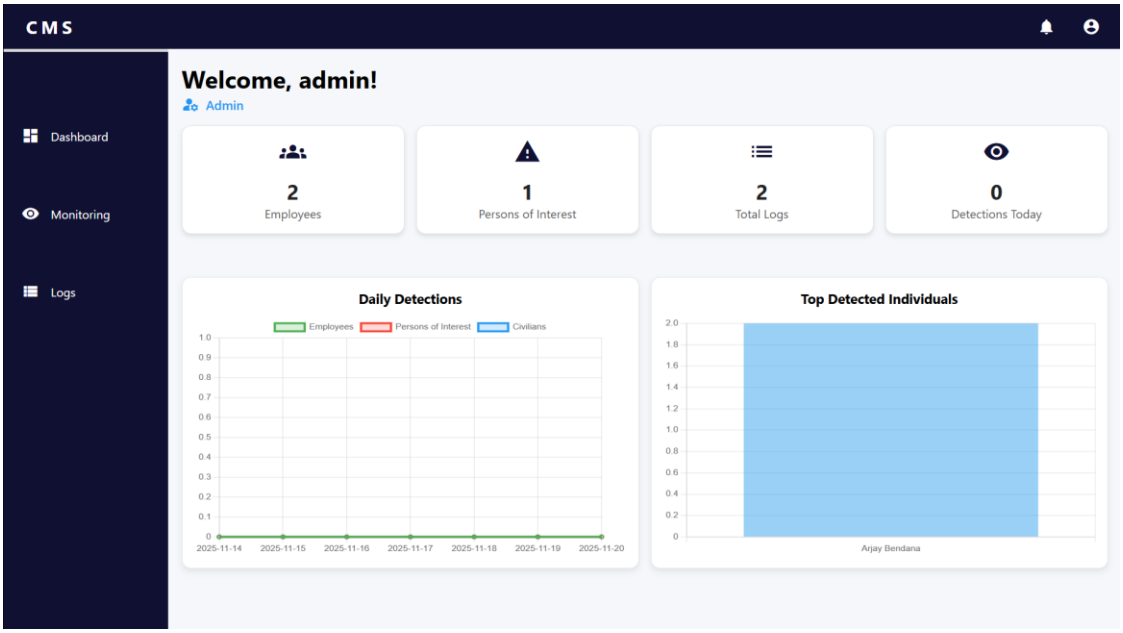
Sample Input / Output

Login



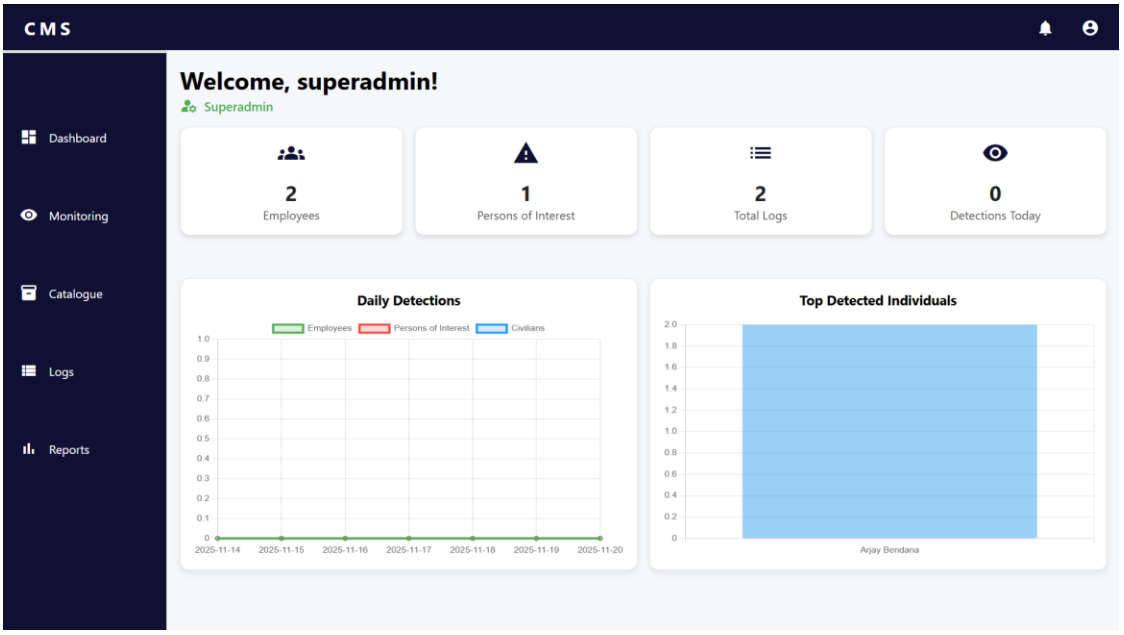
This is where admin/super admin can use their accounts to log in and access the system.

Admin View



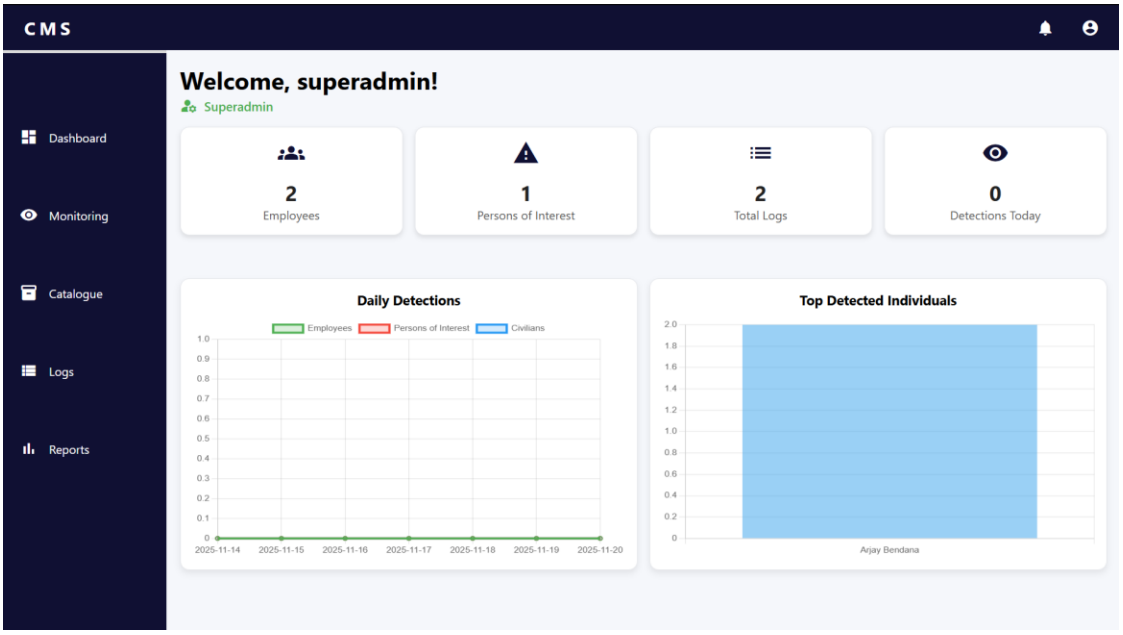
Admin can only access dashboard, monitoring, logs and notifications.

Super Admin View



Super Admin can access all the features in the system, dashboard, monitoring, catalogue, logs, reports and notifications.

Dashboard



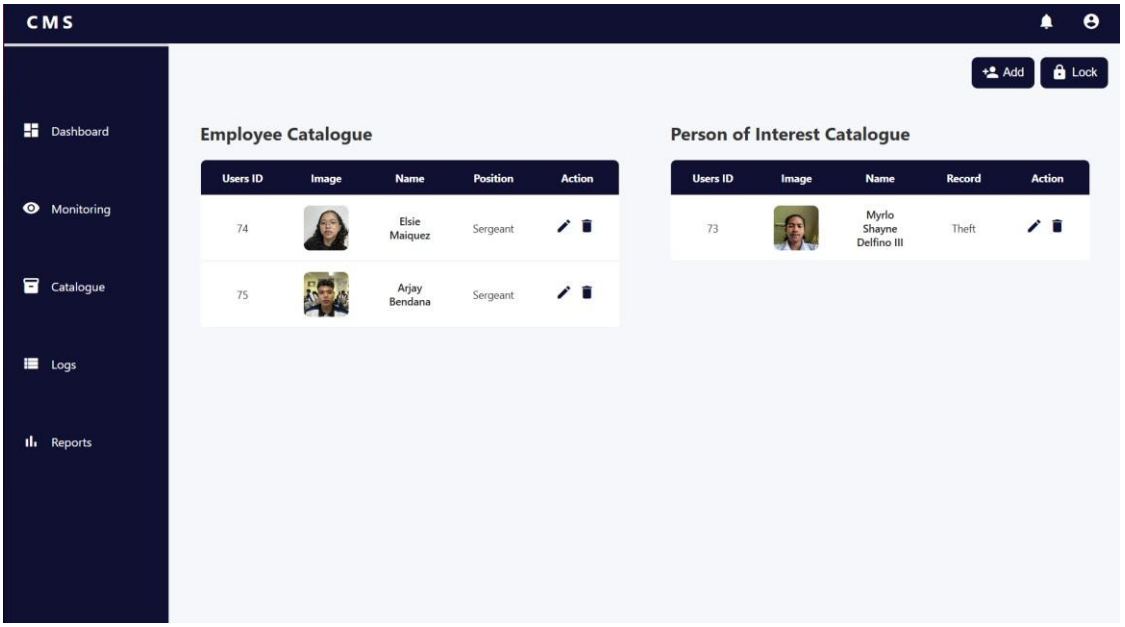
Dashboard contains total employees, total person of interest, total logs, daily detections and graphs.

Monitoring



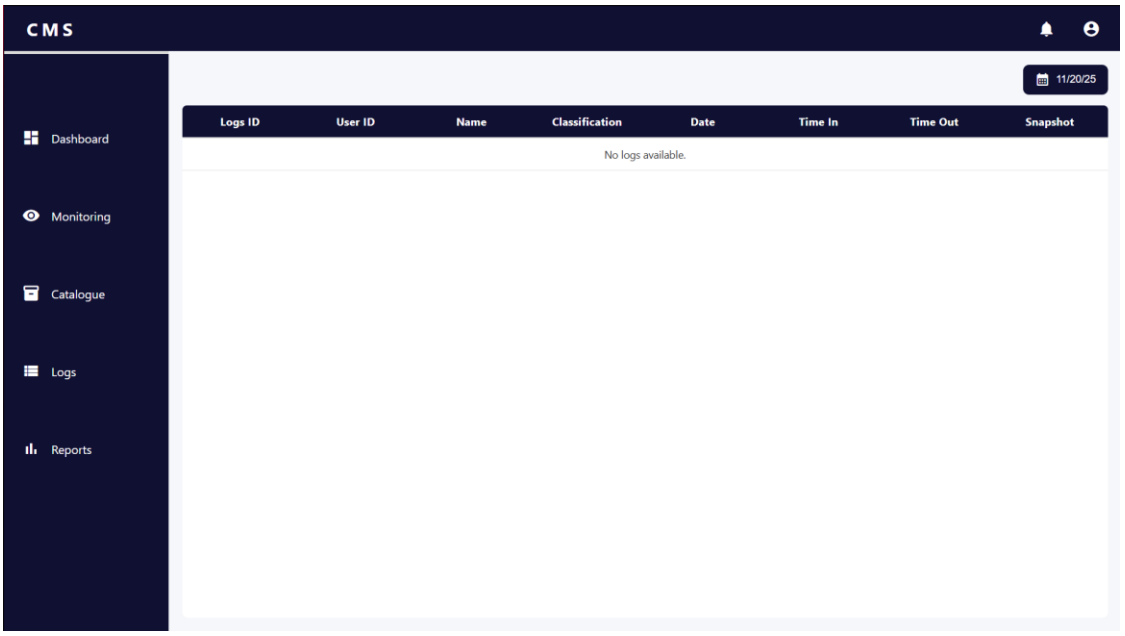
Monitoring Module contains a wide screen view for monitoring individuals.

Catalogue



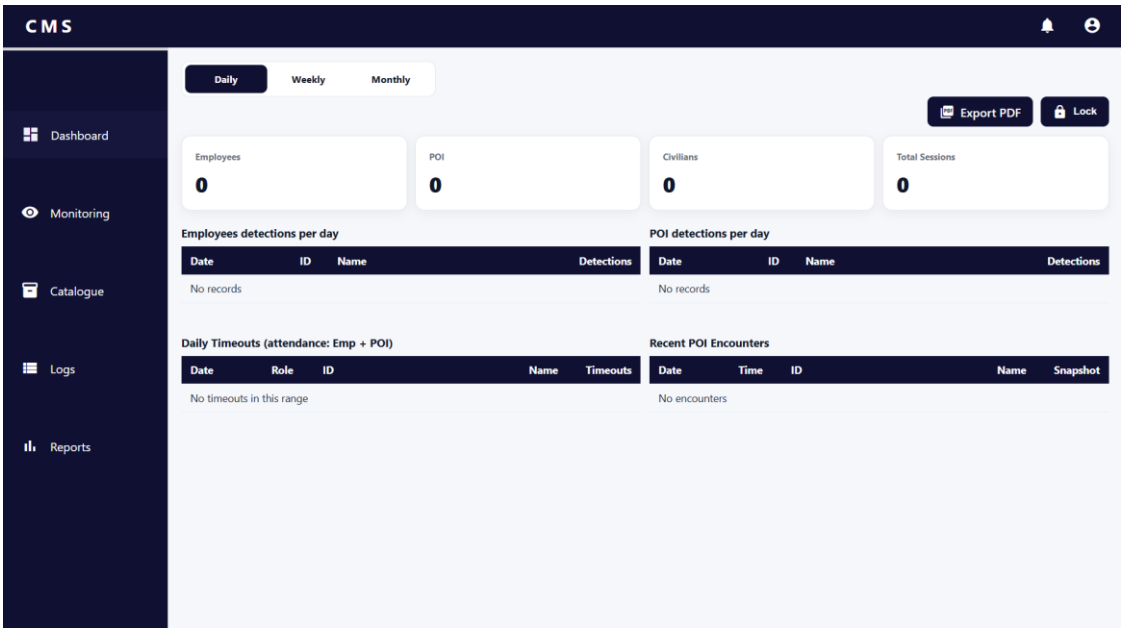
This module contains all the registered individuals in the system.

Logs



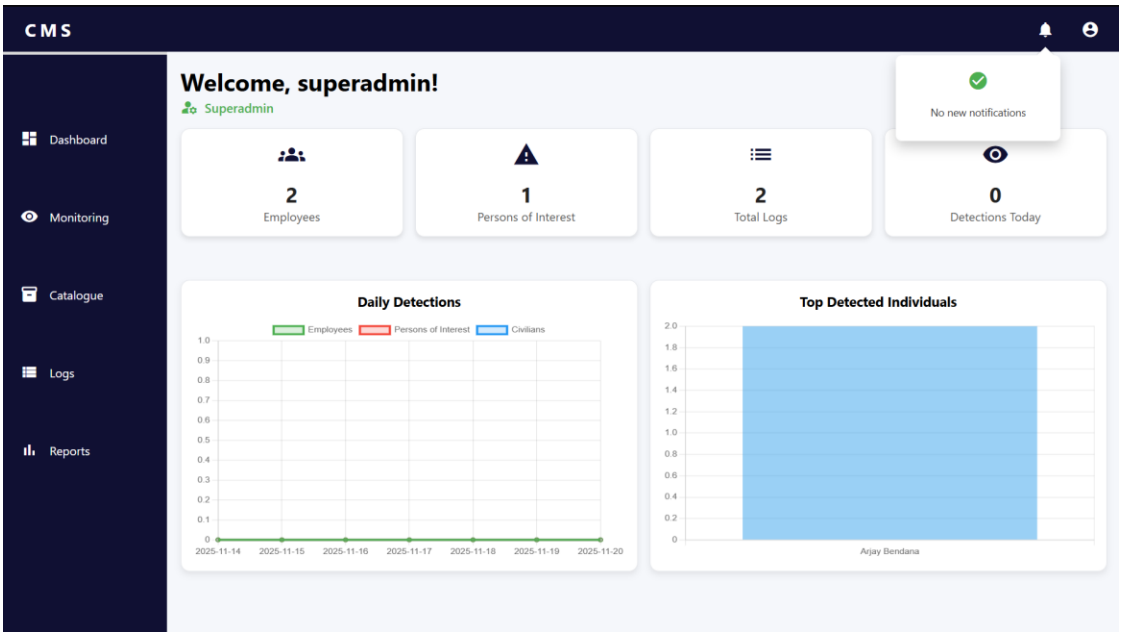
This module displays all the logs on registered individuals once they got detected by the system.

Reports



Reports Module contains all the necessary information. It can be toggled daily, weekly, and monthly. It can also provide a pdf file for reports.

Notification



Once a registered individual got detected, it sends a notification and alarm depending on what role that registered individual have.

APPENDIX F.

Curriculum Vitae



DAEVID LAWRENCE L. ESPORLAS

INFORMATION PERSONAL

Date of Birth: Jan, 25, 2004

Place of Birth: Sto. Tomas, Batangas

Gender: Male

Age: 21

Civil Status: Single

Citizenship: Filipino

Religion: Roman Catholic

Father's Name: Clark Esporlas

Mother's Name: Loida Esporlas

Contact Number: 09266995526

EMAIL: esporlasdaevidlawrence25@gmail.com

EDUCATION

LSPU-LB COLLEGE OF COMPUTER STUDIES DEGREE (WEB AND MOBILE APPLICATION DEVELOPMENT)

2021 - 2025

First Asia Institute of Technology and Humanities (Tanauan) (SENIOR HIGH SCHOOL STEM STRAND)

2019-2020

Sta. Anastacia San Rafael National Highschool

2015-2019

Sta. Anastacia Elementary School

2009-2015

SKILLS

Technical Skills

Programming Language Java, Python C/C++, JS, PHP, SQ



Arjay I. Bendaña

PERSONAL INFORMATION

Date of Birth: Oct, 16, 2002

Place of Birth: Antipolo City, Rizal.

Gender: Male

Age: 23

Civil Status: Single

Citizenship: Filipino

Religion: Born Again

Father's Name: Randy Bendaña

Mother's Name: Marife Bendaña

Contact PHONE: 09504581805

EMAIL:
bendanajhay8@gmail.com

EDUCATION

LSPU-LB COLLEGE OF COMPUTER STUDIES DEGREE (WEB AND MOBILE APPLICATION DEVELOPMENT)

2022 - 2025

DepEd Tayo Los Baños Lalakay Senior High School

2020-2021

Kaysakat National High School

2015-2019

San Joseph Elementary School

2010-2015

SKILLS

Technical Skills

- Microsoft Excel
- HTML, PHP
- Video Editing
- Comunication



MYRLO SHAYNE V. DELFINO III

PERSONAL INFORMATION

Date of Birth: Aug, 26, 2003

Place of Birth: Sta. Cruz Laguna

Gender: Male

Age: 22

Civil Status: Single

Citizenship: Filipino

Religion: Roman Catholic

Father's Name: Sherwin T. Delfino

Mother's Name: Jane V. Delfino

Contact PHONE: 09532961839

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EDUCATION

LSPU-LB COLLEGE OF COMPUTER STUDIES DEGREE (WEB AND MOBILE APPLICATION DEVELOPMENT)

2022 - 2025

Trace College (Los Banos) (SENIOR HIGH SCHOOL GAS STRAND)

2019-2021

Los Banos National High School

2015-2019

Los Banos Central Elementary School

2009-2015

SKILLS

Technical Skills

- Microsoft excel

Communication

APPENDIX G.

Publishable Paper

CAMP MACARIO SAKAY WEB SECURITY SYSTEM: IMPLEMENTATION OF FACIAL IDENTIFICATION AND RECOGNITION USING CONVOLUTIONAL NEURAL NETWORK (CNN)

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User Evaluation, Camp Macario Sakay.

ABSTRACT

This study developed and evaluated a Web-Based Facial Recognition Security System for Camp Macario Sakay to strengthen identity verification, improve monitoring accuracy, and modernize security operations within the camp. The system incorporates real-time facial detection, automated logging, and a web- based interface designed to replace traditional manual verification methods. To measure its technical quality and user acceptance, the study applied the ISO 25010 Software Quality Model and the Technology Acceptance Model (TAM). IT experts rated the system as excellent in functional suitability, reliability, performance efficiency, and portability, with usability and security identified as areas needing further enhancement. End-user evaluation through TAM revealed strong agreement regarding the system's usefulness, ease of use, positive attitude toward use, and intention to adopt the platform. Overall, the results confirm that the system is effective, user-friendly, and capable of supporting improved and efficient camp security operations. Recommendations for refining security and usability features are provided to guide future system development

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I. INTRODUCTION

In today's world, security systems are essential for protecting an establishment from potential threats, ensuring the safety of individuals and assets, and maintaining smooth business operations (Martins Ajaero, 2020). Information Technologies (IT) and electronics opened up new opportunities for surveillance, increasing its effectiveness and resilience. There are cameras everywhere. They can be found, among other places, at retail establishment, manufacturing floors, residences, apartments, landbank, schools, financial organizations, and transit hubs. This security camera uses include anything from quality control monitoring to basic security safety. (Arshiya Jabeen, 2013).

The Philippines had one of the highest rates of the imprisonment per hundred thousand people. Poorer neighborhoods and locations with more inhabitants and higher unemployment rates have notably high crime rates. By Christy Balita on Statista (2023). According to Vito Barcelo from Manila Standard (2024) Filipino families and other establishments who were victimized by common crimes increased to 6.1% in September 2024, according to the latest Social Weather Station survey. The survey said that 6.1 percent of Filipinos reported to be victimized by any of the common crimes (pickpocketing or robbery of personal property, break- ins, carnapping, and physical violence) within the past six months. This is 2.3 points above the 3.8% in June 2024 and the highest since the 8.1% in September 2023, the survey said by Social Weather Stations (SWS, 2024). Nowadays, every establishment is facing serious security concerns, to achieve the level of security they require many staff with specialized training, according to the Philippine Association of Detective & Protective Agency Operators (PADPAO, 2021). Because they are human , these employees make mistakes that might compromise the security level.

A Face Recognition Security

System, which can identify intrusions into restricted or high security locations and reduce human error, is one suggested solution to the aforementioned problem. (Michel Owayjan, et al., 2023). Face recognition is an important function of surveillance systems, with the help of Convolutional Neural Network (CNN). Verifying and identifying individuals who appear in a scene that is frequently caught by a dispersed network of cameras, where the identity can now be precisely determined (Yahia Said, et al., 2020). Using Convolutional Neural Network (CNN) as In essence, face recognition software compares an individual's geometric and facial attributes. The conventional approach stores people's photos in a database so they may be compared to photos that were taken. A variety of characteristics were retrieved and utilized as face features, such as corners, edges, texture descriptors, etc. For categorization, these characteristics were used with machine learning methods. (Ranjeeth K., et al., 2023).

The Camp Macario Sakay faces challenges when it comes to identifying every person that was entering the establishment. These challenges include the lack of a centralized identification system, limited manpower for manual verification, the possibility of human error in recognizing individuals, and the absence of real-time monitoring capabilities. Having an open gate for everyone, the camp wants to have another layer of security to accurately distinguish between employees, civilians, and even recorded criminals to prevent entry by unrecognized or unverified individuals and ensure the safety and integrity of the premises.

The development of a web security system using Convolutional Neural Network (CNN) for facial identification and recognition adds another layer of security in Camp Macario Sakay by providing an intelligent, automated solution for identifying and categorizing individuals entering the premises. This system serves as a reliable tool to distinguish employees, civilians, and recorded criminals in real time, helping security personnel make informed decisions quickly. With the use of a standard webcam installed at the entrance, the system can seamlessly scan faces. By integrating this technology, the camp enhances its security infrastructure, ensures efficient access management, and minimizes potential threats through advanced Convolutional Neural Network-based surveillance.

II. METHODOLOGY

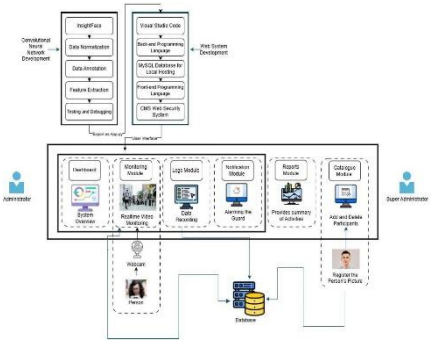


Figure 2. System Architecture

Figure 2 presents the architecture of the Camp General Macario Sakay Web Security System. The pipeline starts with Face Detection and Recognition, where the live camera feed is processed by a lightweight Python service using InsightFace (pre-trained buffalo_l bundle) to detect, align, and embed faces, then perform similarity matching against the Inventory. Rather than training a custom model, the system relies on these pre-trained models and a fixed cosine-similarity threshold with brief smoothing to stabilize labels.

Materials

This section displays the software and hardware used to develop and finish the Camp Macario Sakay Web Security System. It consists of the tools and platforms employed during development, such as programming language, frameworks, and databases.

Insight Face

Insight Face is a high-performance face analysis toolkit built on pre-trained models (e.g., ArcFace backbones bundled in the buffalo_l pipeline) for detection, alignment, embedding, and recognition. In our system, InsightFace processes the live camera feed to detect faces, generate normalized embeddings, and compare them against the Catalogue Module. When a match is found, the individual is identified as an Employee or Person of Interest based on the stored record; if no match exists, the person is labeled a Civilian. This approach enables accurate, real-time recognition without manual dataset collection or custom model training.

Software

The software required to create the application, including its features and descriptions, are essential for building, training, deploying, and managing the facial recognition system. These tools ensure smooth development workflows, system functionality, and integration of the CNN model into the web-based security platform.

Visual Studio Code. A lightweight and powerful code editor developed by Microsoft, widely used for web and software development. In the development of this system, Visual Studio Code served as the primary environment for writing and organizing the project's source code, including the frontend (HTML, CSS, JavaScript) and backend (PHP, SQL) components. Its integration features and support for extensions allowed the researchers to efficiently manage code, debug, and collaborate throughout the system development lifecycle.

XAMPP. A free and open-source web server solution that includes Apache, MySQL, and PHP, providing a complete local environment for developing, hosting, and storing data for web applications. In this project, XAMPP was used to locally host the backend of the facial recognition security system and to store all system data, including employee records, criminal profiles, facial recognition logs, and notifications. It enabled the researchers to run PHP scripts, manage the MySQL database, and ensure that all information captured by the system was securely stored on a local server. This setup allowed smooth interaction between the facial recognition model and system modules such as user registration, monitoring, logs, and notifications during development and deployment phases.

Python. It is used for implementing the core facial recognition features and machine learning processes. It supports powerful libraries such as InsightFace and OpenCV, which handle face detection, embedding generation, and recognition. Python also allows fast development and easy integration with both frontend and backend components.

JavaScript. It is used to add dynamic and interactive features to the web interface. It enables real-time updates, camera integration for live feeds, and responsive interaction between the user and the system. It also handles client-side validations and improves the overall user experience. In the context of this project, JavaScript plays a crucial role in displaying facial recognition results instantly after processing. It also supports the seamless communication between the frontend and backend, ensuring smooth user interaction with the system modules.

HTML. It provides the structure and layout of the web pages for the system. It defines how content such as login forms, dashboards, and user interfaces are organized and displayed in a browser. HTML works in combination with CSS and JavaScript to build a complete and functional frontend.

CSS. It is used to style the HTML elements and make the user interface visually appealing. It controls the layout, color schemes, typography, and responsiveness of the pages. With CSS, the system can maintain a clean and modern design that works well across different devices.

PHP. It is used to handle the server-side operations of the system. It processes requests from the frontend, manages user sessions, and communicates with the database to fetch or store data. PHP is essential for handling the logic behind login authentication, user access, and secure data flow.

SQL (Structured Query Language). It is used to manage and manipulate the system's database. It stores critical data such as employee and visitor profiles, facial recognition results, and access logs. SQL enables efficient queries for inserting, updating, deleting, and retrieving data needed by the system. It ensures that all stored information is organized, accessible, and secure for real-time system operations.

Project Development

Figure 4 presents the Agile Scrum model, which was the methodology applied in this study for the development of the web-based facial recognition security system. This approach allowed the project to progress iteratively, enabling the researchers to build and improve the system in manageable phases. The development cycle followed key Scrum phases, including the creation of a product backlog that outlined all required system features, followed by sprint planning meetings to determine task priorities and assignments. During each sprint, specific features such as facial recognition, monitoring, logging, and notifications were developed and tested.

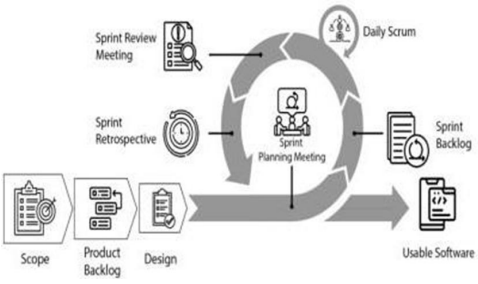


Figure 4. Agile Scrum Model

Evaluation Procedure

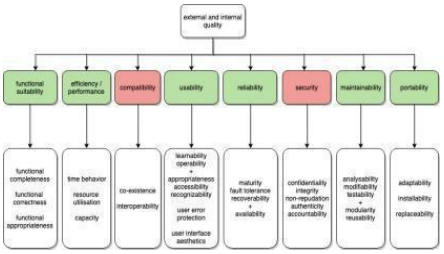


Figure 11. ISO 25010 Quality Characteristics

Figure 11 illustrates the ISO 25010 standard that was used by IT experts to evaluate the technical quality of the developed web-based facial recognition security system. This evaluation focused on key quality characteristics including Functional Suitability, Reliability, Performance Efficiency, Usability, Security, Compatibility, Maintainability, and Portability. ISO 25010 is widely recognized for assessing both the functional and non-functional aspects of software systems, including internal and external system behaviors and architecture. The IT experts assessed whether the system components met these standards to ensure the system's stability and effectiveness before it was tested by end users. They verified that each module such as monitoring, logs, notifications, and participant management functioned as intended and delivered accurate information. A 4-point Likert scale was used to rate each criterion, providing measurable results that reflect the overall quality and readiness of the system for deployment

Standards such as ISO 25010 are applied to ensure the quality of the Camp Macario Sakay Web-System. The project team used a Likert scale to gather feedback from Camp Macario Sakay and IT specialists, evaluating important factors such as usability, efficiency, and reliability. The assessment followed the ISO 25010 quality model, a recognized standard for software quality measurement, focusing on critical aspects including functionality, security, maintainability, and portability.

In addition, the evaluation also considered other quality characteristics from the ISO 25010 model such as performance efficiency, compatibility, interaction capability (usability), reliability, flexibility, and safety. These include sub-factors like functional completeness, time behavior, interoperability, learnability, confidentiality, scalability, and risk identification, ensuring a holistic assessment of the system's quality

Population of the Study

The study involves 33 participants who contributed to examining and assessing the viability of creating a web-based security system. This includes 3 I.T. specialists who provided technical insights on system development, integration, and assessment, 30 Personnel of Camp Macario Sakay who evaluated the system's relevance to facility management. Their combined feedback played a vital role in ensuring the system's functionality, practicality, and alignment with the camp's security needs. A specialist in information technology with experience in online appointment systems also supports the evaluation process by providing insights on system performance. Additionally, they offer recommendations to enhance and optimize the system further.

Category of Respondents	Sample
IT Expert	3
Camp Personnel	30
Total	33

Table 11. Respondents of the Research Project

This study shows the combination of IT Experts and Camp Macario Sakay Personnel provided a well-rounded view of the system's strengths and areas for improvement. Their feedback played a big role in shaping the final version of the platform and making sure it works for the Camp Macario Sakay Security.

III. RESULT AND DISCUSSION

Camp Macario Sakay Web Security System

Figure 13 shows the dashboard also the main interface of Camp Macario Sakay

Web Security System (CMS), highlighting its core features designed to navigate all the important information easily. The interface shows the overall total of employee and person of interest, total logs, detections daily, and graphs. The design emphasizes accessibility and ease of navigation for both admin and super admin.

Figure 13. User Interface for dashboard



Figure 14 section (a) shows the Login Module, where admin and super admin are able to log in and access the system. The system allows the admin and super admin to access features depends on their role. Section (b) System View of Admin displays the accessible features which is limited for this role. It can only access Dashboard Module, Monitoring Module, Logs Module and Notification Module. Section (c) System View of Super Admin displays all the modules that Admin can access, and addition to it, Catalogue Module and Reports Module is now accessible for Super admin role. Section (d) Notification Module provides notification to notify the admin whenever there's detected individual. It comes with an alarm where it rings a sound depends on what role that individual.

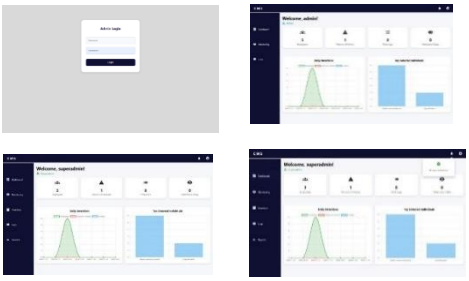


Figure 14. (a) Login Module, (b) System View of Admin, (c) System View of Super Admin, and (d) Notification Module

Figure 15 section (a) shows Monitoring Module, where admin monitors every individual that enters the premises. Section (b) InsightFace Detection, where it detects faces and compare if that individual is registered in Catalogue Module, and if so, it shows a colored line box depends on what role that individual has. The detection comes with a feature called "Anti Spoofing Security" that needed to blink twice to know if that person facing on the camera is a real person or not. If an individual didn't blink and showed an image from phone or printed image, its categorized as "Suspect Spoof" or "Suspect: too small". Section (c) Role Detection Sound, whenever an Employee or Person of Interest got detected, it rangs an alarm depends on what role that individual has when it got detected.



Figure 16 section (a) 4 Pin Lock Security displays a security feature for Catalogue Module and Reports Module. It makes the system more secured even if the user is the super admin where it can access everything in the system. It also has customed 4 numerical pin to use for accessing Catalogue Module and Reports Module. Section (b) Catalogue Module shows two different in tables, Employee Catalogue and Person of Interest Catalogue. Both tables show Users ID, Image, Name, Position/Record, Action (Edit and Delete). Section (c) Add Person/Edit Person Form shows a form to add/edit person. The form contains role selection, full name text box, position text box and image picker. Once submitted, person is added to its desired catalogue or its credentials is edited. It also comes with a popup message for confirmation for add/edit and deleting credential of individual registered in the system.

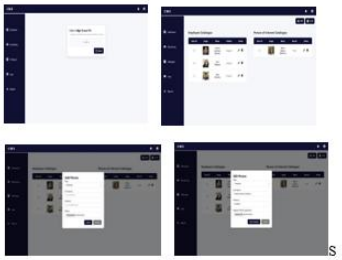


Figure 16. (a) 4 Pin Lock Security, (b) Catalogue Module, (c) Add Person/Edit Person Form

Figure 17 section (a) Logs Module displays the table where every detected individual with role either Employee or Person of Interest is logged. The table contains Logs Id, Users Id, Name, Classification/Role, Date, Time In, Time Out, and Snapshot. Section (b) Date Picker shows a mini calendar to choose what date is desired. By clicking on the date, it can choose what month and year and upon choosing, it shows all the existing logs on that date, if there's no log on that date it shows a plain table.



Figure 17. (a) Logs Module, (b) Date Picker

Figure 18 section (a) Reports Module, it displays the counters for Employee, Civilian, Person of Interest and overall session. It also have table for Employee Detection per day, Person of Interest Detection per day, Daily Timeouts Attendance for Employee and Person of Interest, and lastly Recent Person of Interest Encounters. Section (b) Report to Pdf, upon clicking the Export Pdf button, it makes a pdf file that shows all the counters, all the tables, and for the Person of Interest Encounters, it shows all the snapshots that it got from the Logs Module. The exporting can be toggled daily, weekly and monthly.



Figure 18. (a) Reports Module, (b) Report to Pdf

Table 17. Results of Perceived Usefulness

Test and evaluate the web-based system using the ISO 25010

Table 16 shows the evaluated system by the IT experts using the ISO 25010 software quality model. The results show outstanding performance across most quality attributes. Functional suitability, reliability, performance efficiency, and portability all received perfect mean scores, indicating excellent operation and system responsiveness.

Compatibility and maintainability were also rated highly, reflecting good adaptability and ease of updates. Although usability and security still performed well, they were noted as areas requiring further improvement to better support user experience and data protection. Overall, the composite mean score of 4.88 signifies that the system demonstrates very good quality and meets the required standards effectively.

Table 16. Results of survey answered by the IT expert

Categories	IT1	IT2	IT3	Mean	SD	Verbal interpretation
Functional Suitability	5	4	5	5.00	0.00	EXCELLENT
Reliability	5	5	5	5.00	0.00	EXCELLENT
Performance Efficiency	5	5	5	5.00	0.00	EXCELLENT
Usability	5	5	5	5.00	0.00	FAIR
Security	5	5	5	5.00	0.00	NEEDS IMPROVEMENT
Compatibility	4	4	5	4.33	0.58	EXCELLENT
Maintainability	5	4	4	4.80	0.31	GOOD
Portability	5	5	5	5.00	0.00	EXELLENT
Composite Mean				**4.88**	**0.22**	**GOOD**

The IT experts rated the system with a high overall quality, reflected in a composite mean of 4.88 (SD = 0.22) and a verbal interpretation of Good. The system achieved excellent scores in several key areas, including functional suitability, reliability, performance efficiency, and portability, indicating that it performs its intended functions accurately, operates consistently, and can run smoothly across different environments**.**

Technology Acceptance Model (TAM)

The findings show's that respondents strongly agree that the system is useful in enhancing camp security operations, with a composite mean of 3.66. Users particularly value its ability to efficiently track and log individuals (3.80), while also recognizing that it accurately detects faces, speeds up identity verification, and improves reliability in security tasks (3.63-3.67). Overall, responses are consistent, indicating a positive perception of the system's benefits in security management**.**

Table 17. Results of Perceived Usefulness

ITEM	MEAN	STANDARD DEVIATION	INTERPRETATION
The system accurately detect Faces.	3.67	±0.48	Strongly Agree
The system enhances camp security	3.63	0.56	Strongly Agree
The system makes identityerification faster.	3.67	±0.55	Strongly Agree
The system tracks and logsindividual efficiency.	3.80	±0.41	Strongly Agree
The system improves reliability insecurity operation	3.67	±0.48	Strongly Agree
COMPOSITE MEAN	**3.6**	**±0.49**	**Strongly Agree**

The results shows that respondents strongly agree that the system is easy to use, with a composite mean of 3.68 and a low standard deviation, indicating consistent feedback. Users particularly appreciate the interface design and setup process, both receiving perfect mean scores of 4.00. Learning to use the system and understanding its instructions received slightly lower, yet still strongly positive ratings (3.60), suggesting minor improvements may further enhance usability. Overall, the findings confirm that the system is user-friendly and manageable even for first-time users.

IV. CONCLUSION AND FUTURE WORKS

The Camp General Macario Sakay Web Security System has successfully established a web-based facial recognition technology that employs CNN and InsightFace to improve security, automate identification, and eliminate human error. To guarantee the safe and effective administration of workers, people of interest, and citizens, the system incorporates real-time monitoring, logging, notifications, and role-based access. The project, which was developed using Agile Scrum, used reliable hardware and software tools, such as Python, PHP, MySQL, and XAMPP, and prioritized important features through iterative sprints. Multiple levels of testing verified the operation, performance, and dependability of the system.

Overall, the experiment shows that in a regulated setting like Camp Macario Sakay, a pre-trained CNN-based facial recognition system may efficiently enhance security operations, expedite access management, and support decision- making.

RECOMMENDATION

Based on the evaluation results and the identified limitations of the Web-Based Facial Recognition Security System for Camp General Macario Sakay, the researchers recommend several improvements to enhance its performance and reliability. First, the system's usability may be further improved by refining the interface and providing clearer instructions or guided tutorials to support new users. Strengthening the system's security features is also necessary, particularly through the implementation of stronger data encryption, multi-factor authentication, and regular security audits to protect sensitive facial data.

To address challenges in recognizing faces under poor lighting or partial obstructions, the researchers recommend expanding the dataset and enhancing the facial recognition model with more advanced techniques to improve accuracy. Automating the registration process, such as through bulk uploads or ID scanning, may also help streamline data entry and reduce manual workload.

Moreover, the system would benefit from the integration of mobile access or multi-channel notifications to ensure real-time monitoring and faster response from security personnel. Future work should also consider testing the system in other locations to validate its performance in different environments and to gather broader user feedback. Finally, compliance with data privacy regulations must be maintained through clear policies on consent, storage, and data protection to ensure ethical and responsible system use.

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